MATH 3402 Notes

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Chapter 1

Topology

1.1 Topological Spaces

Functional analysis is a study of topological spaces.

Definition 1.1.1: Topology

Let X be a set. A topology $\mathcal{T} \subseteq \mathcal{P}(X)$ is a collection of subsets of X such that:

(i) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$.

(ii) If $U_\alpha \in \mathcal{T}$ for all α, then $\bigcup_\alpha U_\alpha \in \mathcal{T}$ (closed under arbitrary unions).

(iii) If $U_1, \ldots, U_n \in \mathcal{T}$, then $\bigcap_{k=1}^n U_k \in \mathcal{T}$ (closed under finite intersections).

A set X endowed with a topology is called a **topological space**, denoted by $(X, \mathcal{T})$.

A set $U \in \mathcal{T}$ is called an **open set**. A set $V \subseteq X$ is **closed** if its complement $X \setminus V$ is in $\mathcal{T}$.

Using this terminology, one can say that a **topological space** is a set X together with a collection of subsets of X, called *open sets*, such that $\emptyset$ and X are both open, and such that arbitrary unions and finite intersections of open sets are open.

A set can be both open and closed, e.g., $\emptyset, X$.

Example.

- Trivial topology: $\mathcal{T} = \{\emptyset, X\}$.
- Discrete topology: $\mathcal{T} = \mathcal{P}(X)$ (the power set of X).

Example.

Suppose X is an infinite set. The **finite complement topology** is defined as:

$$\mathcal{T} = \{V \subseteq X \mid X \setminus V \text{ is finite}\} \cup \{\emptyset\}$$

Alternative way: Let X be a set, let $\mathcal{T}_f$ be the collection of all subsets U of X such that $X - U$ is either finite or is all of X. Then $\mathcal{T}_f$ is a topology on X, called the *finite complement topology*.

Proof. Both X and $\emptyset$ are in $\mathcal{T}_f$, since $X - X$ is finite and $X - \emptyset$ is all of X.

If $\{U_\alpha\}$ is an indexed family of nonempty elements of $\mathcal{T}_f$, to show that $\bigcup U_\alpha$ is in $\mathcal{T}_f$, we compute

$$X - \bigcup U_\alpha = \bigcap (X - U_\alpha).$$

The latter set is finite because each set $X - U_\alpha$ is finite.

If $U_1, \ldots, U_n$ are nonempty elements of $\mathcal{T}_f$, to show that $\bigcap U_i$ is in $\mathcal{T}_f$, we compute

$$X - \bigcap U_i = \bigcup_{i=1}^{n} (X - U_i).$$

The latter set is a finite union of finite sets and, therefore, finite.

□

Definition 1.1.2: Metric Spaces

Let X be a set, and let $d : X \times X \to \mathbb{R}$ be a function satisfying:

(i) $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$ (non-degeneracy).

(ii) $d(x, y) = d(y, x)$ (symmetry).

(iii) $d(x, y) \leq d(x, z) + d(z, y)$ (triangle inequality).

Let (X, d) be a metric space. A metric ball is defined as:

$$B(x, \epsilon) = \{y \in X \mid d(x, y) < \epsilon\}$$

The metric topology is given by:

$$\mathcal{T}_d = \{U \subseteq X \mid \forall x \in U, \exists \epsilon > 0 \text{ such that } B(x, \epsilon) \subseteq U\}$$

1.2 Basis for a topology

Definition 1.2.1: Subbasis

A **subbasis** $\mathcal{B}$ of a topological space $(X, \mathcal{T})$ is a collection of subsets of X such that:

$$X = \bigcup_{B \in \mathcal{B}} B$$

The *topology generated by the subbasis* $\mathcal{S}$ is defined to be the collection $\mathcal{T}$ of all unions of finite intersections of elements of $\mathcal{S}$.

Definition 1.2.2: Basis

A **basis** $\mathcal{B}$ is a subbasis such that for all $B_1, B_2 \in \mathcal{B}$ and for all $x \in B_1 \cap B_2$, there exists $B_3 \in \mathcal{B}$ such that: $x \in B_3 \subseteq B_1 \cap B_2$. B is called basis elements.

Definition 1.2.3

The topology $\mathcal{T}_\mathcal{B}$ generated by $\mathcal{B}$ is defined as: $U \in \mathcal{T}_\mathcal{B}$ if $\forall x \in U, \exists B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Theorem 1.2.4

$\mathcal{T}_\mathcal{B}$ is a topology on X.

Proof. We must show that $\mathcal{T}_\mathcal{B}$ is closed under arbitrary unions and finite intersections (axiom (1) is clear).

If $\{U_i\}_{i \in I} \subseteq \mathcal{T}_\mathcal{B}$, let $A := \bigcup_{i \in I} U_i$. Let $x \in A$. By definition, $\exists i_0 \in I$ such that $x \in U_{i_0}$. Now, $U_{i_0} \in \mathcal{T}_\mathcal{B}$, so $\exists B \in \mathcal{B}$ such that $x \in B \subseteq U_{i_0}$. Since $U_{i_0} \subseteq A$, this shows that $A \in \mathcal{T}_\mathcal{B}$ by the definition of $\mathcal{T}_\mathcal{B}$.

For finite intersections, let us take $U_1, U_2 \in \mathcal{T}_{\mathcal{B}}$, so let $A := U_1 \cap U_2$, and let $x \in A$. By definition of $\mathcal{T}_{\mathcal{B}}$, $\exists B_1, B_2 \in \mathcal{B}$ such that

$$x \in B_1 \subseteq U_1, \quad x \in B_2 \subseteq U_2.$$

$$\Rightarrow x \in B_1 \cap B_2 \subseteq U_1 \cap U_2.$$

By definition of basis, $\exists B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

$$\Rightarrow x \in B_3 \subseteq A.$$

Thus, $A \in \mathcal{T}_{\mathcal{B}}$.

By induction, it follows that the same holds for $U_1 \cap \cdots \cap U_n$. □

Theorem 1.2.5

Given a subbasis $\mathcal{S}$ for a set X, the collection $\mathcal{B} := \{S_1 \cap \cdots \cap S_r : S_i \in \mathcal{S}, r \in \mathbb{Z}_{>0}\}$ is a basis.

Proof. From the above remark, we know that $\mathcal{S} \subseteq \mathcal{B}$, so $\mathcal{B}$ clearly covers X. Now let

$$\mathcal{B}_1 = S_1 \cap \cdots \cap S_m \in \mathcal{B}, \quad \mathcal{B}_2 = S_1' \cap \cdots \cap S_n' \in \mathcal{B}.$$

Then by construction, $\mathcal{B}_3 = \mathcal{B}_1 \cap \mathcal{B}_2 = (S_1 \cap \cdots \cap S_m) \cap (S_1' \cap \cdots \cap S_n') \in \mathcal{B}$. Thus, $\mathcal{B}$ is a basis. □

Here are three topology generated by their basis.

Definition 1.2.6

The **standard topology** on $\mathbb{R}$ is the topology generated by the basis:

$$\mathcal{B} = \{(a, b) \mid -\infty < a < b < \infty\}$$

Definition 1.2.7

The **lower limit topology** on $\mathbb{R}$ is generated by the basis $\mathcal{B}_l = \{[a, b) \mid a < b\}$. $\mathbb{R}$ endowed with $\mathcal{T}_{\mathbb{R}_l}$ is denoted as $\mathbb{R}_l$.

Definition 1.2.8

Let $K := \{1/n : n \in \mathbb{Z}_{>0}\} \subseteq \mathbb{R}$. The topology generated by $\mathcal{B}'' = \{(a, b) : a, b \in \mathbb{R}\} \cup \{(a, b) \setminus K : a, b \in \mathbb{R}\}$ is called the *K-topology* and $\mathbb{R}_K := (\mathbb{R}, \mathcal{T}_{\mathcal{B}''})$.

Definition 1.2.9

Suppose that $\mathcal{T}$ and $\mathcal{T}'$ are two topologies on a given set X. If $\mathcal{T}' \supseteq \mathcal{T}$, we say that $\mathcal{T}'$ is *finer* than $\mathcal{T}$; if $\mathcal{T}'$ *properly* contains $\mathcal{T}$, we say that $\mathcal{T}'$ is *strictly finer* than $\mathcal{T}$.

Theorem 1.2.10

Three results about the above mentioned topologies:

1. *The lower limit topology is strictly finer than the standard topology.*
2. *The K-topology is strictly finer than the standard topology.*
3. *$\mathbb{R}_\ell$ (lower limit topology) and $\mathbb{R}_K$ (K-topology) are not comparable, i.e., $\mathcal{T}_{\mathcal{B}}' \nsubseteq \mathcal{T}_{\mathcal{B}}''$ and $\mathcal{T}_{\mathcal{B}}'' \nsubseteq \mathcal{T}_{\mathcal{B}}'$.*

Proof. We proceed in order:

1. Take $U \subseteq \mathbb{R}$ a standard open subset, and let $x \in U$. $\exists a, b \in \mathbb{R}$ such that $x \in (a, b) \subseteq U$ (i.e., $U \in \mathcal{T}_{\mathcal{B}_{\text{std}}}$). Let $c \in (a, x)$. Then, $x \in [c, b) \subseteq (a, b) \subseteq U$, meaning U is open in $\mathbb{R}_\ell$. However, while $[a, b)$ is open in $\mathbb{R}_\ell$, it is not open in $\mathbb{R}$ (i.e., there is no open interval (a, b) such that $x \in (a, b)$ and $(a, b) \subseteq [c, b)$). Therefore, $\mathcal{T}_{\mathcal{B}_{\text{std}}} \subset \mathcal{T}_{\mathcal{B}'}$.

2. Similarly, for a standard open subset $U \subseteq \mathbb{R}$ and $x \in U$, $U \in \mathcal{T}_{\mathcal{B}''}$, so U is open in $\mathbb{R}_K$. Meanwhile, given $B = (-1, 1) \setminus K \in \mathcal{B}$ and $x = 0 \in B$, there is no open interval (a, b) such that $0 \in (a, b)$ and $(a, b) \subseteq B$. Thus, $\mathcal{T}_{\mathcal{B}_{\text{std}}} \subset \mathcal{T}_{\mathcal{B}''}$.

3. Let $U = [a, b) \in \mathcal{T}_{\mathcal{B}'}$ and pick $x = a \in U$. Notice that there is no $B \in \mathcal{B}''$ such that $x \in B$ and $B \subseteq [a, b)$, so $U \notin \mathcal{T}_{\mathcal{B}''}$. At the same time, $\mathbb{R} \setminus K$ is open in $\mathbb{R}_K$ (clearly), but not in $\mathbb{R}_\ell$, since every $V \in \mathcal{T}_{\mathcal{B}'}$ containing 0 contains at least one element of K. The result follows.

□

1.3 Closed Sets and Limit Points

Definition 1.3.1: Neighborhood

Let $(X, \mathcal{T})$ be a topological space. We declare $U \in \mathcal{T}$ to be a neighborhood of $x \in X$ if $x \in U$.

Example.

> If (X, d) is a metric space, a neighborhood of $x \in X$ is given by:
>
> $$B(x, \epsilon) = \{y \in X : d(x, y) < \epsilon\}$$

If $X = \mathbb{C}$ is endowed with the trivial topology $\{\emptyset, X\}$, then any subset $\Omega \subseteq X$ that $\Omega \neq \emptyset$ and $\Omega \neq X$

$$\Rightarrow \Omega \text{ will be neither open nor closed.}$$

Definition 1.3.2

Let $A \subseteq (X, \mathcal{T})$. A point x is called:

(a) **An interior point of A:** if $x \in A$ and there exists a neighborhood U of x such that $U \subseteq A$. The set of interior points of A is denoted A° or $\text{int}(A)$.

(b) **A limit point of A:** if $x \in X$ for every neighborhood U of x, we have $U \cap A \neq \emptyset$. The set of limit points of A corresponds to the **closure** $\overline{A}$ or $\text{cl}(A)$.

(c) **An accumulation point of A:** if for every neighborhood U of x, we have $U \cap (A \setminus \{x\}) \neq \emptyset$. The set of accumulation points is denoted by A'.

Example.

> Let $\mathbb{R}$ have the standard topology and define $A = [0, 1) \cup \{2\}$.

Then, $\text{int}(A) = (0, 1)$, closure of $A = \overline{A} = [0, 1] \cup \{2\}, A' = [0, 1]$

1.4 Product Topology

Definition 1.4.1: Product Topology

Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. The **product topology** on $X \times Y$ is the topology generated by:

$$\mathcal{B}_{X \times Y} = \{U \times V \mid U \in \mathcal{T}_X, V \in \mathcal{T}_Y\}$$

Theorem 1.4.2

$\mathcal{B}_{X\times Y}$ is a basis.

Proof. Let us check that $\mathcal{B}$ is a basis. The first condition is trivial, since $X \times Y$ is itself a basis element. The second condition is almost as easy, since the intersection of any two basis elements $U_1 \times V_1$ and $U_2 \times V_2$ is another basis element. For

$$(U_1 \times V_1) \cap (U_2 \times V_2) \;=\; (U_1 \cap U_2) \;\times\; (V_1 \cap V_2),$$

and the latter set is a basis element because $U_1 \cap U_2$ and $V_1 \cap V_2$ are open in X and Y, respectively.

$$(x, y) \in U_3 \times V_3 = (U_1 \times V_1) \cap (U_2 \times V_2).$$

□

If $\mathcal{B}_X, \mathcal{B}_Y$ are bases for $\mathcal{T}_X, \mathcal{T}_Y$ respectively, then $\mathcal{B} := \{A \times B : A \in \mathcal{B}_X, B \in \mathcal{B}_Y\}$ is a basis for the product topology. The product topology is sometimes denoted by $\mathcal{T}_X \times \mathcal{T}_Y$.

Definition 1.4.3

The standard topology on $\mathbb{R}^n$ is the product of the standard topologies on $\mathbb{R}$.

1.5 The Subspace Topology

Definition 1.5.1: Subspace topology

Let $Y \subseteq (X, \mathcal{T})$. The **subspace topology** on Y, induced from $\mathcal{T}_X$, is the topology:

$$\mathcal{T}_Y = \{U \cap Y \mid U \in \mathcal{T}_X\}$$

$\mathcal{T}_Y$ is a topology.

Proof. It contains $\emptyset$ and Y because

$$\emptyset = Y \cap \emptyset \quad \text{and} \quad Y = Y \cap X,$$

where $\emptyset$ and X are elements of $\mathcal{T}$. The fact that it is closed under finite intersections and arbitrary unions follows from the equations

$$(U_1 \cap Y) \cap \cdots \cap (U_n \cap Y) = (U_1 \cap \cdots \cap U_n) \cap Y,$$

$$\bigcup_{\alpha \in J} (U_\alpha \cap Y) = \left(\bigcup_{\alpha \in J} U_\alpha \right) \cap Y.$$

□

Lemma 1.5.2

If $\mathcal{B}$ is a basis for $\mathcal{T}_X$, then the collection $\mathcal{B}' := \{B \cap Y : B \in \mathcal{B}\}$ is a basis for the subspace topology on Y.

Proof. Since $\mathcal{B}$ covers X, clearly $\mathcal{B}'$ covers Y. Now for all $\mathcal{B}'_1, \mathcal{B}'_2 \in \mathcal{B}'$, $\exists \mathcal{B}_1, \mathcal{B}_2 \in \mathcal{B}$ such that

$$\mathcal{B}'_1 \cap \mathcal{B}'_2 = (B_1 \cap Y) \cap (B_2 \cap Y) = (B_1 \cap Y) \cap (Y \cap B_2) = (B_1 \cap B_2) \cap Y \in \mathcal{B}',$$

where $B_1 \cap B_2 \in \mathcal{B}$ because $\mathcal{B}$ is a basis. Thus, $\mathcal{B}'$ is a basis.

□

Example.

> Let $Y := [0,1) \subseteq \mathbb{R}$ and consider on Y the subspace topology induced from the standard topology on $\mathbb{R}$.

Proof. Then, any open interval $(a,b) \subseteq Y$ is open in Y since $(a,b) \cap [0,1) = (a,b)$.

Let $c \in (0,1)$. Then: $[0,c) = (-1,c) \cap [0,1)$, where $[0,c)$ and $(-1,c)$ are open in $\mathbb{R}$ and $[0,1)$ is the Y. All the sets $[0,c)$, $0 < c < 1$, are open in Y! □

Example.

> Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, and consider on $A \subseteq X$ and $B \subseteq Y$ the respective subspace topologies $\mathcal{T}_A$ and $\mathcal{T}_B$. Let $\mathcal{T}_{A\times B}$ denote the subspace topology on $A \times B \subseteq X \times Y$ induced by the product topology $\mathcal{T}_X \times \mathcal{T}_Y$ on $X \times Y$. Then, $\mathcal{T}_{A\times B} = \mathcal{T}_A \times \mathcal{T}_B$.

Proof.

$$\begin{aligned}
\mathcal{T}_{A\times B} &= \{W \cap (A \times B) : W \in \mathcal{T}_X \times \mathcal{T}_Y\} \\
&= \{(U \times V) \cap (A \times B) : U \in \mathcal{T}_X, V \in \mathcal{T}_Y\} \\
&= \{(U \cap A) \times (V \cap B) : U \in \mathcal{T}_X, V \in \mathcal{T}_Y\} \\
&= \mathcal{T}_A \times \mathcal{T}_B.
\end{aligned}$$

□

1.6 Continuity

1.6.1 Continuity of a Function

Definition 1.6.1

Let X, Y be sets, and let $f : X \to Y$ be a map. Let $A \subseteq X$, $B \subseteq Y$.

- $f(A) := \{y \in Y \mid y = f(a) \text{ for some } a \in A\}$ is the image of A.
- $f^{-1}(B) := \{x \in X \mid f(x) \in B\}$ is the preimage of B.

Proposition 1.6.2

Let $\{A_i\} \subseteq X$, $\{B_j\} \subseteq Y$.

1. $f^{-1}\left(\bigcup_j B_j\right) = \bigcup_j f^{-1}(B_j)$; $\quad f\left(\bigcup_i A_i\right) = \bigcup_i f(A_i)$
2. $f^{-1}\left(\bigcap_j B_j\right) = \bigcap_j f^{-1}(B_j)$; $\quad f\left(\bigcap_i A_i\right) \subseteq \bigcap_i f(A_i)$
3. $f^{-1}(B_1 \setminus B_2) = f^{-1}(B_1) \setminus f^{-1}(B_2)$; $\quad f(A_1) \setminus f(A_2) \subseteq f(A_1 \setminus A_2)$
4. $B_1 \subseteq B_2 \Rightarrow f^{-1}(B_1) \subseteq f^{-1}(B_2)$; $\quad A_1 \subseteq A_2 \Rightarrow f(A_1) \subseteq f(A_2)$
5. $f(f^{-1}(B)) \subseteq B$; $\quad A \subseteq f^{-1}(f(A))$

Definition 1.6.3

Let X, Y be topological spaces. A map $f : X \to Y$ is **continuous** if for every open set $V \subseteq Y$, $f^{-1}(V) \subseteq X$ is open.

Definition 1.6.4

We say that $f : X \to Y$ is **continuous at** $x \in X$ if for every open set $V \subseteq Y$ with $f(x) \in V$, there exists an open neighborhood U of x such that $f(U) \subseteq V$.

Continuity of f depends on the map f but also on the topologies specified for its domain and range.

Example.

> Let X be endowed with the discrete topology, then every map $f : X \to Y$ is continuous.
>
> $$V \subseteq Y \text{ open } \Rightarrow f^{-1}(V) \subseteq X \text{ automatically open.}$$

In particular, if $\mathbb{N}$ is endowed with the discrete topology, then any sequence of real numbers $\{a_n\}$ can be viewed as a continuous function $a : \mathbb{N} \to \mathbb{R}$.

Example.

> Let $f : \mathbb{R}_{\text{std}} \to \mathbb{R}_\ell$ be the identity map, where $f(x) = x$. ($\mathbb{R}_\ell$ represents the lower limit topology). Then f is not continuous. There are open sets in $\mathbb{R}_\ell$ that are not open in $\mathbb{R}_{\text{std}}$, e.g., $[a, b)$, and $f^{-1}([a, b)) = [a, b)$ is not open in $\mathbb{R}_{\text{std}}$.

Definition 1.6.5: (ϵ, δ)-continuity

Let $f : \mathbb{R} \to \mathbb{R}$, continuous at $x \in \mathbb{R}$ if and only if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that } |x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon.$$

Theorem 1.6.6

(ϵ, δ)-continuity $\Leftrightarrow$ continuity for $f : \mathbb{R}_{\text{std}} \to \mathbb{R}_{\text{std}}$

Proof. $\Rightarrow$: suppose $f : \mathbb{R} \to \mathbb{R}$ is (ϵ, δ)-continuous at $x \in \mathbb{R}$. Let $V \subseteq \mathbb{R}$ be an open neighborhood of $f(x)$. Since open intervals (a, b) form a basis for the standard topology,

$$\Rightarrow \exists \epsilon > 0 \text{ such that } (f(x) - \epsilon, f(x) + \epsilon) \subseteq V.$$

$$(\epsilon, \delta)\text{-continuous} \Rightarrow \exists \delta > 0 \text{ such that}$$
$$f((x - \delta, x + \delta)) \subseteq (f(x) - \epsilon, f(x) + \epsilon) \subseteq V$$

$\Leftarrow$: almost the same. □

Theorem 1.6.7

Let $f : X \to Y$ be a map of topological spaces. Then the following are equivalent:

1. f is continuous,
2. f is continuous at every point $x \in X$,
3. $\forall A \subseteq X, \quad f(\overline{A}) \subseteq \overline{f(A)}$ *(sequential continuity)*,
4. $\forall B \subseteq Y$ closed set, $f^{-1}(B)$ is closed in X.

Proof. (1) $\Rightarrow$ (2): let $x \in X$, and let $V \subseteq Y$ be an open neighborhood of $f(x)$. From (1), $U := f^{-1}(V)$ is open in X, and moreover, $x \in U$. Finally,

$$f(U) = f(f^{-1}(V)) \subseteq V.$$

(2) $\Rightarrow$ (3): let $f(x) \in f(\overline{A})$, with $x \in \overline{A}$. Proceed by contradiction and suppose that $f(x) \notin \overline{f(A)}$, i.e., $f(x)$ is not a limit point for $f(A) \subseteq Y$. Then $\exists V \subseteq Y$ open, a neighborhood of $f(x)$, such that $V \cap f(A) = \emptyset$. From (2), $\exists U \subseteq X$ an open neighborhood of x such that $f(U) \subseteq V$.

$$\Rightarrow f(U \cap A) \subseteq f(U) \cap f(A) \subseteq V \cap f(A) = \emptyset.$$

But $x \in \overline{A} \Rightarrow U \cap A \neq \emptyset$, which is a contradiction.

(3) $\Rightarrow$ (4): note that a set is closed if and only if it is equal to its closure. Let $B \subseteq Y$ be a closed set, and define $A := f^{-1}(B)$. WTS that $A = \overline{A} \iff \overline{A} \subseteq A$. From (3), we have:

$$f(\overline{A}) \subseteq \overline{f(A)}$$

$$\overline{f(A)} = \overline{f(f^{-1}(B))} \subseteq \overline{B} = B.$$

Thus,

$$\overline{A} \subseteq f^{-1}(f(\overline{A})) \subseteq f^{-1}(\overline{f(A)}) \subseteq f^{-1}(B) = A.$$

$$\Rightarrow \overline{A} = A \Rightarrow A \text{ is closed.}$$

(4) $\Rightarrow$ (1): if $V \subseteq Y$ is open, then $Y \backslash V$ is closed. Therefore, $X \backslash f^{-1}(V) = f^{-1}(Y) \backslash f^{-1}(V) = f^{-1}(Y \backslash V)$ is closed by (4).

$$\Rightarrow f^{-1}(V) \text{ is open.}$$

□

Theorem 1.6.8

Let $f : X \to Y$ be a function between topological spaces.

1. *If f is constant, i.e., $\exists y_0 \in Y$ such that $f(x) = y_0$ for all $x \in X$, then it is continuous.*
2. *If $A \subseteq X$ is endowed with the subspace topology, then the* ***inclusion map*** *$i : A \to X$, $a \mapsto a$, is continuous.*
3. *If $f : X \to Y$ and $g : Y \to Z$ are continuous, then so is $g \circ f : X \to Z$.*

Theorem 1.6.9

Let $f : X \to Y$ be a function between topological spaces.

1. *If $f : X \to Y$ is continuous, and $A \subseteq X$ has the subspace topology, then $f|_A : A \to Y$ is continuous.*
2. *If $f : X \to Y$ is continuous and $Y \subseteq Z$ has the subspace topology, then $f : X \to Z$ is continuous.*
3. *If $X = \bigcup_{i \in I} U_i$ is a union of open sets and $f : X \to Y$ is such that $f|_{U_i}$ is continuous for all $i \in I$, then f is continuous.*

1.6.2 Homeomorphisms

Definition 1.6.10

We say that $f : X \to Y$ is **open** if $f(U) \in \mathcal{T}_Y, \forall U \in \mathcal{T}_X$.

Definition 1.6.11

If $f : X \to Y$ is a bijection, continuous, and open, then f is called a **homeomorphism**.

Homeomorphism is isomorphism in the category of topological spaces.

In particular, topological invariants are preserved by homeomorphisms. Some common topological invariants include: Connectivity, Compactness, Dimension, Degree, fundamental group, homology groups...

Proposition 1.6.12

Let $f : X \to Y$ be a continuous bijection. The following are equivalent:

1. *f is a homeomorphism*
2. *f is an open map*
3. *f is a closed map*
4. *X is compact and Y is Hausdorff*

Example.

Continuous bijection but not homeomorphic.

Proof. Let $f : \mathbb{R}_{\text{std}} \to \mathbb{R}_l$ be the identity map. Forward direction is continuous (lower-limit topology is *finer*). Inverse is *not* continuous $\Rightarrow$ not a homeomorphism □

Example.

$$\mathbb{R} \cong (-1, 1) \quad \text{via} \quad f(x) = \frac{x}{1 - x^2}.$$

$\cong$ is homeomorphic

1.7 Connected Spaces

Definition 1.7.1

Let $(X, \mathcal{T})$ be a topological space. We say X is **disconnected** if there exist non-empty open sets $U, V \subset X$ such that $X = U \cup V$ and $U \cap V = \emptyset$. We say X is **connected** if X is not disconnected.

Definition 1.7.2

A subset $A \subseteq X$ is connected if it is connected with respect to the induced topology (subspace topology).

Proposition 1.7.3

For a topological space X the following conditions are equivalent:

1. X is connected, that is, it cannot be divided into two disjoint non-empty open sets.
2. The only subsets of X which are both open and closed are X and the empty set.

Example.

$X = \{a, b\}$, $\mathcal{T} = \{\emptyset, X\} \Rightarrow X$ is connected.

Example.

Subset $X = [-1, 0) \cup (0, 1] \subset \mathbb{R}_{\text{std}}$.

Proof. Let $U = (-2, 0) \cap X$, $V = X \cap (0, 2)$. Both U and V are open. Since $X = U \cup V$ and $U \cap V = \emptyset$, we conclude that X is disconnected. □

Example.

The rationals $\mathbb{Q}$ are not connected. $\mathbb{Q} \hookrightarrow \mathbb{R}_{\text{std}}$, the connected subsets (connected components) are singletons $\{q\}$, for $q \in \mathbb{Q}$.

Proof. If Y is a subspace of $\mathbb{Q}$ containing two points p and q, one can choose an irrational number a lying between p and q, and write Y as the union of the open sets

$$Y \cap (-\infty, a) \quad \text{and} \quad Y \cap (a, +\infty).$$

□

Example.

Let $X = \{(x, 0) \mid x > 0\} \cup \{(x, 1/x) \mid x > 0\} \subseteq \mathbb{R}^2$. Then X is disconnected

Proof.

$$U = \{(x, y) \mid y < 1/3x, x > 0\}, \quad V = \{(x, y) \mid y > 1/2x, x > 0\}$$

□

Lemma 1.7.4

If the sets C and D form a separation of X, and if Y is a connected subspace of X, then Y lies entirely within either C or D.

Proposition 1.7.5

If $A \subseteq X$ is connected, and $B \subseteq X$ such that $A \subseteq B \subseteq \bar{A}$, then B is connected.

Proof. Let A be connected and let $A \subset B \subset \bar{A}$. Suppose that $B = C \cup D$ is a separation of B. By Lemma above, the set A must lie entirely in C or in D; suppose that $A \subset C$. Then $\bar{A} \subset \bar{C} \Rightarrow A \subset B \subset \bar{A} \subset \bar{C}$

Since $\bar{C}$ and D are disjoint, B cannot intersect D. This contradicts the fact that D is a nonempty subset of B.

□

Proposition 1.7.6

Let $\{Y_\alpha\}_{\alpha \in A}$ be connected subsets of X with $\bigcap_{\alpha \in A} Y_\alpha \neq \emptyset$. Then $\bigcup_{\alpha \in A} Y_\alpha$ is connected.

Proof. Let $\{A_\alpha\}$ be a collection of connected subspaces of a space X; let p be a point of $\bigcap A_\alpha$. We prove that the space $Y = \bigcup A_\alpha$ is connected. Suppose that $Y = C \cup D$ is a separation of Y.

The point p is in one of the sets C or D; suppose $p \in C$. Since A_α is connected, it must lie entirely in either C or D, and it cannot lie in D because it contains the point p of C. Hence $A_\alpha \subset C$ for every α, so that $\bigcup A_\alpha \subset C$, contradicting the fact that D is nonempty.

□

Theorem 1.7.7

Let A be a connected subset of X, and let $f : X \to Y$ be continuous. Then $f(A)$ is connected.

Proof. Since $f|_A : A \to f(A)$, assume $f : X \to Y$ is surjective, hence $Y = f(A)$. Suppose Y is disconnected, meaning there exist disjoint non-empty open sets $U, V \subset Y$ such that $Y = U \cup V$ and $U \cap V = \emptyset$. Then:

$$X = f^{-1}(Y) = f^{-1}(U \cup V) = f^{-1}(U) \cup f^{-1}(V).$$

Since f is continuous, $f^{-1}(U)$ and $f^{-1}(V)$ are open in X. Since $U \cap V = \emptyset$ and f is surjective, it follows that:

$$f^{-1}(U) \cap f^{-1}(V) = \emptyset.$$

This contradicts the assumption that X is connected, hence proving the theorem.

□

Theorem 1.7.8

The finite product of connected sets is connected under the product topology.

Theorem 1.7.9

The connected open subsets of $\mathbb{R}$ are given by the intervals (a, b), where $-\infty \leq a < b \leq +\infty$.

If not open like (a,b], then it is closed connected of R.

Proof. Proof by contradiction. Suppose $(a, b) = U \cup V$ for some non-empty open sets $U, V \subset (a, b)$, with $U \cap V = \emptyset$. Let $x \in U$ and $y \in V$. Assume without loss of generality that $x < y$. Define: $c = \sup\{u \in U : u < y\}$. Since c contains x, it is non-empty. There are two cases, depending on whether $c \in U$ or $c \in V$.

Case 1: Suppose $c \in U$. Since U is open, there exists $\varepsilon > 0$ such that $(c - \varepsilon, c + \varepsilon) \subseteq U$. This implies:

$$(c - \varepsilon, c + \varepsilon) \cap V \subseteq U \cap V = \emptyset.$$

Thus, $y \notin (c - \varepsilon, c + \varepsilon)$, which means:

$$c + \frac{\varepsilon}{2} \in U, \quad c + \frac{\varepsilon}{2} < y.$$

This contradicts the definition of c.

Case 2: Suppose $c \in V$. Let $\varepsilon > 0$ such that $(c - \varepsilon, c + \varepsilon) \subseteq V \cap (a, y)$. For any $u \in U \cap (a, y)$ with $u < y$, we have $u < c$. Since $U \cap (c - \varepsilon, c + \varepsilon) = \emptyset$, we also have $u \leq c - \varepsilon$. This again contradicts the definition of c as $\sup\{u \in U : u < y\}$.

□

Definition 1.7.10

We say a topological space X is path-connected if for all $x, y \in X$, there exists a continuous map: $\gamma : [0, 1] \to X$ such that $\gamma(0) = x$ and $\gamma(1) = y$.

Theorem 1.7.11

A path-connected space X is connected

Proof. Suppose $X = A \cup B$ is a separation of X. Let $f : [a, b] \to X$ be any path in X. Being the continuous image of a connected set, the set $f([a, b])$ is connected, so that it lies entirely in either A or B. Therefore, there is no path in X joining a point of A to a point of B, contrary to the assumption that X is path connected.

□

Example.

> There exist spaces that are connected but not path-connected. For example:
>
> $$\bar{S} = \{(0, y) \in \mathbb{R}^2 : -1 \leq y \leq 1\} \cup \{(x, \sin(1/x)) : 0 < x \leq 1\}.$$

Proof. See tute sheet 3. □

1.8 Compact Spaces

Definition 1.8.1: Open Covering

An **open covering** of a topological space X is a collection of open subsets $\{U_\alpha\}_{\alpha \in A}$ of X such that:

$$X = \bigcup_{\alpha \in A} U_\alpha.$$

Definition 1.8.2: Subcover

If $\{U_\beta\}_{\beta \in B}$ is a subcollection of $\{U_\alpha\}_{\alpha \in A}$, i.e., $B \subseteq A$, then $\{U_\beta\}_{\beta \in B}$ is called a **subcover** if:

$$X = \bigcup_{\beta \in B} U_\beta.$$

Definition 1.8.3

A topological space X is said to be

- compact if every open covering has a finite subcover.
- sequentially compact if every sequence $(x_n)_{n \in \mathbb{N}}$ in X has a convergent subsequence (whose limit is in X).

Example.

> 1. If X is finite $\Rightarrow$ it is compact.
> 2. $X = \mathbb{R}$ is **not** compact: take $U_n = (-n, n) \subseteq \mathbb{R}$. Then $\{U_n\}_{n \in \mathbb{N}}$ is an open covering, without any open subcovering. (Same for any $X \subseteq \mathbb{R}$ which is unbounded.)
> 3. $X = \{0\} \cup \{1/n \mid n \in \mathbb{N}\} \subseteq \mathbb{R}$ is compact. Indeed, the open set $U \cap X$ (where $U \subseteq \mathbb{R}$ is open) containing $\{0\}$ will also contain all but finitely many of the elements in X.
> 4. $(0, 1] \subseteq \mathbb{R}$ is **not** compact: take $\{(1/n, 1] \mid n \in \mathbb{N}\}$ as an open covering.

Theorem 1.8.4

If $f : X \to Y$, $f \in C^0$, X is compact $\Rightarrow f(X)$ is compact.

Proof. Let $\{U_\alpha\}_{\alpha \in A}$ be a covering of $f(X)$, where $U_\alpha \in \mathcal{T}_Y$. Then $\{f^{-1}(U_\alpha)\}_{\alpha \in A}$ is an open (by continuous) covering of X. Since X is compact, there is a finite index set B such that $X = \bigcup_{\beta \in B} f^{-1}(U_\beta)$

$$\Rightarrow f(X) \subseteq f\left(\bigcup_{\beta \in B} f^{-1}(U_\beta)\right) = \bigcup_{\beta \in B} f\left(f^{-1}(U_\beta)\right) \subseteq \bigcup_{\beta \in B} U_\beta$$

which gives a finite subcover of $f(X)$. □

Theorem 1.8.5

Let $A \subseteq X$ with A closed and X compact. Then, A is compact.

Proof. Let $\{U_i\}_{i\in I}$ be an open cover of A, and write $U_i = V_i \cap A$, with V_i open in X. Then $\{X \setminus A\} \cup \{V_i\}_{i\in I}$ is an open covering of X. Since X is compact, there exists a finite subcovering:

$$X = \bigcup_{k=1}^{n} V_{i_k} \cup (X \setminus A).$$

It's clear that $\{U_{i_k}\}_{k=1}^{n}$ is then a finite subcovering for A. □

In general, not every compact subset is closed.

Definition 1.8.6

A topological space X is called a Hausdorff space if if $\forall x, y \in X$, $x \neq y$, $\exists U, V$ open with $x \in U$, $y \in V$ and $U \cap V = \emptyset$.

Example.

> Metric spaces are Hausdorff. Indeed, given $x, y \in X$, $x \neq y$, we have $\varepsilon := d(x, y) > 0$
>
> $$\Rightarrow \mathcal{B}_{\varepsilon/2}(x) \cap \mathcal{B}_{\varepsilon/2}(y) = \emptyset.$$

Lemma 1.8.7

In a Hausdorff space X, points $\{x\}$ are closed sets.

Proof. Let $y \in X \setminus \{x\}$ and take U and V open with $x \in U$, $y \in V$ and $U \cap V = \emptyset$. Then $y \in V \subseteq X \setminus \{x\}$ implies that y is an interior point of $X \setminus \{x\}$, meaning $X \setminus \{x\}$ is open. □

Theorem 1.8.8:

Let X be a Hausdorff space, and let $K \subseteq X$ be compact. Then, K is closed.

Proof. Let us show that $A := X \setminus K$ is open (which then means K is closed, by definition). Let $a \in A$. For each $x \in K$, using the Hausdorff condition, we take neighbourhoods U_x and V_x of x and a, respectively, so that $U_x \cap V_x = \emptyset$. Since $\{U_x\}_{x\in K}$ cover the compact set K, we may extract a finite subcovering:

$$K \subseteq \bigcup_{i=1}^{n} U_{x_i}, \quad x_1, \ldots, x_n \in K.$$

Let $V := \bigcap_{i=1}^{n} V_{x_i}$, an open neighbourhood of a. Then $V \cap U_{x_i} = \emptyset$ for all $i \in \{1, \ldots, n\}$. It follows that $V \cap K = \emptyset$, meaning $a \in V \subseteq X \setminus K$ and so $X \setminus K$ is open. □

Theorem 1.8.9: Tychonoff's Theorem

Let $\{K_i\}_{i\in I}$ be a family of compact spaces. Then, $K := \prod_{i\in I} K_i$ is compact with the product topology.

Proof. Beyond this course. □

Theorem 1.8.10

Let X and Y be topological spaces, with $f : X \to Y$ a bijective continuous function. If X is compact and Y is Hausdorff, then f is a homeomorphism.

Proof. Let $A \subseteq X$ be closed; then A is compact. Therefore, $f(A)$ is compact because f is continuous. Since Y is Hausdorff, $f(A)$ is closed in Y.

□

1.8.1 Compactness in Metric Spaces

Lemma 1.8.11

Closed intervals $[a, b] \subseteq \mathbb{R}$ are compact.

Theorem 1.8.12

A metric space (X, d) is compact if and only if it is sequentially compact.

Definition 1.8.13

A subset S of a metric space (X, d) is bounded if $\exists r > 0$ such that $\forall x, y \in S$, we have $d(x, y) < r$. Equivalently, if $\sup\{d(x, y) : x, y \in X\}$ is finite. The metric space (X, d) is a bounded metric space (or d is a bounded metric) if X is bounded as a subset of itself.

Definition 1.8.14

Let (X, d) be a metric space. $\forall \varepsilon > 0$, we have an open covering of X by ε-balls $\{B_\varepsilon(x)\}_{x \in X}$. We say X is totally bounded if there exists a finite subcovering $\{B_\varepsilon(x_i)\}_{i=1}^n$.

Observe that compact metric spaces are totally bounded and that total boundedness implies boundedness.

Lemma 1.8.15

Every compact metric space is also complete

Proof. Let $\{x_n\}_{n \in \mathbb{N}}$ be a Cauchy sequence. Let $\{x_{n_k}\}_{k \in \mathbb{N}}$ be a convergent subsequence, say $x_{n_k} \to \bar{x} \in X$. We then claim that $x_n \to \bar{x}$. Indeed, given $\varepsilon > 0$, let N such that $d(x_n, x_m) < \varepsilon/2, \quad \forall n, m \geq N$. Choose also $K \in \mathbb{N}$ so that

$$d(x_{n_k}, \bar{x}) < \varepsilon/2, \quad \forall k \geq K.$$

By enlarging K if necessary, we may also assume that $n_k \geq N$ for all $k \geq K$. Thus, for all $n \geq N$, take any $n_k \geq N$:

$$d(x_n, \bar{x}) \leq d(x_n, x_{n_k}) + d(x_{n_k}, \bar{x}) < \varepsilon.$$

Thus, $x_n \to \bar{x}$. □

Theorem 1.8.16

A metric space is compact if and only if it is complete and totally bounded.

In $\mathbb{R}^n$, the above conditions can be simplified.

Theorem 1.8.17: Heine-Borel

A subset $K \subseteq \mathbb{R}^n$ is compact if and only if it is closed and bounded.

Proof. $\Rightarrow$ Since $\mathbb{R}^n$ is Hausdorff (it is a metric space) and K is compact, it must be closed. Regarding boundedness, let $B_R(\bar{0}) := \{x \in \mathbb{R}^n \mid d(x, \bar{0}) < R\}, R > 0, \bar{0} = (0, \dots 0)$. The family $\{B_n(\bar{0})\}_{n\in\mathbb{N}}$ is an open covering for K. Extracting a finite subcovering $\{B_{n_k}(\bar{0})\}_{k=1}^m$ with $n_1 < \cdots < n_m$, we get

$$K \subseteq \bigcup_{k=1}^{m} B_{n_k}(\bar{0}) = B_{n_m}(\bar{0}).$$

Thus, K is bounded.

$\Leftarrow$ Let $R > 0$ such that $K \subseteq B_R(\bar{0})$. Then, since $B_R(\bar{0}) \subseteq [-R, R]^n$, we have $K \subseteq [-R, R]^n$. Now, $[-R, R] \subseteq \mathbb{R}$ is compact $\Rightarrow [-R, R]^n$ is compact as well. Since K is a closed subset of a compact set, K is compact itself. $\square$

Compactness can also be understood more explicitly in other metric spaces. For instance, consider the space

$$X := C([a, b], \mathbb{R}) = \{f : [a, b] \to \mathbb{R} \mid f \text{ continuous}\}$$

endowed with the supremum metric $d_\infty(f, g) := \sup\{|f(x) - g(x)| : x \in [a, b]\}$.

Definition 1.8.18

A subspace Y of a topological space $(X, \mathcal{T})$ is precompact if $\overline{Y}$ (closure of Y) is compact.

Definition 1.8.19

$K \subseteq (X, d_\infty)$ is equicontinuous if $\forall \varepsilon > 0\ \exists \delta > 0$ s.t. $|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon, \quad \forall x, y \in [a, b],\ \forall f \in K$.

Lemma 1.8.20

In complete metric space, K is pre-compact if and only if it is totally bounded.

Proof. This is directly from the above theorem. Note that K is totally bounded if and only if $\bar{K}$ is totally bounded. $\square$

Theorem 1.8.21: Arzelà-Ascoli

A set $K \subseteq C([a, b], \mathbb{R}, d_\infty)$ is pre-compact if and only if it is bounded and equicontinuous.

We only prove $\Rightarrow$ direction.

Proof. First assume that $\bar{K}$ is compact. Then it is totally bounded and then bounded. To prove that K is equicontinuous, take an $\varepsilon > 0$. Total boundedness gives a finite $\varepsilon/3$-net $N = \{f_1, f_2, \dots, f_n\} \subset K$. Then for any $f \in K$, select $f_i \in N$ such that

$$\|f - f_i\| = \sup\{|f(x) - f_i(x)| : x \in X\} < \varepsilon/3.$$

Since each $f_i \in N \subset C([a.b], \mathbb{R})$ is uniformly continuous, for $i = 1, 2, 3, \dots, n$, there exists $\delta_i > 0$ such that

$$|x - y| < \delta_i \implies |f_i(x) - f_i(y)| < \varepsilon/3, \quad x, y \in X.$$

Let $\delta = \min\{\delta_1, \delta_2, \dots, \delta_n\}$. Then for any $x, y \in [a, b]$

$$\begin{aligned} |x - y| < \delta \implies & |f(x) - f(y)| \le |f(x) - f_i(x)| + |f_i(x) - f_i(y)| + |f_i(y) - f(y)| \\ & < \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon. \end{aligned}$$

□

1.9 The Countability Axioms

Definition 1.9.1

A set S is countable if S is finite or if there exists a bijection $f : S \to \mathbb{N}$.

Example.

- $\mathbb{N}$, $\mathbb{Q}$ are countable.
- $\mathbb{R}$ is not countable.

Definition 1.9.2: Neighborhood Basis

Given a topological space $(X, \mathcal{T})$, a set $\mathcal{B}_x$ of neighborhoods of $x \in X$ is a **neighborhood basis** if for every $U \in \mathcal{T}$ with $x \in U$, there exists $B \in \mathcal{B}_x$ such that $x \in B \subseteq U$.

Definition 1.9.3

A subset $A \subseteq X$ is said to be **dense** in X if $\overline{A} = X$.

In general, the easiest (and most common) way to prove the density of a subset $A \subseteq X$ is by showing that every non-empty open subset of X has a non-empty intersection with A.

Definition 1.9.4

Let $(X, \mathcal{T})$ be a topological space. We say X is

1. **First countable** if $\forall x \in X$, there is a countable neighbourhood basis (i.e., B_x is countable).
2. **Second countable** if there is a basis $\mathcal{B}$ of $\mathcal{T}$ that is countable.
3. **Separable** if there exists $A \subseteq X$ that is countable and dense.
4. **Lindelöf** if every open cover has a countable subcover.

Theorem 1.9.5

N_2 implies N_1

Proof. If $\mathcal{B}$ is a countable basis for the topology of X, then the subset of $\mathcal{B}$ consisting of those basis elements containing the point x is a countable basis at x.

□

Example.

$\mathbb{R}_{\text{std}}$ is second countable: $\{(a, b) : a, b \in \mathbb{Q}\}$ is a countable basis. $\mathbb{R}^n$ is also second countable: the collection of all products of intervals having rational end points.

Theorem 1.9.6

Let $(X, \mathcal{T})$ be a second countable topological space. Then X is separable and Lindelöf.

Proof. Let X be N_2 and $\mathcal{B} = \{B_n\}_{n\in\mathbb{Z}_{>0}}$ a countable basis. For every $n \in \mathbb{Z}_{>0}$ such that $B_n \neq \emptyset$, pick $x_n \in B_n$ and let $\mathcal{D} := \{x_n : n \in \mathbb{Z}_{>0}\}$. Then $\overline{\mathcal{D}} = X$. Indeed, $\forall U \in X$, $\exists B_n \in \mathcal{B}$ such that $B_n \in U$. Thus, $U \cap D \neq \emptyset$.

Given an open covering $\{U_i\}_{i\in I}$ of X, for all $n \in \mathbb{Z}_{>0}$ choose (if possible) an $i_n \in I$ so that $\mathcal{B}_n \subseteq U_{i_n}$. Then for a subset $\mathcal{P} \subseteq \mathbb{Z}_{>0}$, we have a subcovering $\{U_{i_n}\}_{n\in\mathcal{P}}$. Indeed, if $x \notin \bigcup_{n\in\mathbb{Z}_{>0}} U_{i_n}$, then choose $i \in I$ with $x \in U_i$, and by definition of a basis, $\exists \mathcal{B}_n \in \mathcal{B}$ such that $x \in \mathcal{B}_n \subseteq U_{i_n}$, meaning $x \in U_{i_n}$, a contradiction. Therefore, X is Lindelöf.

□

Theorem 1.9.7

For a metric space, $N_2 \Leftrightarrow$ separable $\Leftrightarrow$ Lindelöf.

Proof. A metric space is a topological space, so we have already shown that being N_2 implies separability and being Lindelöf. It remains to show that separability implies being Lindelöf and that being Lindelöf implies being N_2. We begin with the former. Let $\mathcal{A}$ be a countably dense subset of X. We claim that the set of open balls in X below is a basis for X:

$$\mathcal{B} = \{B_{1/n}(a) \subseteq X : a \in \mathcal{A}, n \in \mathbb{Z}_{>0}\}.$$

Note that $\mathcal{B}$ is countable since it is in bijection with $\mathcal{A} \times \mathbb{Z}_{>0}$. Thus, let $x \in U \subseteq X$ for U open; since (X, d) is a metric space, $x \in B_\varepsilon(x) \subseteq U$ for some $\varepsilon > 0$. Let n be such that $1/n < \varepsilon/2$. Then, since $\mathcal{A}$ is dense, some $a \in \mathcal{A}$ is contained in $B_{1/n}(x)$, and conversely $x \in B_{1/n}(a)$. By the triangle inequality, $x \in B_{1/n}(a) \subseteq U$, so $\mathcal{B}$ is a basis, as desired.

Now the latter. The set of open balls

$$\mathcal{B}_n = \{B_{1/n}(x) \subseteq X : x \in X\}$$

is an open cover of X for each $n \in \mathbb{Z}_{>0}$; since X is Lindelöf, it has a countable subcover $\mathcal{B}'_n$. We claim that $\mathcal{B} = \bigcup_{n\in\mathbb{Z}_{>0}} \mathcal{B}'_n$ is a basis for X. Note that it is countable since it is a countable union of countable sets. Thus, let $x \in B_\varepsilon(x) \subseteq U$ for some $\varepsilon > 0$, as before, and let n be such that $1/n < \varepsilon/2$. Then, $\exists x' \in X$ such that $B_{1/n}(x') \in \mathcal{B}'_n$ contains x. By the triangle inequality, $x \in B_{1/n}(x') \subseteq U$, so indeed, $\mathcal{B}$ is a (countable) basis for X and therefore X is N_2.

□

Theorem 1.9.8

$\mathbb{R}_l$ is N_1, not N_2. It is also separable and Lindelöf.

Proof. Given $x \in \mathbb{R}_\ell$, the set of all basis elements of the form $[x, x + 1/n)$ is a countable basis at x. To show that $\overline{\mathbb{Q}} = \mathbb{R}_\ell$, it suffices to show that $\mathbb{Q} \cap B \neq \emptyset$ for all basis elements $B \in \mathcal{B}'$ (basis of the lower limit topology). Indeed, these elements are $[a, b)$ with $a < b$, so since $\mathbb{Q} \cap (a, b) \neq \emptyset$ for all $a, b \in \mathbb{R}$ with $a < b$, clearly $\mathbb{Q} \cap [a, b) \neq \emptyset$.

To see that $\mathbb{R}_\ell$ has no countable basis, let $\mathcal{B}$ be a basis for $\mathbb{R}_\ell$. Choose for each $x \in X$ an element $\mathcal{B}_x \in \mathcal{B}$ such that $x \in \mathcal{B}_x \subseteq [x, x + 1)$. Now consider another point $y \in X$. If $x \neq y$, then $\mathcal{B}_x \neq \mathcal{B}_y$, since $x = \inf \mathcal{B}_x$ and $y = \inf \mathcal{B}_y$. Therefore, $|\mathcal{B}| \geq |\mathbb{R}| > \aleph_0$, meaning $\mathcal{B}$ is uncountable. □

1.9.1 First-Countable Spaces - Sequences

Theorem 1.9.9

Metric spaces are N_1

Proof. $B^d_{\frac{1}{n}}(x) = \{y \in X : d(x, y) < \frac{1}{n}\}$ □

Definition 1.9.10

Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in a topological space $(X, \mathcal{T})$. We say (x_n) **converges** to $x \in X$, denoted $\lim_{n\to\infty} x_n \to x$ (often shortened to $x_n \to x$), if $\forall U$ neighbourhoods of x, $\exists N \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq N$.

Note that limits need not be unique. For instance, if $X = \{a, b\}$ and $\mathcal{T} = \{\emptyset, X\}$, then every sequence in X converges to both a and b. The limits of sequences will be unique only when X is *Hausdorff*.

Theorem 1.9.11

Let $(X, \mathcal{T})$ be an N_1 topological space and $A \subseteq X$. Then

$$\overline{A} = \{x \in X : \exists (x_n)_{n\in\mathbb{N}} \subseteq A \text{ such that } x_n \to x\}.$$

Theorem 1.9.12

Let X and Y be topological spaces with X being N_1. Then $f : X \to Y$ is continuous if and only if $\forall (x_n)_{n\in\mathbb{N}}$ with $x_n \to x \in X$, we have $f(x_n) \to f(x)$.

1.10 The Separation Axioms

Definition 1.10.1

Let X be a topological space. We say X is of class:

(1) $\mathbf{T_1}$, if points $\{x\}$ are closed in X. (Equivalently, if $\forall x, y \in X, x \neq y \Rightarrow \exists U, V$ open, $x \in U, y \in V$, $x \notin V, y \notin U$.)

(2) $\mathbf{T_2}$ (or **Hausdorff**), if $\forall x, y \in X, x \neq y$, $\exists U, V$ open with $U \cap V = \emptyset$ such that $x \in U, y \in V$.

(3) $\mathbf{T_3}$ (or **regular**), if it is T_2 and in addition $\forall x \in X, \forall F \subseteq X$ closed, $x \notin F$, $\exists U, V$ open, $U \cap V = \emptyset$, such that $x \in U$ and $F \subseteq V$.

(4) $\mathbf{T_4}$ (or **normal**), if it is T_1 and in addition $\forall F_1, F_2 \subset X$ closed sets with $F_1 \cap F_2 = \emptyset$, $\exists U, V \subseteq X$ open, $U \cap V = \emptyset$, such that $F_1 \subseteq U$ and $F_2 \subseteq V$.

*Clearly, $T_n \Rightarrow T_{n-1}$ $\forall n = 2, 3, 4$. But **none** of the converse assertions hold.*

Theorem 1.10.2

Let $k \in \{2, 3\}$. Then, any subspace of a T_k space is T_k (in the subspace topology), and any product of T_k spaces is T_k (in the product topology).

Example.

> Consider finite-complement topology $\mathcal{T}$, where: $\mathcal{T} = \{A \subseteq X \mid X \setminus A \text{ is finite}\} \cup \{\emptyset\}$.

Proof. Then, X is T_1: indeed, by definition, finite sets are closed. However, X is not T_2: Any two non-empty open sets intersect non-trivially. Indeed, $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$ is finite. □

Example.

> $\mathbb{R}_K = (\mathbb{R}, \mathcal{T}_K)$, where a basis for $\mathcal{T}_K$ is $\mathcal{B}_K = \{(a, b) : a, b \in \mathbb{R}\} \cup \{(a, b) \setminus K : a, b \in \mathbb{R}\}$, where $K = \{1/n : n \in \mathbb{Z}_{>0}\}$. Then $\mathbb{R}_K$ is T_2 but not T_3.

Proof. Indeed, $\mathbb{R}_{\text{std}}$ is T_2 and $\mathbb{R}_K$ is finer, thus it is also T_2. However, $\mathbb{R}_K$ has more closed sets; for instance, $K \subseteq \mathbb{R}_K$ is closed. But we cannot separate $\{0\}$ from K with disjoint open sets. Indeed, suppose that $\exists U, V$ open and disjoint such that $0 \in U$ and $K \subseteq V$. Choose a basis element containing 0 and lying in U. It must be of the form $(a, b) \setminus K$, since each basis element of the form (a, b) containing 0 intersects K. Choose n such that $1/n \in (a, b)$. Then choose a basis element about $1/n$ contained in V; it must be a basis element of the form (c, d). Finally, choose z such that $z < 1/n$ and $z > \max\{c, 1/(n+1)\}$. Then $z \in U$ and $z \in V$, so $U \cap V \neq \emptyset$, a contradiction. □

Example.

$\mathbb{R}_\ell = (\mathbb{R}, \mathcal{T}_\ell)$, where $\mathcal{B}_\ell = \{[a, b) : a, b \in \mathbb{R}\}$ is a basis for $\mathcal{T}_\ell$. We claim that $\mathbb{R}_\ell$ is normal (T_4).

Proof. Let $A, B \in \mathbb{R}_\ell$ be closed and $A \cap B = \emptyset$; then $\mathbb{R}_\ell \setminus A$ and $\mathbb{R}_\ell \setminus B$ are open. For $\forall a \in A$, take a basis element $[a, x_a) \subseteq \mathbb{R}_\ell \setminus B$. Same for each $b \in B$: $[b, x_b) \subseteq \mathbb{R}_\ell \setminus A$. Then set

$$U := \bigcup_{a \in A} [a, x_a), \quad V := \bigcup_{b \in B} [b, x_b).$$

Since for each pair $[a, x_a) \cap [b, x_b) = \emptyset$ (otherwise either $b \in [a, x_a)$ or $a \in [b, x_b)$, which cannot hold by construction), we have $U \cap V = \emptyset$. □

Example.

$\mathbb{R}_\ell^2 := \mathbb{R}_\ell \times \mathbb{R}_\ell$ (product topology) is T_3. However, it can be shown that it's **not** T_4. This shows that

$$T_3 \not\Rightarrow T_4,$$

and that the product of normal spaces need not be normal.

Theorem 1.10.3

Metric spaces are T_4.

Proof. Let (X, d) be a metric space, $A, B \subseteq X$ closed and $A \cap B = \emptyset$. For each $a \in A$, $a \in X \setminus B$ (since $A \cap B = \emptyset$), so $\exists \varepsilon_a > 0$ such that $B_{\varepsilon_a}(a) \subseteq X \setminus B$ (i.e., $B_{\varepsilon_a}(a) \cap B = \emptyset$). Similarly for $b \in B$.

Now set

$$U := \bigcup_{a \in A} B_{\varepsilon_a/2}(a), \quad V := \bigcup_{b \in B} B_{\varepsilon_b/2}(b).$$

They are open and disjoint: if $x \in U \cap V$, then $x \in B_{\varepsilon_a/2}(a) \cap B_{\varepsilon_b/2}(b)$. Without loss of generality, suppose $\varepsilon_a \leq \varepsilon_b$. Then

$$d(a, b) \leq d(x, a) + d(x, b) < \frac{\varepsilon_a}{2} + \frac{\varepsilon_b}{2} \leq \varepsilon_b \quad \Longrightarrow \quad a \in B_{\varepsilon_b}(b) \quad \Longrightarrow \quad A \cap B_{\varepsilon_b}(b) \neq \emptyset,$$

which is a contradiction. □

Chapter 2

Completeness: Banach and Hilbert spaces

2.1 Normed Space

We will consider vector spaces over $\mathbb{K}$, $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

Definition 2.1.1

A **topological vector space** is a vector space V endowed with a topology such that the operations

$$+ : V \times V \to V, \qquad \cdot : \mathbb{K} \times V \to V,$$
$$+ : (x, y) = x + y, \qquad \cdot : (\lambda, y) = \lambda y,$$

are continuous.

Example.

> $\mathbb{R}^n$, $\mathbb{C}^n$ with their usual metric topologies (induced by the Euclidean distance), are topological vector spaces.

Definition 2.1.2

A subset $W \subseteq V$ of a topological vector space is called a **subspace** if it is a linear subspace and it is closed for the topology.

Linear subspace: satisfy $+$ and $\cdot$. Closed: contains all the limit points.

Definition 2.1.3

Given $\Omega \subseteq V$, the **linear span** of Ω is the vector subspace

$$\operatorname{span}(\Omega) := \left\{ \sum_{k=1}^{m} \lambda_k v_k \ : \ v_k \in \Omega, \ \lambda_k \in \mathbb{K}, \ m \in \mathbb{N} \right\}.$$

Its closure $\overline{\operatorname{span}(\Omega)}$ is the **closed linear span** of Ω. We say Ω is **total** in V if $\overline{\operatorname{span}(\Omega)} = V$.

It will be useful to assume a stronger link between the topology and the algebraic structure of V.

Definition 2.1.4

Let E be a vector space over $\mathbb{K} = \mathbb{R}, \mathbb{C}$. A function $\|\cdot\| : E \to \mathbb{R}$ is called a **semi-norm** if:

1. $\|x\| \geq 0, \quad \forall x \in E.$
2. (*Homogeneity*) $\|\lambda x\| = |\lambda| \cdot \|x\|, \quad \forall \lambda \in \mathbb{K}, \forall x \in E.$
3. (*Triangle inequality*) $\|x + y\| \leq \|x\| + \|y\|, \quad \forall x, y \in E.$

If in 1) we have in addition $\|x\| = 0$ iff $x = 0$, $\|\cdot\|$ is called a **norm**.
Then, the pair $(E, \|\cdot\|)$ is called a (real or complex) **normed space**.

3) is equivalent to 3') $\|x - y\| \geq |\|x\| - \|y\||$

Proposition 2.1.5

Let $(E, \|\cdot\|)$ be a normed space, and define

$$d : E \times E \to \mathbb{R}, \quad d(x, y) := \|x - y\|.$$

Then, (E, d) is a metric space (sometimes d is denoted by $d_{\|\cdot\|}$). □

Proof. Let $x, y, z \in E$. The fact that $d(x, y) \geq 0$ for all $x, y \in E$ follows immediately from the first property of a norm. We check the other three properties of a metric:

1. $d(x, y) = 0 \Leftrightarrow x = y$. $\|\cdot\|$ is a norm, so $\|x - y\| = 0 \Leftrightarrow x - y = 0 \Leftrightarrow x = y$.
2. $d(x, y) = d(y, x)$. We observe that
$$\|x - y\| = \|-(y - x)\| = \|y - x\|.$$
3. $d(x, y) \leq d(x, z) + d(z, y)$. Using the triangle inequality for norms,
$$\|x - y\| = \|(x - z) + (z - y)\| \leq \|x - z\| + \|z - y\|.$$

□

Example.

> The discrete metric on $\mathbb{R}$, given by
>
> $$d(x, y) = \begin{cases} 1, & x \neq y \\ 0, & x = y \end{cases} \quad \text{doesn't come from a norm (no homogeneity).}$$

The topology induced by a norm $\|\cdot\|$ on E is by definition the metric topology induced by $d_{\|\cdot\|}$. In particular,

$$\lim_{n\to\infty} x_n = x \quad \Leftrightarrow \quad \|x_n - x\| \xrightarrow[n\to\infty]{} 0.$$

for all $(x_n) \in E$.

Proposition 2.1.6

Let $(E, \|\cdot\|)$ be a normed space.

1. The norm $\|\cdot\| : E \to \mathbb{R}$ is a continuous function.
2. $(E, \|\cdot\|)$ is a topological vector space.

Proof. Since E is a metric space, it is Hausdorff and so we may use sequences. (if Hausdorff, then sequence only has one limit.)

1. Let $(x_n)_{n\in\mathbb{N}}$ be in E, $x_n \to x$ implies $\|x_n - x\| \to 0$. But by triangle inequality,

$$0 \leq |\|x_n\| - \|x\|| \leq \|x_n - x\| \quad \forall n$$

$$\Rightarrow \|x_n\| \xrightarrow[n\to\infty]{} \|x\|.$$

2. We first show that $+ : E \times E \to E$ is continuous. Take a sequence $((x_n, y_n))_{n\in\mathbb{N}} \subseteq E \times E$ with $(x_n, y_n) \to (x, y)$ as $n \to \infty$. Then $x_n \to x$ and $y_n \to y$ under the product topology. Thus, by the triangle inequality,

$$\|(x_n + y_n) - (x + y)\| \leq \underbrace{\|x_n - x\|}_{\to 0} + \underbrace{\|y_n - y\|}_{\to 0} \implies x_n + y_n \to x + y.$$

For $\cdot : \mathbb{K} \times E \to E$, take $(x_n)_{n\in\mathbb{N}} \subseteq E$ such that $x_n \to x$ as $n \to \infty$ and let $k \in \mathbb{K}$. Then

$$\|kx_n - kx\| = |k| \cdot \underbrace{\|x_n - x\|}_{\to 0} \implies kx_n \to kx,$$

as desired.

□

2.2 Banach Spaces

Definition 2.2.1

A metric space is **Cauchy complete** (or just **complete**) if every Cauchy sequence of points has a limit that is contained in the metric space. (every Cauchy sequence is convergent)

Definition 2.2.2

We say a normed space is **complete** if the metric space formed using the norm's induced metric is Cauchy complete. A **Banach space** is a complete normed space.

Example.

Examples include $\mathbb{R}^n$, $\mathbb{C}^n$, but we will see many others in the coming lecture.

Definition 2.2.3

We say two normed spaces $(E_1, \|\cdot\|_1)$ and $(E_2, \|\cdot\|_2)$ are *isomorphic*, written $E_1 \simeq E_2$, if there exists a linear isometry between them; that is, a linear bijection $T : E_1 \to E_2$, such that

$$\|Tx\|_2 = \|x\|_1, \qquad \forall\, x \in E_1.$$

Definition 2.2.4

In a topological space X, a subset $A \subseteq X$ is said to be **dense** in X if:

- **Equivalent Definition 1:** The closure of A is equal to X, i.e., $\overline{A} = X$. This means that for any point $x \in X$, either $x \in A$, or x is a limit point of A.
- **Equivalent Definition 2:** For any non-empty open set $U \subseteq X$, we have $U \cap A \neq \emptyset$. In other words, A is "nowhere missing"—every non-empty open set intersects A.

Theorem 2.2.5

Let $(E, \|\cdot\|_E)$ be a normed space. Then, there exists another normed space $\widetilde{E}$ and an injective linear map $\iota : E \to \widetilde{E}$ such that:

1. $\widetilde{E}$ is complete;
2. ι is an isometric embedding, i.e., $\|\iota(x)\|_{\widetilde{E}} = \|x\|_E$;
3. $\iota(E)$ is dense in $\widehat{E}$.

Moreover, such $\widetilde{E}$ is unique up to linear isometries.

Proof. Too long. Omitted. □

Let E be a Banach space, $F \subseteq E$ a vector subspace (not necessarily closed). Then, the norm on E induces a norm on F, and F becomes a normed space. However, this restricted norm is not necessarily complete; in fact, we have

$$\widetilde{F} \simeq \overline{F} \quad \text{as normed spaces.}$$

Theorem 2.2.6

A closed subspace of a Banach space is also a Banach space.

2.3 Hilbert Space

Before moving on to examples, we discuss a particular class of Banach spaces:

Definition 2.3.1

Let $\mathcal{H}$ be a complex vector space. A **Hermitian inner product** on $\mathcal{H}$ is a map

$$\langle \cdot, \cdot \rangle_{\mathcal{H}} : \mathcal{H} \times \mathcal{H} \to \mathbb{C}$$

(also denoted $(\cdot, \cdot)$ sometimes) such that:

1. $\langle x, x \rangle \geq 0$ for all $x \in \mathcal{H}$ and $\langle x, x \rangle = 0 \Leftrightarrow x = 0$.
2. $\langle \cdot, \cdot \rangle$ is $\mathbb{C}$-linear in the 1st coordinate:
$$\langle \lambda_1 x_1 + \lambda_2 x_2, y \rangle = \lambda_1 \cdot \langle x_1, y \rangle + \lambda_2 \cdot \langle x_2, y \rangle, \quad \forall x_1, x_2, y \in \mathcal{H}, \forall \lambda_1, \lambda_2 \in \mathbb{C}.$$
3. $\langle x, y \rangle = \overline{\langle y, x \rangle}$ for all $x, y \in \mathcal{H}$, where the bar denotes complex conjugate.

If $\mathcal{H}$ is real, we replace (3) by $\langle x, y \rangle = \langle y, x \rangle$.

Definition 2.3.2

$\langle \cdot, \cdot \rangle$ is then known simply as an **inner product**. $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ is called a (real or complex) **inner product space** (or **pre-Hilbert space**). If $\mathcal{H}$ is real and finite dimensional, $(\mathcal{H}, \langle \cdot, \cdot \rangle)$ is a **Euclidean space**.

Theorem 2.3.3

Let $(\mathcal{H}, \langle\cdot,\cdot\rangle)$ be an inner product space. Then:

1. $\langle x, y_1 + \lambda y_2\rangle = \langle x, y_1\rangle + \overline{\lambda}\cdot\langle x, y_2\rangle, \quad \forall x, y_1, y_2 \in \mathcal{H}, \forall \lambda \in \mathbb{K}.$
   If $\mathbb{K} = \mathbb{R}$, $\lambda = \overline{\lambda}$.
2. *(Cauchy-Schwartz Inequality).*
$$|\langle x, y\rangle|^2 \leq \langle x, x\rangle \cdot \langle y, y\rangle, \quad \forall x, y \in \mathcal{H} \quad \Leftrightarrow \quad |\langle x, y\rangle| \leq \|x\| \cdot \|y\| \quad \forall x, y \in \mathcal{H}$$
   with
$$|\langle x, y\rangle|^2 = \langle x, x\rangle \cdot \langle y, y\rangle \Leftrightarrow x = \lambda y, \quad \lambda \in \mathbb{K}.$$

Proof. 1. We have
$$\langle x, y_1 + \lambda y_2\rangle = \overline{\langle y_1 + \lambda y_2, x\rangle} = \overline{\langle y_1, x\rangle + \lambda\langle y_2, x\rangle} = \overline{\langle y_1, x\rangle} + \overline{\lambda}\cdot\overline{\langle y_2, x\rangle} = \langle x, y_1\rangle + \overline{\lambda}\cdot\langle x, y_2\rangle.$$

2. If $x = 0$ or $y = 0$, the result is obvious. Let $y \neq 0$. We find λ such that $\langle x - \lambda y, y\rangle = 0$. This implies that $\langle x, y\rangle - \lambda\langle y, y\rangle = 0$, whence $\lambda = \langle x, y\rangle/\langle y, y\rangle$. Then by (1) in the definition of $\langle\cdot,\cdot\rangle$, we have
$$\begin{aligned} 0 \leq \langle x - \lambda y, x - \lambda y\rangle &= \langle x - \lambda y, x\rangle - \overline{\lambda}\underbrace{\langle x - \lambda y, y\rangle}_{=0} \\ &= \langle x, x\rangle - \lambda\langle y, x\rangle = \frac{1}{\langle y, y\rangle}\left(\langle x, x\rangle\langle y, y\rangle - \langle x, y\rangle\overline{\langle x, y\rangle}\right). \end{aligned}$$
   Now
$$0 \leq \frac{1}{\langle y, y\rangle}\left(\langle x, x\rangle\langle y, y\rangle - \langle x, y\rangle\overline{\langle x, y\rangle}\right) = \langle x, x\rangle - \frac{|\langle x, y\rangle|^2}{\langle y, y\rangle}.$$
   The result follows.

□

Theorem 2.3.4

Given an inner product space $(\mathcal{H}, \langle\cdot,\cdot\rangle)$, we can define
$$\|\cdot\| : \mathcal{H} \to \mathbb{R}, \quad \|x\| := \sqrt{\langle x, x\rangle}.$$
This is a norm.

Proof. The only non-trivial point is the triangle inequality:
$$\|x + y\|^2 = \langle x + y, x + y\rangle = \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle = \langle x, x\rangle + \langle x, y\rangle + \overline{\langle x, y\rangle} + \langle y, y\rangle.$$
Observing that $\langle x, y\rangle + \overline{\langle x, y\rangle} = 2\,\mathrm{Re}(\langle x, y\rangle)$ and $\mathrm{Re}(z) \leq |z|$ for all $z \in \mathbb{C}$, we have
$$\|x + y\|^2 = \|x\|^2 + \|y\|^2 + 2\,\mathrm{Re}(\langle x, y\rangle) \leq \|x\|^2 + \|y\|^2 + 2|\langle x, y\rangle|.$$
Now by the Cauchy-Schwarz inequality,
$$\|x + y\|^2 \leq \|x\|^2 + \|y\|^2 + 2\cdot\|x\|\cdot\|y\| = (\|x\| + \|y\|)^2 \Longrightarrow \|x + y\| \leq \|x\| + \|y\|,$$
since $\|x\| \geq 0$ for all $x \in \mathcal{H}$, as desired. □

Definition 2.3.5

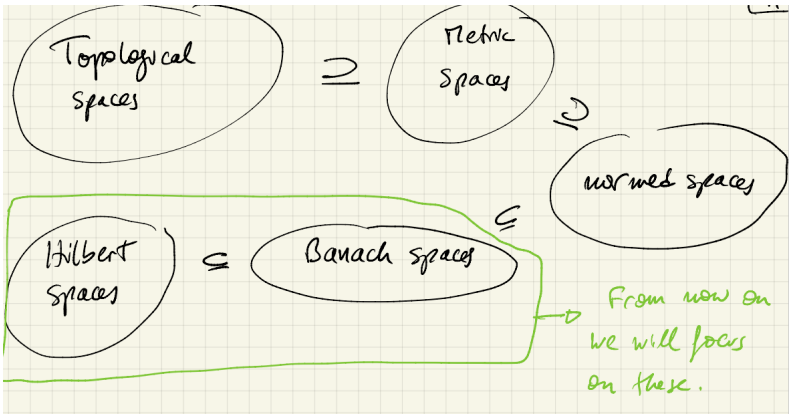
An inner product space $(H, \langle \cdot, \cdot \rangle)$ which is complete in the norm $\|x\| = \sqrt{\langle x, x \rangle}$ is called a *Hilbert space*.

Theorem 2.3.6

If H is an inner product space, there exists a Hilbert space $\widetilde{H}$ and a linear isometry $i : H \to \widetilde{H}$ such that H is dense in $\widetilde{H}$.

Definition 2.3.7

$\widetilde{H}$ is called the *completion* of H; it is unique up to linear isometry.



A natural question is: When is a Banach space also a Hilbert space? i.e., given a norm, when does it come from an inner product?

Proposition 2.3.8

Let $(E, \|\cdot\|)$ be a normed space. Then, there exists an inner product $\langle \cdot, \cdot \rangle$ such that $\|\cdot\| = \sqrt{\langle \cdot, \cdot \rangle}$ if and only if $\|\cdot\|$ satisfies the parallelogram identity:

$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2), \quad \forall x, y \in E.$$

Proof. ($\Rightarrow$). We prove fully only this direction:

$$\begin{aligned}
\|x+y\|^2 + \|x-y\|^2 &= \langle x+y, x+y \rangle + \langle x-y, x-y \rangle \\
&= 2\left(\langle x, x \rangle + \langle y, y \rangle\right) + \langle x, y \rangle + \langle y, x \rangle - \langle y, x \rangle - \langle x, y \rangle \\
&= 2\left(\|x\|^2 + \|y\|^2\right).
\end{aligned}$$

($\Leftarrow$). In the real case, we can define

$$\langle x, y \rangle := \frac{1}{2}\left(\|x+y\|^2 - \|x\|^2 - \|y\|^2\right), \quad x, y \in E,$$

from which the result follows (from three conditions). $\square$

2.4 Examples

2.4.1 Finite Dimension Vector space

On $\mathbb{R}^n$ and $\mathbb{C}^n$ we have inner products defined by

$$(x, y) := \sum_{i=1}^{n} x_i \cdot \overline{y_i}, \quad \text{for } x, y \in \mathbb{R}^n \text{ or } \mathbb{C}^n.$$

Then, these are both Hilbert spaces ($\mathbb{R}^n$ and $\mathbb{C}^n$ are complete). The scalar product gives rise to the norm:

$$\|x\|_2 := \sqrt{(x, x)} = \left(\sum_{i=1}^{n} |x_i|^2 \right)^{1/2}.$$

Another norm on $\mathbb{R}^n$ or $\mathbb{C}^n$ could be defined by:

$$\|x\|_1 := \sum_{i=1}^{n} |x_i|.$$

It turns out that $\|\cdot\|_1$ is equivalent to $\|\cdot\|_2$.

Definition 2.4.1

Two norms $\|\cdot\|_a$, $\|\cdot\|_b$ are *equivalent* if there exist constants $c, C > 0$ such that

$$c\|x\|_a \leq \|x\|_b \leq C\|x\|_a, \quad \forall x \in E.$$

For the examples above, we have:

$$\|x\|_2^2 = \sum_{i=1}^{n} |x_i|^2 \leq \left(\sum_{i=1}^{n} |x_i| \right)^2 = \|x\|_1^2 \quad \Rightarrow \quad \|x\|_2 \leq \|x\|_1$$

Also,

$$\begin{aligned}
\|x\|_1^2 &= \left(\sum_{i=1}^{n} |x_i| \cdot 1 \right)^2 = \left(\sum_{i=1}^{n} |x_i| \cdot |1| \right)^2 \\
&\leq \sum_{i=1}^{n} \sum_{j=1}^{n} |x_i||x_j| \leq \sum_{i,j=1}^{n} \frac{1}{2} \left(|x_i|^2 + |x_j|^2 \right) \\
&= \frac{1}{2} \left(n \sum_{i=1}^{n} |x_i|^2 + n \sum_{j=1}^{n} |x_j|^2 \right) = n \sum_{i=1}^{n} |x_i|^2 = n\|x\|_2^2
\end{aligned}$$

$$\Rightarrow \quad \frac{1}{\sqrt{n}} \|x\|_1 \leq \|x\|_2$$

More generally, we have:

Theorem 2.4.2

Let E be a finite-dimensional vector space. Then, all norms on E are equivalent. E is complete with respect to any norm.

2.4.2 The Uniform Metric and Norm

Definition 2.4.3

Let X be a topological space and set

$$\mathbb{R}^X := \{f : X \to \mathbb{R}\} \qquad \text{(a vector space)}$$

$$\mathcal{B}(X) := \mathcal{B}(X, \mathbb{R}) := \{f : X \to \mathbb{R} : f \text{ bounded}\}$$

$$\mathcal{BC}(X) := \mathcal{BC}(X, \mathbb{R}) := \{f : X \to \mathbb{R} : f \text{ bounded and continuous}\}.$$

We observe that $\mathcal{BC}(X) \subseteq \mathcal{B}(X) \subseteq \mathbb{R}^X$ are vector subspaces. On $\mathcal{B}(X)$ we define the **uniform metric** d_∞ and **uniform norm** $\|\cdot\|_\infty$ (also called the *sup metric* and *sup norm*, respectively) defined by

$$d_\infty(f, g) := \|f - g\|_\infty := \sup\{|f(x) - g(x)| : x \in X\}, \quad \forall f, g \in \mathcal{B}(X).$$

This is well-defined, since f and g are bounded by definition.

This example holds more generally if $\mathbb{R}$ is replaced by a Banach space E.

Theorem 2.4.4

$(\mathcal{B}(X), \|\cdot\|_\infty)$ is a Banach space.

Uniform Cauchy $\Rightarrow$ pointwise Cauchy in $\mathbb{R}$

Proof. Let (f_n) be a Cauchy sequence, where $f_n : X \to \mathbb{R}$, $n \in \mathbb{N}$. For each $x \in X$, we have

$$|f_n(x) - f_m(x)| \leq \|f_n - f_m\|_\infty.$$

Then:

- $(f_n(x))_{n\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$,
- $\exists \lim_{n\to\infty} f_n(x) =: f(x), |f(x) - f_n(x)| \to 0$ as $n \to \infty$

We claim that $f_n \to f$ in the uniform topology. Let $\varepsilon > 0$. Choose N such that

$$\|f_n - f_m\|_\infty \leq \varepsilon/2, \quad \forall n, m \geq N. \tag{1}$$

Fix now $n \geq N$, $\forall x \in X$. Then by (1),

$$|f_n(x) - f_m(x)| \leq \varepsilon/2, \quad \forall m \geq N.$$

Taking $m \to \infty$, we obtain

$$|f_n(x) - f(x)| \leq \varepsilon/2 < \varepsilon. \tag{*}$$

Firstly, this implies

$$|f(x)| \leq |f(x) - f_n(x)| + |f_n(x)| \leq \varepsilon + C, \quad \forall x \in X,$$

for some constant C since $f_n \in \mathcal{B}(X)$.

$\Rightarrow f \in \mathcal{B}(X)$. Also, since $(*)$ holds $\forall x \in X$, $\forall n \geq N$,

$$\|f_n - f\|_\infty < \varepsilon, \quad \forall n \geq N,$$

and this proves that $f_n \to f$ uniformly. □

In analysis, convergence with respect to the norm $\|\cdot\|_\infty$ is usually called uniform convergence.

Theorem 2.4.5: Uniform Limit Theorem

$\mathrm{BC}(X)$ is closed subspace of $\mathrm{B}(X)$. In particular, $(\mathrm{BC}(X), \|\cdot\|_\infty)$ is a Banach space.

Proof. Let (f_n) be a sequence in $\mathrm{BC}(X)$ with $f_n \to f$ uniformly. We want to show that $f : X \to \mathbb{R}$ is continuous. To that end, let $y_0 = f(x_0) \in \mathbb{R}$, $\varepsilon > 0$, and let us find an open neighborhood U of x_0 such that

$$f(U) \subseteq (y_0 - \varepsilon, y_0 + \varepsilon).$$

Let $N \in \mathbb{N}$ such that

$$|f_n - f| = \|f_n - f\|_\infty \leq \varepsilon/3, \quad \forall n \geq N. \tag{1}$$

Using continuity of f_N, let U be a neighborhood of x_0 such that

$$f_N(U) \subseteq (f_N(x_0) - \varepsilon/3, f_N(x_0) + \varepsilon/3). \tag{2}$$

Then, for any $x \in U$, we have:

$$\begin{aligned} |f(x) - y_0| &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)| \\ &< \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon. \end{aligned}$$

$\Rightarrow f \in \mathrm{BC}(X)$. □

Corollary 2.4.6

Let K be a compact topological space. Then $(\mathcal{C}(K), \|\cdot\|_\infty)$ is a Banach space.

Proof. $\mathcal{C}(K) = \mathcal{BC}(K)$, since any continuous function $f : X \to \mathbb{R}$ with compact domain must be bounded (because f continuous means $f(K)$ is also compact, and so bounded in $\mathbb{R}$ by Heine-Borel). □

2.4.3 p-norms and p-spaces

Finite Dimensions

Definition 2.4.7

Let $p \in \mathbb{R}_{\geq 1}$. On $\mathbb{K}^n$ ($\mathbb{K} = \mathbb{R}, \mathbb{C}$) we can define the **$p$-norm** (also called the ℓ^p or ℓ_p) as

$$\|x\|_p := \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}.$$

If $p = 1$, this is the taxi-cab norm, while if $p = 2$, it is the Euclidean norm. As $p \to \infty$, we approach $\|\cdot\|_\infty$ for finite vector spaces (as mentioned above).

Lemma 2.4.8: Young's Inequality

Let $p, p' \in (1, \infty)$ be **dual** to each other, i.e. $\frac{1}{p} + \frac{1}{p'} = 1$. Then

$$ab \leq \frac{a^p}{p} + \frac{b^{p'}}{p'}, \quad \forall a, b \in \mathbb{R}_{\geq 0}.$$

Proof. Let $f(x) = e^x$, which is convex. Then by the definition of convexity:

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y).$$

Take

$$ab = e^{\log a + \log b} = e^{\frac{1}{p}\log a^p + \frac{1}{q}\log b^q} \leq \frac{1}{p}e^{\log a^p} + \frac{1}{q}e^{\log b^q} = \frac{1}{p}a^p + \frac{1}{q}b^q.$$

□

Lemma 2.4.9: Hölder's Inequality

Let $p, p' \in [1, \infty]$ be dual (if $p = 1$, $p' = \infty$). Then

$$\sum_{i=1}^{n} |x_i| \cdot |y_i| \leq \|x\|_p \cdot \|y\|_{p'} = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p} \left(\sum_{i=1}^{n} |y_i|^{p'}\right)^{1/p'}.$$

Or let $1 < p, q < \infty$, with $\frac{1}{p} + \frac{1}{q} = 1$. If $u \in L^p(K)$, $v \in L^q(K)$, then:

$$\int_K |uv| \, dx \leq \|u\|_p \|v\|_q = \left(\int_K |u|^p\right)^{1/p} \left(\int_K |v|^q\right)^{1/q}$$

Proof. By homogeneity, we assume that $\|u\|_p = \|v\|_q = 1$. Young's inequality then gives

$$\int_K |uv| \, dx \leq \frac{1}{p} \int_K |u|^p \, dx + \frac{1}{q} \int_K |v|^q \, dx = \frac{1}{p} + \frac{1}{q} = 1 = \|u\|_p \|v\|_q.$$

□

Theorem 2.4.10

$\|\cdot\|_p$ is a norm on $\mathbb{K}^n$ for all $p \in (1, \infty)$.

This is just the Minkowski's Inequality.

Proof. The only non-trivial property is the triangle inequality. Let $x, y \in \mathbb{K}^n \setminus \{0\}$ and let p' be the dual of p. Then

$$\begin{aligned}
\|x + y\|_p^p &= \sum_{i=1}^{n} |x_i + y_i|^p = \sum_{i=1}^{n} |x_i + y_i| \cdot |x_i + y_i|^{p-1} \\
&\leq \sum_{i=1}^{n} \left(|x_i| \cdot |x_i + y_i|^{p-1} + |y_i| \cdot |x_i + y_i|^{p-1}\right) \\
&\overset{\text{Hölder}}{\leq} \|x\|_p \left(\sum_{i=1}^{n} |x_i + y_i|^{p'(p-1)}\right)^{1/p'} + \|y\|_p \left(\sum_{i=1}^{n} |x_i + y_i|^{p'(p-1)}\right)^{1/p'} \\
&= (\|x\|_p + \|y\|_p) \cdot \left(\sum_{i=1}^{n} |x_i + y_i|^p\right)^{(p-1)/p} \\
&= (\|x\|_p + \|y\|_p) \cdot \|x + y\|_p^{p-1},
\end{aligned}$$

where $p'(p-1) = p$ in the second line since

$$\frac{1}{p} + \frac{1}{p'} = 1 \Longrightarrow \frac{1}{p'} = 1 - \frac{1}{p} \Longrightarrow p' = \frac{p}{p-1}.$$

Thus, if $x + y \neq 0$, we get $\|x + y\|_p \leq \|x\|_p + \|y\|_p$, as desired. □

- *We have*
$$\|x\|_q \leq \|x\|_p \leq n^{\frac{q-p}{pq}} \|x\|_q, \qquad \forall x \in \mathbb{K}^n, \ 1 \leq p \leq q \leq \infty.$$
- *Having shown that $\|\cdot\|_p$ is a norm, we observe that $\left(\mathbb{K}^n, \|\cdot\|_p\right)$ is a Banach space. This is unsurprising, since all these norms are equivalent to the Euclidean norm $\|\cdot\|_2$ on a finite-dimensional vector space.*

Infinite Dimensions

Definition 2.4.11

Let $S = \mathbb{K}^{\mathbb{N}}$ denote the space of sequences in $\mathbb{K}$. It is a vector space; for instance, we define addition as $(x_n) + (y_n) = (x_n + y_n)$. The p-norms on S are defined as follows:

$$\|(x_n)\|_p := \begin{cases} \left(\sum_{n=0}^{\infty} |x_n|^p\right)^{1/p}, & 1 \le p < \infty, \\ \sup\{|x_n| : n \in \mathbb{N}\}, & p = \infty. \end{cases}$$

Of course, the sequence may be unbounded, or the above series divergent. We consider the following subspaces:

$$\ell_p := \ell_p(\mathbb{K}) := \{(x_n) \in S : \|(x_n)\|_p < \infty\}, \quad 1 \le p \le \infty$$

$$c_0 := c_0(\mathbb{K}) := \left\{(x_n) \in S : \lim_{n\to\infty} |x_n| = 0\right\}.$$

Theorem 2.4.12

Let $1 \le p \le \infty$. Then $(\ell_p, \|\cdot\|_p)$ and $(c_0, \|\cdot\|_\infty)$ are Banach spaces. Moreover, for $1 < p < q < \infty$, have

$$\ell_1 \subset \ell_p \subset \ell_q \subset c_0 \subset \ell_\infty.$$

Finally, $\|x\|_q \le \|x\|_p$ for all $x \in \ell_p$ if $1 \le p < q \le \infty$.

Cauchy in l_p:

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ s.t.\ \left\|x^{(k)} - x^{(m)}\right\|_p = \left(\sum_{n=0}^{\infty} \left|x_n^{(k)} - x_n^{(m)}\right|^p\right)^{1/p} < \varepsilon, \quad \forall k, m \ge N.$$

Proof. (Outline). To prove that they are normed spaces, one needs to verify the triangle inequality, which follows from taking $n \to \infty$.

Regarding inclusions, if $\sum_{n\ge 0} |x_n|^p$ converges, then $|x_n| \to 0$, meaning $\ell_p \subseteq c_0$. The inclusion becomes strict due to the harmonic sequence $(1/n)_{n\in\mathbb{Z}_{>0}} \in c_0 \setminus \ell_p$, for all $p \in [1, \infty)$. $\ell_1 \subset \ell_p$: Absolutely summable sequences also have their p-th powers summable, so the inclusion holds. $\ell_p \subset \ell_q \quad (p < q)$: if $x \in \ell_p$, then $|x_n|^q = |x_n|^p \cdot |x_n|^{q-p} \le |x_n|^p \|x\|_\infty^{q-p}$, and the latter is summable, so the series converges. The sequence $x_n = n^{-1/p}$ lies in ℓ_q but not in ℓ_p.

For completeness, there are three steps. Let $\left(x^{(k)}\right)_{k\in\mathbb{N}} \subseteq \ell_p$ be Cauchy (i.e., a sequence of sequences):

$$x^{(0)} = \left(x_0^{(0)}, x_1^{(0)}, \dots\right), \quad x^{(1)} = \left(x_0^{(1)}, x_1^{(1)}, \dots\right), \quad \dots$$

1. For each fixed n, the components $\left(x_n^{(k)}\right)$ form a Cauchy sequence in $\mathbb{K}$ since for $x \in \mathcal{S}$, $|x_i| \le \|x\|_p$ (as for $y \in \mathbb{K}^n$), meaning they converge: $x_n^{(k)} \to \bar{x}_n$ as $k \to \infty$. Thus, we have a candidate $(\bar{x}_n) \in \mathcal{S}$ for the limit.

2. Show that $\left\|x^{(k)} - \bar{x}\right\|_p \to 0$ as $k \to \infty$. In particular, $\exists k$ such that $x^{(k)} - \bar{x} \in \ell_p$.

3. Since ℓ_p is a vector space and $x^{(k)}, x^{(k)} - \bar{x} \in \ell_p$, we have

$$\underbrace{x^{(k)}}_{\in \ell_p} + \underbrace{x^{(k)} - \bar{x}}_{\in \ell_p} = \bar{x} \in \ell_p.$$

□

2.4.4 The Lebesgue Space

Definition 2.4.13

Let $K \subset \mathbb{R}$ be a compact set, $K \subset [a, b]$, consider the function $\|\cdot\|_p$ on $\mathcal{C}(K)$:

$$\|f\|_p = \left(\int_K |f(x)|^p \, dx\right)^{\frac{1}{p}}.$$

Then define $L^p(K) = \overline{\mathcal{C}(K)}^{\|\cdot\|_p} \leftarrow$ (completion of $\mathcal{C}(K)$ with respect to $\|\cdot\|_p$)

Lemma 2.4.14: Minkowski's Inequality

Assume $1 \leq p \leq \infty$ and $u, v \in L^p(U)$. Then

$$\|u + v\|_{L^p(U)} \leq \|u\|_{L^p(U)} + \|v\|_{L^p(U)}.$$

Proof.

$$\begin{aligned}
\|u+v\|^p_{L^p(U)} &= \int_U |u+v|^p \, dx \\
&\leq \int_U |u+v|^{p-1}\big(|u| + |v|\big) \, dx \\
&\leq \left(\int_U |u+v|^p \, dx\right)^{\frac{p-1}{p}} \left(\left(\int_U |u|^p \, dx\right)^{\frac{1}{p}} + \left(\int_U |v|^p \, dx\right)^{\frac{1}{p}}\right) \\
&= \|u+v\|^{p-1}_{L^p(U)}\big(\|u\|_{L^p(U)} + \|v\|_{L^p(U)}\big).
\end{aligned}$$

The last inequality is obtained by using Holder inequality twice. Set $p = p$ and $p' = \frac{p}{p-1}$. □

Theorem 2.4.15

$\|\cdot\|_p$ is a norm on $\mathcal{C}(K)$ for all $1 \leq p < \infty$

This is just the Minkowski's Inequality.

Proof. To show that $\|\cdot\|_p$ is a norm:

$$\|f\|_p = 0 \iff f = 0$$

$$\int_K |f(x)|^p \, dx = 0 \Rightarrow f(x) = 0 \quad \forall x \in K \quad \text{because } f \text{ is continuous.}$$

$$\|\lambda f\|_p = \left(\int_K |\lambda f(x)|^p \, dx\right)^{\frac{1}{p}} = |\lambda| \left(\int_K |f(x)|^p \, dx\right)^{\frac{1}{p}} = |\lambda| \|f\|_p \quad \Rightarrow \text{homogeneous}$$

$$\begin{aligned}
\|u+v\|^p_p = \int_K |u+v|^p \, dx &= \int_K |u+v||u+v|^{p-1} \, dx \\
&\leq \int_K |u||u+v|^{p-1} \, dx + \int_K |v||u+v|^{p-1} \, dx \\
\leq \left(\int_K |u|^p \, dx\right)^{1/p} \left(\int_K |u+v|^p \, dx\right)^{(p-1)/p} &+ \left(\int_K |v|^p \, dx\right)^{1/p} \left(\int_K |u+v|^p \, dx\right)^{(p-1)/p} \\
&= \|u\|_p \|u+v\|^{p-1}_p + \|v\|_p \|u+v\|^{p-1}_p \\
&\Rightarrow \|u+v\|_p \leq \|u\|_p + \|v\|_p
\end{aligned}$$

□

Example.

$L^p(K)$ is a Banach space for $1 \leq p < \infty$. But $L^1(K)$ and $L^\infty(K)$ are also Banach spaces.

2.4.5 The Sobolev space

On $C^\infty([a,b])$, consider the norm $\|\cdot\|_{k,p} \quad (k \in \mathbb{Z}_{>0},\ p \in [1,\infty])$ defined by

$$\|f\|_{k,p} := \left(\sum_{i=0}^{k} \left\| f^{(i)} \right\|_p^p \right)^{1/p}, \qquad \forall f \in C^\infty([a,b]),$$

where $f^{(i)}$ denotes the ith derivative of f. The completion of $C^\infty([a,b])$ with respect to $\|\cdot\|_{k,p}$ is called a **Sobolev space**, usually denoted by $W_{k,p}([a,b])$. It is Hilbert for $p = 2$. These are extremely important in the study of partial differential equations.

Chapter 3

Continuous Linear Functionals

3.1 Almost Orthogonal Vector

Lemma 3.1.1: Riesz lemma

Let $Y \subsetneq X$ be a proper closed subspace of X. For any $\varepsilon \in (0,1)$, there exists $z \in X$ with $\|z\| = 1$, $\|z - y\| \geq 1 - \varepsilon$, $\forall y \in Y$.

Proof. A3 □

Lemma 3.1.2

Finite-dimensional linear normed spaces are isomorphic if and only if they have equal dimensions.

Theorem 3.1.3

Let X be a normed space. Then the unit ball $B(0,1) = \{x \in X : \|x\| \leq 1\}$ is compact if and only if $\dim(X) < \infty$.

Proof. $\Leftarrow$:

If X is finite dimensional, then $X \simeq \mathbb{R}^n$. The compact subsets of $\mathbb{R}^n$ are closed and bounded. Thus, $B(0,1)$ is compact.

$\Rightarrow$:

Suppose X has infinite dimensions.

We will construct a sequence $\{x_v\}_{v \in \mathbb{N}} \subseteq B(0,1)$ such that $\|x_v - x_\mu\| \geq \frac{1}{2}$ for all $v \neq \mu$. Proceed by contradiction and suppose $B(0,1)$ is compact. Then since topology comes from a metric, i.e., every sequence has a convergent subsequence, let $\{x_{v_k}\}$ be the convergent subsequence, converging to $x \in B(0,1)$. Then

$$\|x_{v_k} - x_{v_\ell}\| \leq \|x_{v_k} - x\| + \|x - x_{v_\ell}\| \to 0$$

Let us now construct the sequence $\{x_v\}$. We will do this inductively.

Start by picking a point x_1, so suppose we have $x_1, \ldots, x_n$ such that $\|x_j - x_k\| \geq \frac{1}{2}$ for all $1 \leq j \neq k \leq N$. We can choose $x_1, \ldots, x_N$ to be linearly independent. Let $Y_n = \operatorname{span}\{x_1, \ldots, x_N\}$, then Y is a proper closed subspace.

The Riesz lemma $\Rightarrow \exists x_{N+1}$ such that $\|x_{N+1}\| = 1$ and $\|x_{N+1} - x_k\| \geq \frac{1}{2}$ $\forall k$. Contradiction. Thus, X has finite dimensions.

□

Corollary 3.1.4

Every finite dimension space is closed.

Corollary 3.1.5

In finite dimension spaces, every linear transformation is bounded.

Proof. Finite dimension $\Rightarrow$ isomorphic with $\mathbb{K}^n$ and unit ball is compact $\Rightarrow$ bounded. $\square$

3.2 The Hahn-Banach Theorem

3.2.1 Linear functionals

Definition 3.2.1

A linear transformation is a function $T : \mathbb{R}^n \to \mathbb{R}^m$ which satisfies:

1. $T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$
2. $T(c\mathbf{x}) = cT(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$ and $c \in \mathbb{R}$

Definition 3.2.2

Let E be a normed space over $\mathbb{K} = \mathbb{R}, \mathbb{C}$. A linear functional on E (sometimes called just a *functional*) is a linear transformation $\ell : E \to \mathbb{K}$. We say ℓ is bounded if $\exists c \in \mathbb{R}$ such that

$$|\ell(x)| \leq c \cdot \|x\| \quad \text{for all } x \in E$$

(equivalently, ℓ maps bounded sets in E to bounded sets in $\mathbb{K}$).

Theorem 3.2.3

Let E be a normed space. For a linear functional $\ell : E \to \mathbb{K}$, the following are equivalent:

1. ℓ is bounded.
2. ℓ is continuous at 0.
3. ℓ is continuous.
4. ℓ is uniformly continuous.

Whenever we talk about "bounded functionals", these are also continuous functionals.

Proof. (4) $\Rightarrow$ (3) and (3) $\Rightarrow$ (2) are obvious.
(2) $\Rightarrow$ (1): By (2), given $\varepsilon = 1$, $\exists \delta > 0$ such that if $\|x\| < \delta$ then $|\ell(x)| < 1$. To prove boundedness, let $y \in E$, $y \neq 0$, and set $x := \delta y/(2\|y\|)$. Then

$$\|x\| = \frac{\delta\|y\|}{2\|y\|} = \frac{\delta}{2} < \delta.$$

Thus, we have

$$\frac{\delta}{2\|y\|} \cdot |\ell(y)| = |\ell(x)| < 1 \quad \Longrightarrow \quad |\ell(y)| < \frac{2}{\delta} \cdot \|y\|.$$

Therefore, if we take $c = 2/\delta$, then the condition for boundedness holds.

(1) $\Rightarrow$ (4). Let $c \in \mathbb{R}$ be such that ℓ is bounded by it. Then $\forall x, y \in E$,

$$|\ell(x) - \ell(y)| = |\ell(x - y)| \leq c \cdot \|x - y\|,$$

with c independent of x and y. This implies uniform continuity for ℓ. $\square$

Example.

Let E be a finite dimensional Banach space (e.g. $\mathbb{R}^n$) with a basis $\{e_1, \ldots, e_n\}$. Then every $x \in E$ admits a unique representation in the form

$$x = \sum_{k=1}^{n} x_k e_k.$$

It is then easy to see that the function $\ell : E \to \mathbb{K}$ defined by

$$\ell(x) := \sum_{k=1}^{n} \ell_k x_k, \qquad \ell_1, \ldots, \ell_n \in \mathbb{K}$$

is a bounded and continuous linear functional ($\mathbb{K} = \mathbb{R}, \mathbb{C}$, with $\mathbb{K} = \mathbb{R}$ if $E = \mathbb{R}^n$). In particular, to see boundedness, take $c = n \cdot \max_{1<k<n} |\ell_k|$.

Example.

$\ell : \mathcal{C}([0,1]) \to \mathbb{R}$ defined by

$$\ell(f) := \int_0^1 f(t)\, dt, \quad \forall f \in \mathcal{C}([0,1])$$

is a linear functional. Consider on $\mathcal{C}([0,1])$ the uniform norm $\|\cdot\|_\infty$. Then

$$|\ell(f)| = \left| \int_0^1 f(t)\, dt \right| \leq \int_0^1 |f(t)|\, dt \leq \int_0^1 \|f\|_\infty\, dt = \|f\|_\infty,$$

so ℓ is bounded (namely, $c = 1$).

Example.

Let $a := (a_n)_{n\in\mathbb{N}} \in \ell_\infty(\mathbb{R})$ be a fixed sequence. We define $F_a : \ell_1 \to \mathbb{R}$ by

$$F_a((b_n)) := \sum_{n\geq 0} a_n \cdot b_n.$$

F_a is a bounded functional. Indeed (limit can be moved outside because $|\cdot| : \mathbb{R} \to \mathbb{R}$ is continuous),

$$|F_a((b_n))| = \left| \sum_{n=0}^{\infty} a_n b_n \right| = \left| \lim_{N\to\infty} \sum_{n=0}^{N} a_n b_n \right| = \lim_{N\to\infty} \left| \sum_{n=0}^{N} a_n b_n \right| \leq \lim_{N\to\infty} \sum_{n=0}^{N} |a_n b_n| \leq \|a\|_\infty \sum_{n=0}^{\infty} |b_n|.$$

But $\sum_{n\geq 0} |b_n| = \|b\|_1$, so we have

$$|F_a((b_n))| \leq \|a\|_\infty \cdot \|b\|_1.$$

Therefore, setting $c = \|a\|_\infty$, we see that F_a is bounded, as desired.

3.2.2 Dual space

Definition 3.2.4

Let E be a vector space and $\ell : E \to \mathbb{K}$ a bounded functional. Its **norm** (sometimes called the **operator norm**) is defined as

$$\|\ell\| := \sup_{x \neq 0} \frac{|\ell(x)|}{\|x\|} = \sup_{x \in \mathcal{S}_1(0)} |\ell(x)|,$$

where $\mathcal{S}_1(0) := \{x \in E : \|x\| = 1\}$ is the unit sphere centred on 0.

Observe that $|\ell(x)| \leq \|\ell\| \cdot \|x\|$ for all $x \in E$ and that since $\mathbb{K}$ is itself a vector space, functionals on E form a vector space, so that if $\ell, m : E \to \mathbb{K}$ are functionals, we define

$$(\ell + m)(x) := \ell(x) + m(x), \quad (\lambda \cdot \ell)(x) := \lambda \cdot \ell(x), \quad \forall x \in E, \lambda \in \mathbb{K}.$$

Bounded functionals are then a vector subspace of this space. Endowed with the operator norm, it becomes a normed space;

Proof.

$$\|\ell\| \geq 0 \quad \text{and} \quad \|\ell\| = 0 \iff \ell \equiv 0.$$

$$\|\lambda \ell\| = |\lambda| \cdot \|\ell\| \quad \text{for any scalar } \lambda.$$

$$|(\ell + m)(x)| \leq |\ell(x)| + |m(x)| \leq (\|\ell\| + \|m\|) \cdot \|x\| \quad \Longrightarrow \quad \|\ell + m\| = \sup_{x \neq 0} \frac{|(\ell + m)(x)|}{\|x\|} \leq \|\ell\| + \|m\|,$$

□

Definition 3.2.5

Given a normed space E, the space of bounded functionals on E with the operator norm is called the **dual space** of E (more specifically, *dual to the space* E) and is denoted by E' (or E^*).

Theorem 3.2.6

The dual space E' is a Banach space (even if E is not).

Proof. Let $(\ell_n) \subseteq E'$ be Cauchy and fix $x \in E$. Then

$$|\ell_n(x) - \ell_m(x)| = |(\ell_n - \ell_m)(x)| \leq \|\ell_n - \ell_m\|_{E'} \cdot \|x\|_E,$$

which means $(\ell_n(x))_{n \in \mathbb{N}}$ is Cauchy in $\mathbb{K}$. $\mathbb{K}$ is complete, so $\exists \ell(x) := \lim_{n \to \infty} \ell_n(x)$. There are now three properties to prove:

1. ℓ is linear. Indeed, $\forall x, y \in E$ and $\forall \lambda \in \mathbb{K}$,

$$\ell(x) + \lambda \ell(y) = \lim_{n \to \infty} \ell_n(x) + \lambda \cdot \lim_{n \to \infty} \ell_n(y) = \lim_{n \to \infty} (\ell_n(x) + \lambda \cdot \ell_n(y)) = \lim_{n \to \infty} \ell_n(x + \lambda y) = \ell(x + \lambda y).$$

2. ℓ is bounded. Since (ℓ_n) is Cauchy, it is bounded: $\exists c \in \mathbb{R}$ such that $\|\ell_n\| \leq c$ for all $n \in \mathbb{N}$. Thus,

$$|\ell(x)| = \lim_{n \to \infty} |\ell_n(x)| \leq \lim_{n \to \infty} \|\ell_n\| \cdot \|x\| \leq c \cdot \|x\|.$$

3. $\ell_n \to \ell$ in the operator norm topology. Let $\varepsilon > 0$. Then since (ℓ_n) is Cauchy, $\exists N \in \mathbb{N}$ such that

$$\|\ell_n - \ell_m\|_{E'} < \varepsilon, \quad \forall n, m \geq N \Longrightarrow |(\ell_n(x) - \ell_m(x))| < \varepsilon \cdot \|x\|, \quad \forall x \in E.$$

Taking $n \to \infty$, we get

$$|\ell(x) - \ell_m(x)| < \varepsilon \cdot \|x\| \implies \|\ell - \ell_m\| = \sup_{x \neq 0} \frac{|(\ell(x) - \ell_m(x))|}{\|x\|} < \varepsilon \implies \lim_{n\to\infty} \ell_n = \ell$$

in the operator norm topology (i.e. $\|\ell - \ell_n\| \to 0$), as desired.

□

3.2.3 Hahn-Banach Theorem

Main question: we are interested in with regards to dual spaces is what their elements are, if they have any at all. In finite dimensions, we know that E' is isomorphic (as a vector space) to E (although not in a canonical way, since the isomorphism depends on the choice of a basis for E).

To see that there are a lot of bounded linear functionals, we will show that we can *extend* a functional defined on a vector subspace $G \subseteq E$ to all of E. That is, given $\ell \in G'$, we look for $L \in E'$ such that $L|_G = \ell$. Notice that, by definition, if such an extension exists, then

$$\|L\| = \sup_{x \in E \setminus \{0\}} \frac{|L(x)|}{\|x\|} \geq \sup_{x \in G \setminus \{0\}} \frac{|L(x)|}{\|x\|} = \|\ell\|.$$

However, we will require that the extension does *not* increase the norm: $\|L\|_{E'} = \|\ell\|_{G'}$. In the below theorem, we first show that we can assume G is closed, i.e. we extend ℓ to the closure of G.

Theorem 3.2.7: Extension by Continuity

Let E be a normed space and $G \subseteq E$ a dense linear subspace (i.e. $\overline{G} = E$). Given $\ell \in G'$, $\exists! L \in E'$ such that $L|_G = \ell$ and $\|L\| = \|\ell\|$.

$\exists!$ means "there exists unique" or "there exists exactly one".
In metric space, convergent sequence is Cauchy

Proof.

- Uniqueness: $L \in E'$ so L is bounded, meaning it is continuous. Given $x \in E$, take $(g_n) \subseteq G$, with $g_n \to x$ as $n \to \infty$ (since G is dense), and we must have $L(x) := \lim_{n\to\infty} \ell(g_n)$ by continuity. L is therefore unique, since limit is unique.

- Existence: the limit exists because $(\ell(g_n))$ is Cauchy in $\mathbb{K}$ $(= \mathbb{R}, \mathbb{C})$, which is complete (Cauchy sequence then convergent):

$$|\ell(g_n) - \ell(g_m)| = |\ell(g_n - g_m)| \leq \|\ell\| \cdot \|g_n - g_m\| \xrightarrow{n,m\to\infty} 0,$$

  where the last term goes to zero since (g_n) is Cauchy (since converges).

- It is also well-defined: If $(\bar{g}_n)$ is another sequence in G with $\bar{g}_n \to x$, then

$$|\ell(g_n) - \ell(\bar{g}_n)| \leq \|\ell\| \cdot \|g_n - \bar{g}_n\| \leq \|\ell\| \cdot (\|g_n - x\| + \|x - \bar{g}_n\|).$$

  The two terms in the final bracket both go to zero, so it follows that

$$\lim_{n\to\infty} \ell(g_n) = \lim_{n\to\infty} \ell(\bar{g}_n)$$

- It extends ℓ since if $x \in G$ and we choose $g_n = x$ for all $n \in \mathbb{N}$, then

$$L(x) = \lim_{n\to\infty} \ell(g_n) = \ell(x).$$

- Linearity is clear: Let $x, y \in E$, and choose sequences $g_n \to x$, $h_n \to y$.

$$L(\alpha x + \beta y) = \lim \ell(\alpha g_n + \beta h_n) = \alpha \lim \ell(g_n) + \beta \lim \ell(h_n) = \alpha L(x) + \beta L(y).$$

- Regarding bounds,

$$|L(x)| = \left|\lim_{n\to\infty} \ell(g_n)\right| = \lim_{n\to\infty} |\ell(g_n)| \leq \lim_{n\to\infty} \|\ell\| \cdot \|g_n\| = \|\ell\| \cdot \lim_{n\to\infty} \|g_n\| = \|\ell\| \cdot \|x\|$$

$$\Rightarrow \|L\| = \sup \frac{|L(x)|}{\|x\|} \leq \|\ell\|,$$

where we can move the limit around because $|\cdot| : \mathbb{R} \to \mathbb{R}$ and $\|\cdot\| : E \to \mathbb{R}$ are continuous. But we know that $\|\ell\| \leq \|L\|$, so $\|\ell\| = \|L\|$, as desired.

□

We begin with a weaker version of the Hahn-Banach Theorem

Lemma 3.2.8: Weak Hahn-Banach Theorem

Let E be a normed space and $G \subseteq E$ a subspace, $\ell \in G'$ and suppose that for some $y \in E \setminus G$, $E = span(G, y)$. Then $\exists L \in E'$ such that $L|_G = \ell$ and $\|L\|_{E'} = \|\ell\|_{G'}$.

Proof. $\forall x \in E$, $\exists \lambda \in \mathbb{R}, g \in G$ such that $x = g + \lambda y$. We set $L(y) = c \in \mathbb{R}$ (to be determined later) and $L(g) = \ell(g)$ for all $g \in G$. Then $L(x) = \ell(g) + \lambda c$. We want to choose c so that $\|L\| = \|\ell\|$. It is enough to have $\|L\| \leq \|\ell\|$, i.e.

$$|L(x)| \leq \|L\|\|x\| \leq \|\ell\| \cdot \|x\| \quad \text{for all } x \in E.$$

This is equivalent to

$$|\ell(g) + \lambda c| \leq \|\ell\| \cdot \|g + \lambda y\|, \quad \forall g \in G, \lambda \in \mathbb{R}.$$

Say $\lambda > 0$ (if $\lambda < 0$, then $-x = -g + (-\lambda)y$, where $(-g) \in G$ and $(-\lambda) \in \mathbb{R}$). Then

$$|\ell(g) + \lambda c| \leq \|\ell\| \cdot \|g + \lambda y\| \iff -\|\ell\| \cdot \|g + \lambda y\| \leq \ell(g) + \lambda c \leq \|\ell\| \cdot \|g + \lambda y\|$$

$$\iff -\|\ell\| \cdot \left\|\frac{g}{\lambda} + y\right\| \leq \ell(g/\lambda) + c \leq \|\ell\| \cdot \left\|\frac{g}{\lambda} + y\right\|$$

$$\iff -\|\ell\| \cdot \left\|\frac{g}{\lambda} + y\right\| - \ell(g/\lambda) \leq c \leq \|\ell\| \cdot \left\|\frac{g}{\lambda} + y\right\| - \ell(g/\lambda).$$

Let $h = g/\lambda$. We want c to satisfy these inequalities $\forall h \in G$. To see that this is possible, we claim:

$$\sup_{h\in G} \left(-\|\ell\| \cdot \|h + y\| - \ell(h)\right) \leq \inf_{h\in G} \left(\|\ell\| \cdot \|h + y\| - \ell(h)\right).$$

It is enough to show that

$$-\|\ell\| \cdot \|h_1 + y\| - \ell(h_1) \leq \|\ell\| \cdot \|h_2 + y\| - \ell(h_2), \quad \forall h_1, h_2 \in G,$$

which is equivalent to

$$\ell(h_2) - \ell(h_1) \leq \|\ell\| \cdot \left(\|h_2 + y\| + \|h_1 + y\|\right), \quad \forall h_1, h_2 \in G.$$

And indeed,

$$\ell(h_2 - h_1) \leq |\ell(h_2 - h_1)| \leq \|\ell\| \cdot \|h_2 - h_1\| = \|\ell\| \cdot \|(h_2 + y) - (h_1 + y)\| \leq \|\ell\| \cdot \left(\|h_2 + y\| + \|h_1 + y\|\right).$$

From this claim, we can choose $c \in \mathbb{R}$ "in between", and the lemma follows. □

With this simpler version, we are able to extend functionals "one dimension at a time". If E were to have a countable basis, then an easy induction argument would finish the job. However, what if E had an uncountable basis?

Definition 3.2.9

A relation $\prec$ on a set X is a **partial ordering** if it is:

1. *Reflexive*: $x \prec x$ for all $x \in X$.
2. *Transitive*: $x \prec y$ and $y \prec z$ implies $x \prec z$.
3. *Anti-symmetric*: if $x \prec y$ and $y \prec x$ implies $x = y$.

The pair $(X, \prec)$ is then called a **partially ordered set** (or *poset*).

Example.

$\leq$ on $\mathbb{R}$ and $\subseteq$ on $\mathcal{P}(X)$ for some set X are both partial orderings.

Definition 3.2.10

Let $(X, \prec)$ be a partially ordered set. We say $m \in X$ is **maximal** if $\forall x \in X$, $m \prec x$ implies $m = x$. We say m is an **upper bound** for a subset $S \subseteq X$ if $s \prec m$ for all $s \in S$. A **chain** $C \subseteq X$ is a subset such that $\forall x, y \in C$, we have $x \prec y$ or $y \prec x$ (i.e. C with the induced order is a totally ordered set).

Lemma 3.2.11: Zorn's Lemma

Let $(X, \preceq)$ be a partially ordered set such that each chain in X has an upper bound. Then, X has a maximal element.

This result can be seen to be equivalent to the axiom of choice.

Theorem 3.2.12: Hahn-Banach Theorem

Let E be a normed space and $G \subseteq E$ a subspace. Then, $\forall \ell \in G'$, $\exists L \in E'$ such that $L|_G = \ell$ and $\|L\|_{E'} = \|\ell\|_{G'}$.

Proof. We prove only the real case with $\mathbb{K} = \mathbb{R}$; the complex case is similar. Let X be the set of all extensions $L : \mathcal{D}(L) \to \mathbb{R}$ of ℓ, where $\mathcal{D}(L) \supseteq G$ is the domain of L, $L|_G = \ell$, and $\|L\| = \|\ell\|$. We have $X \neq \emptyset$, since $\ell \in X$. For $L_1, L_2 \in X$, we say $L_1 \prec L_2$ if L_2 is an extension of L_1 as above; namely, $\mathcal{D}(L_1) \subseteq \mathcal{D}(L_2)$. Then $\prec$ defines a partial ordering on X.

Let $C = \{L_\alpha : \alpha \in A\}$ (A is an index set) be a chain in X. Set

$$\mathcal{D}(\overline{L}) := \bigcup_{\alpha \in A} \mathcal{D}(L_\alpha)$$

and define

$$\overline{L} : \mathcal{D}(\overline{L}) \to \mathbb{R}, \quad \overline{L}(x) = L_\alpha(x) \quad \text{if } x \in \mathcal{D}(L_\alpha).$$

$\overline{L}$ is well-defined: if $x \in \mathcal{D}(L_\alpha) \cap \mathcal{D}(L_\beta)$, $\alpha, \beta \in A$, then since C is a chain, we have $\alpha \prec \beta$ or vice versa. Say $\alpha \prec \beta$. Then L_β extends L_α, so $L_\beta(x) = L_\alpha(x)$ for $x \in \mathcal{D}(L_\alpha)$. It is also not hard to see that $\overline{L}$ is linear and $\|\overline{L}\| = \|\ell\|$. Then $\overline{L} \in X$ extends all of the L_α's, i.e., $\overline{L}$ is an upper bound for C.

Hence, by Zorn's Lemma, $\exists L : \mathcal{D}(L) \to \mathbb{R}$ maximal in X. We claim that $\mathcal{D}(L) = E$. If not, then take $y \in E \setminus \mathcal{D}(L)$ and by the weaker version, L can be extended to $\operatorname{span}\{\mathcal{D}(L), y\}$, contradicting maximality. Therefore, $L \in E'$ is an extension of ℓ with $\|L\| = \|\ell\|$. $\square$

3.2.4 Applications of the Hahn-Banach Theorem

Some Corollary

Corollary 3.2.13

Let $G \subseteq E$ be a subspace of a normed space E. $\forall y \notin G$, $\exists \ell \in E'$ such that $\|\ell\| = 1$, $\ell(y) = \operatorname{dist}(y, G)$ and $\ell|_G = 0$.

Recall that $dist(y, G) = \inf\{\|y - g\| : g \in G\}$.

Proof. On $F = \operatorname{span}\{G, y\}$, define ℓ via $\ell(g + \lambda y) = \lambda c$ for all $g \in G$, where $c = \operatorname{dist}(y, G)$. Then $\|\ell\| = 1$. Indeed,

$$\|\ell\| = \sup_{x \in F \setminus \{0\}} \frac{|\ell(x)|}{\|x\|} = \sup_{F \setminus \{0\}} \frac{|\lambda c|}{\|g + \lambda y\|} = \sup_{g \in G} \frac{c}{\|g/\lambda + y\|} = \frac{c}{\inf_{h \in G} \|h + y\|} = \frac{c}{\operatorname{dist}(y, G)} = 1.$$

By the Hahn–Banach theorem, we can extend this from F to all of E, with the same norm. □

Corollary 3.2.14

$\forall y \in E$, $\exists \ell \in E'$ such that $\|\ell\| = 1$ and $\ell(y) = \|y\|$.

This ℓ *is like a projection onto the line* $\mathbb{R}y$.

Proof. Take $G = \{0\}$ in the previous corollary. □

Corollary 3.2.15

Bounded functionals separate points: if $x_1, x_2 \in E$ and $x_1 \neq x_2$, then $\exists \ell \in E'$ such that $\ell(x_1) \neq \ell(x_2)$.

This says that there are "at least as many" functionals as vectors in E.

Proof. Set $y := x_1 - x_2 \neq 0$ and apply the previous corollary:

$$l(x_1) - l(x_2) = l(x_1 - x_2) = l(y) = \|y\| \neq 0.$$

□

Theorem 3.2.16

Let E be a normed space. A set $M \subseteq E$ is total (i.e., $\overline{\operatorname{span}}(M) = E$) if and only if for $\ell \in E'$, $\ell(M) = \{0\}$ implies $\ell = 0$.

Proof. ($\Rightarrow$) If $\ell(M) = \{0\}$, then $\ell(\operatorname{span}(M)) = \{0\}$ by linearity, which implies that

$$\ell(E) = \ell\left(\overline{\operatorname{span}(M)}\right) = \{0\}$$

by continuity.

($\Leftarrow$) Suppose $G := \overline{\operatorname{span}}(M) \subset E$, let $y \in E \setminus G$ and choose $\ell \in E'$ such that $\|\ell\| = 1$, $\ell|_G = 0$ and $\ell(y) \neq 0$ (so ℓ is not 0). Therefore, $\ell(M) = \{0\}$ but $\ell \neq 0$, a contradiction. It follows that $G = E$.

□

Geometric Applications

Definition 3.2.17

Given $\ell \in E'$, its **kernel** is $\ker \ell = \{x \in E : \ell(x) = 0\}$.

Proposition 3.2.18

Let E be a normed space and $\ell \in E'$ a linear functional. Then $\ker \ell$ is a subspace of E.

Subspace is linear but not necessary closed.

Proof. Linearity is clear:

1. $\ell(0) = 0$, so $0 \in \ker \ell$.
2. If $x, y \in \ker \ell$, then
$$\ell(x+y) = \ell(x) + \ell(y) = 0 + 0 = 0 \quad \Longrightarrow \quad x + y \in \ker \ell.$$
3. If $x \in \ker \ell$, $\alpha \in \mathbb{K}$, then
$$\ell(\alpha x) = \alpha \ell(x) = \alpha \cdot 0 = 0 \quad \Longrightarrow \quad \alpha x \in \ker \ell.$$

The fact that it is closed follows from noticing that $\ker \ell = \ell^{-1}(\{0\})$ and using that ℓ is continuous while $\{0\} \subseteq \mathbb{R}$ is closed (The preimage of a closed set under a continuous map is closed).

□

Definition 3.2.19

Let E be a normed space. For $c \in \mathbb{R}$, we define the **hyperplane** $\Gamma_c \subseteq E$ by
$$\Gamma_c = \{x \in E : \ell(x) = c\} = \ker \ell + z,$$
where $z \in E$ is such that $\ell(z) = c$ (assuming $\ell \neq 0$).

Example.

> For $E = \mathbb{R}^3$, Γ_c are the "regular" planes: take $\ell \in (\mathbb{R}^3)'$ defined by $\ell(x) = a_1x_1 + a_2x_2 + a_3x_3$. Then
> $$\Gamma_c = \left\{x \in \mathbb{R}^3 : a_1x_1 + a_2x_2 + a_3x_3 = c,\ c \in \mathbb{R}\right\}.$$

Definition 3.2.20

Let E be a normed vector space. Given $\Omega \subseteq E$, we call Γ_c a **supporting hyperplane** of Ω at $x_0 \in \Omega$ if $x_0 \in \Gamma_c$ and $\ell(x) \geq c$ or $\ell(x) \leq c$ for all $x \in \Omega$.

Theorem 3.2.21

The unit sphere $S_1(0) = \{x \in E : \|x\| = 1\}$ has a supporting hyperplane at every point.

Proof. Take $x_0 \in S_1(0)$. By Corollary, $\exists \ell \in E'$ such that $\|\ell\| = 1$ and $\ell(x_0) = \|x_0\| = 1$. Then $\Gamma_1 = \{x \in E : \ell(x) = 1\}$ is a supporting hyperplane at x_0. Indeed, $x_0 \in \Gamma_1$ and for $x \in S_1(0)$,
$$\ell(x) \leq |\ell(x)| \leq \|\ell\| \cdot \underbrace{\|x\|}_{=1} = 1.$$

□

General Forms of Functionals

In $\mathbb{R}^n$, every $\ell \in (\mathbb{R}^n)'$ can be written as $\ell(x) = \sum_{i=1}^n \ell_i \cdot x_i, \quad \ell_1, \ldots, \ell_n \in \mathbb{R}$. What about in infinite dimensions?

Theorem 3.2.22

$f \in (\ell^1)'$ if and only if $f(x) = \sum_{n\geq 0} f_n x_n$ for all $x = (x_n) \in \ell^1$ and some $(f_n)_{n\in\mathbb{N}} \in \ell^\infty$. *Namely, the dual of ℓ^1 is ℓ^∞, which is $(\ell^1)' \simeq \ell^\infty$.*

Proof. $\Rightarrow$: construct f_n, then prove boundness.

Let $f \in (\ell^1)'$. Set $\ell_n = (0, \ldots, 0, 1, 0, \ldots, 0) \in \ell^1$, where the 1 is in the nth position. Then

$$f(x) = f((x_0, x_1, \ldots)) = f\left(\lim_{N\to\infty} \sum_{n=0}^N x_n \ell_n\right) = \lim_{N\to\infty} \sum_{n=0}^N f(\ell_n) \cdot x_n = \sum_{n=0}^\infty f(\ell_n) \cdot x_n \quad \forall x \in l^1$$

where we can move the limit outside due to the linearity and continuity of f. Let $f_n := f(\ell_n)$. Then

$$|f_n| \leq \|f\| \cdot \|\ell_n\| = \|f\| \cdot 1 = \|f\|, \quad \forall n \in \mathbb{N} \quad \Rightarrow \quad (f_n) \in \ell^\infty,$$

since $\|f\|_{(\ell^1)'}$ is independent of $n \in \mathbb{N}$.

$\Leftarrow$: prove well-define, linear and bounded.

Conversely, let $(f_n) \in \ell^\infty$. On the subspace

$$G = \left\{(x_n) \in \ell^1 : x_n = 0 \text{ for all but finitely many } n\right\},$$

it is clear that the formula $f(x) = \sum_{n\geq 0} f_n x_n$ defines a functional f whose norm is bounded by $\|(f_n)\|_\infty$. Indeed, $\forall x \in G$, we have

$$|f(x)| = \left|\sum_{n=0}^\infty f_n x_n\right| \leq \|f_n\|_\infty \sum_{n=0}^\infty |x_n| = \|f_n\|_\infty \|x\|_1 \quad \Rightarrow \quad \|f\| \leq \|f_n\|_\infty.$$

Then, we can extend this by continuity to all of ℓ^1 (since G is dense) and obtain $f \in (\ell^1)'$ with the same bound (linearity is clear). □

From the above proof, it follows that

$$\|f\|_{(\ell^1)'} = \|(f_n)_{n\in\mathbb{N}}\|_\infty.$$

Corollary 3.2.23

The map $(\ell^1)' \to \ell^\infty$, $f \mapsto (f_n)_{n\in\mathbb{N}}$, is an isomorphism of normed spaces (isomorphic: linear, bijective, and norm-preserving).

The recommended textbook describes them as isometrically isomorphic, which means the same thing.

Proof. $\Leftarrow$:

$$\ell^\infty \longrightarrow (\ell^1)', \quad (f_n) \mapsto \left[x \mapsto \sum f_n x_n\right].$$

$\Rightarrow$:

$$(\ell^1)' \longrightarrow \ell^\infty, \quad f \mapsto (f(\ell_n))_{n\geq 0}.$$

Thus, linear, bijective, and isometry. □

Fact 3.2.24

- For $p > 1$, $(\ell^p)' = \ell^{p'}$ if $1/p + 1/p' = 1$.
- For $p > 1$, if $1/p + 1/p' = 1$, then $(L^p([a,b]))' = L^{p'}([a,b])$:

$$\forall f \in (L^p)',\ \exists \hat{f} \in L^{p'} \text{ such that } f(g) = \int_a^b \hat{f}(x)g(x)\,dx, \quad \forall g \in L^p([a,b]).$$

Here, $=$ is not equal but isomorphism.

Given Theorem, it is natural to ask whether $(\ell^\infty)' \simeq \ell^1$. In fact, it is not.

Proposition 3.2.25

$(\ell^\infty)' \neq \ell^1$.

Proof. We can see that $\ell^1 \subseteq (\ell^\infty)'$, as in Theorem: for $a = (a_n) \in \ell^1$, we define $f_a : \ell^\infty \to \mathbb{R}$ by

$$f_a((x_n)) := \sum_{n=0}^{\infty} a_n x_n.$$

Since $\sum |a_n| < \infty$ and $|x_n| \leq \|x\|_\infty$, $|f_a(x)| \leq \|a\|_1 \|x\|_\infty$. Therefore, f_a is continuous and linear, with $\|f_a\| \leq \|a\|_1$, hence $f_a \in (\ell^\infty)'$. Thus, every l^1 is embedded into $(\ell^\infty)'$.

WTS $\ell^1 \neq (\ell^\infty)'$. That is $\exists f, \nexists a \in l^1$ such that $a \to f$. Now set

$$G := \{\text{convergent sequences in } \ell^\infty\} \subseteq \ell^\infty$$

and define

$$f : G \to \mathbb{R}, \quad f(x) = \lim_{n\to\infty} x_n, \quad x = (x_n) \in G.$$

Extend f to all of ℓ^∞ by preserving $\|f\| = 1$ (choosing $\underline{1}$). Then $f \in (\ell^\infty)'$ does not come from ℓ^1. If it did, then for any "finite sequence" $(x_n) \in G \subseteq \ell^\infty$ (i.e., $\exists N$ such that $x_n = 0$ for all $n \geq N$),

$$0 = \lim_{n\to\infty} x_n = f((x_n)) = \sum_{n=0}^{N} a_n x_n, \quad \forall (x_n)_{n=0}^N \Rightarrow a_n = 0, \quad \forall n \quad \text{and} \quad f = 0,$$

□

3.2.5 Reflexive Spaces

Let E be a normed space (over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$). We know E' is a Banach space. Consider now $E'' = (E')'$. We will now show that E can be embedded into E''.

Given $x \in E$, let $\phi_x : E' \to \mathbb{K}$ be given by $\phi_x(\ell) = \ell(x)$ for all $\ell \in E'$. Then ϕ_x is a linear functional. Moreover,

$$|\phi_x(\ell)| = |\ell(x)| \leq \|x\|_E \cdot \|\ell\|_{E'} \Rightarrow \|\phi_x\| \leq \|x\|$$

implies that as a map from $(E', \|\cdot\|_{E'})$ to $\mathbb{K}$, ϕ_x is bounded, so $\phi_x \in E''$. In fact, the above computation shows that $\|\phi_x\|_{E''} \leq \|x\|_E$.

Lemma 3.2.26

For a linear map $T : E \to F$, T is injective if and only if $\ker T = \{0\}$

Proof. Linearity gives us

$$T(x) = T(y) \iff T(x) - T(y) = 0 \iff T(x - y) = 0.$$

Hence

$$T \text{ is injective} \iff \big[T(x-y)=0 \implies x-y=0\big] \iff \ker T = \{0\}\,.$$

□

Theorem 3.2.27

Let E be a normed space. The map $\Phi : E \to E''$, $x \mapsto \phi_x$, is an isometric monomorphism (norm-preserving, linear and injective mapping).

Proof. Linearity: $\forall l \in E'$,

$$\Phi(\alpha x + \beta y)(\ell) = \ell(\alpha x + \beta y) = \alpha\ell(x) + \beta\ell(y) = \alpha\Phi(x)(\ell) + \beta\Phi(y)(\ell).$$

For injectivity, suppose $\phi_x = 0$ in E''. Then $\ell(x) = \phi_x(\ell) = 0$ for all $\ell \in E'$. By Corollary, $\forall x \in E$, $\exists \ell \in E'$ such that $\|\ell\| = 1$ and $\ell(x) = \|x\|$, meaning $\|x\| = 0 \Rightarrow x = 0$. Finally, to prove that $\|\phi_x\|_{E''} = \|x\|$, we already know that $\|\phi_x\| \le \|x\|$, and by using the ℓ given by the corollary, we obtain

$$|\phi_x(\ell)| = |\ell(x)| = \|x\| = \underbrace{\|\ell\|}_{=1}\|x\| \Rightarrow \|\phi_x\| \ge \frac{|\phi_x(\ell_0)|}{\|\ell_0\|} = \frac{|\ell_0(x)|}{1} = \|x\|$$

Therefore, $\|\phi_x\| = \|x\|$. □

Definition 3.2.28

We say a normed space E is **reflexive** if $\Phi : E \to E''$ is an isomorphism (i.e. it is surjective).

Example.

> ℓ^p and L^p are reflexive spaces for $p \in (1,\infty)$. However, ℓ^1 and ℓ^∞ are not. Also, $C([0,1])$ is not reflexive.

3.3 The Uniform Boundedness Principle

3.3.1 The Banach-Steinhaus Theorem

Lemma 3.3.1: Nested Balls Lemma

Let E be a Banach space, $(x_n)_{n\in\mathbb{N}} \subseteq E$ and $(r_i)_{i\in\mathbb{N}} \subseteq \mathbb{R}$. Consider a sequence of closed balls in E:

$$\overline{B_{r_0}(x_0)} \supseteq \overline{B_{r_1}(x_1)} \supseteq \cdots, \quad \text{with } r_i \to 0 \text{ as } i \to \infty.$$

Then

$$\bigcap_{i=0}^{\infty} \overline{B_{r_i}(x_i)} \neq \emptyset.$$

Banach space $\Rightarrow$ Complete $\Rightarrow$ Cauchy sequence converges.

Proof. Let $n, m \in \mathbb{N}$ with $n < m$. Then $x_m \in \overline{B_{r_n}(x_n)}$, which means $\|x_n - x_m\| < r_n \to 0$ as $n \to \infty$, i.e. (x_n) is Cauchy. Let $x^* := \lim_{n\to\infty} x_n$. Since $x_m \in \overline{B_{r_n}(x_n)}$ holds for all m and the ball is closed, we have $x^* \in \overline{B_{r_n}(x_n)}$ for all $n \in \mathbb{N}$. Therefore, x^* is in the intersection.

□

Lemma 3.3.2: Baire Category Theorem

Let E be a Banach space and $(C_n)_{n\in\mathbb{N}}$ be a sequence of closed sets in E with $\bigcup_{n\in\mathbb{N}} C_n = E$. Then one of them must have a non-empty interior, i.e. $\exists n \in \mathbb{N}, r > 0$ and $x \in E$ such that $B_r(x) \subseteq C_n$.

Proof. Proof by contradiction: assume that there exists no such open ball and C_n. Then $C_0 \neq E$, so choose $x_0 \in E \setminus C_0$. Since $E \setminus C_0$ is open, $\exists r_0 > 0$ such that

$$\overline{B_{r_0}(x_0)} \subseteq B_{2r_0}(x_0) \subseteq E \setminus C_0.$$

Since $B_{r_0}(x_0) \not\subseteq C_1$, $B_{r_0}(x_0) \cap (E \setminus C_1) \neq \emptyset$ (and open), so for all $x_1 \in B_{r_0}(x_0) \cap (E \setminus C_1)$, $\exists r_1 < r_0/2$ such that

$$\overline{B_{r_1}(x_1)} \subseteq B_{r_0}(x_0) \cap (E \setminus C_1).$$

We continue in this manner: take $x_i \in B_{r_{i-1}}(x_{i-1}) \cap (E \setminus C_i)$ and $r_i < r_{i-1}/2$ such that

$$\overline{B_{r_i}(x_i)} \subseteq B_{r_{i-1}}(x_{i-1}) \cap (E \setminus C_i).$$

In this way, we construct a sequence of nested balls

$$\overline{B_{r_0}(x_0)} \supseteq \overline{B_{r_1}(x_1)} \supseteq \cdots$$

such that $r_n \to 0$ as $n \to \infty$.

By Lemma, $\exists x^* \in E$ such that $x^* \in \overline{B_{r_n}(x_n)}$ for all $n \in \mathbb{N}$, meaning that, by construction, $x^* \notin C_n$ for all $n \in \mathbb{N}$. Then

$$x^* \notin \bigcup_{n\in\mathbb{N}} C_n = E,$$

a contradiction. □

Lemma 3.3.3

Let E be a Banach space. There exist $a \in E$ and $r > 0$ such that the set

$$S_{r,a} := \left\{\ell_n(x) : x \in \overline{B_r(a)},\, n \in \mathbb{N}\right\}$$

is bounded in $\mathbb{R}$.

Proof. Let

$$C_m := \{x \in E : |\ell_n(x)| \leq m,\, \forall n \in \mathbb{N}\} = \bigcap_{n\in\mathbb{N}} \ell_n^{-1}([-m, m]).$$

This is only true when pointwise bounded. Since the ℓ_n are continuous, C_m is closed. By assumption (of the Banach-Steinhaus Theorem), we have $E = \bigcup_{m\in\mathbb{N}} C_m$. Thus, by Baire's theorem, there exists $m \in \mathbb{N}$, $a \in E$ and $r > 0$ such that

$$B_r(a) \subseteq C_m.$$

Thus, since C_m is closed,

$$|\ell_n(x)| \leq m, \quad \forall x \in \overline{B_r(a)},\, \forall n \in \mathbb{N} \quad \Longrightarrow \quad S_{r,a} \subseteq [-m, m],$$

as desired. □

Theorem 3.3.4: Banach–Steinhaus Theorem

Let E be a Banach space. Consider a sequence $(\ell_n)_{n\in\mathbb{N}} \subseteq E'$ and assume that $\forall x \in E$, $\exists c_x > 0$ such that $|\ell_n(x)| \leq c_x$ for all $n \in \mathbb{N}$ (i.e. (ℓ_n) is "pointwise bounded"). Then the sequence (ℓ_n) is bounded, i.e. $\exists c > 0$ such that $\|\ell_n\| \leq c$ for all $n \in \mathbb{N}$.

This is also known as the uniform boundedness principle.

Proof. Recall that

$$\|\ell_n\| = \sup_{x \in B_1(0)} |\ell_n(x)|.$$

It is enough to show that

$$|\ell_n(x)| \leq c, \quad \forall x \in \overline{B_1(0)}, \forall n \in \mathbb{N}.$$

Given $x \in \overline{B_1(0)}$, let $x' := rx + a \in \overline{B_r(a)}$, with r and a as in Lemma. Then $x = (x' - a)/r$ and

$$|\ell_n(x)| = \left|\ell_n\left(\frac{x' - a}{r}\right)\right| \leq \frac{|\ell_n(x')| + |\ell_n(a)|}{r} \leq \frac{2m}{r},$$

where $|\ell_n(x')|, |\ell_n(a)| \leq m$ by Lemma.

Since $2m/r$ does not depend on $n \in \mathbb{N}$ or $x \in E$, we are done. □

Corollary 3.3.5

Let E be a Banach space and $(\ell_n)_{n\in\mathbb{N}} \subseteq E'$ such that it pointwise converges to $\ell : E \to \mathbb{R}$ (i.e. $\ell_n(x) \to \ell(x)$ for all $x \in E$). Then $\ell \in E'$.

Proof. $\ell_n(x) \to \ell(x)$ implies that (ℓ_n) is bounded in $\mathbb{R}$ for each fixed $x \in E$. Thus, we apply the Banach–Steinhaus theorem and conclude that

$$\|\ell_n\| \leq c \quad \text{for all} \quad n \in \mathbb{N},$$

where c is independent of x and n.

This holds if and only if

$$|\ell_n(x)| \leq c \quad \text{for all} \quad x \in B_1(0) \quad \text{and} \quad n \in \mathbb{N}.$$

Taking the limit, we see that

$$|\ell(x)| \leq c \quad \text{for all} \quad x \in B_1(0),$$

and therefore ℓ is bounded with $\ell \in E'$, as desired. □

3.3.2 Weak Convergence

Definition 3.3.6

Let E be a normed space. We say the sequence $(\ell_n)_{n\in\mathbb{N}} \subseteq E'$ **converges weakly** to $\ell \in E'$ if

$$\lim_{n\to\infty} \ell_n(x) = \ell(x), \quad \forall x \in E.$$

By the Banach–Steinhaus Theorem, a weakly convergent sequence in E' is bounded with respect to the operator norm. Here is pointwise convergent.

The above is called weak convergence because convergence in the norm implies weak convergence, but the converse is not true (example below). Indeed, let $\ell_n \to \ell$ (in the norm); then $\|\ell_n - \ell\| \to 0$, so

$$|\ell_n(x) - \ell(x)| \leq \|\ell_n - \ell\| \cdot \|x\| \to 0,$$

for each fixed $x \in E$. Thus, weak convergence is *strictly weaker* than convergence in the norm.

Example.

Let $E = C([-1, 1])$ and $\ell_n : E \to \mathbb{R}$ with $\ell_n(f) = f(1/n)$.

Proof. Then $\ell_n \in E'$ since

$$|\ell_n(f)| = |f(1/n)| \leq \|f\|_\infty.$$

Moreover, $\forall f \in E$,

$$\lim_{n\to\infty} \ell_n(f) = \lim_{n\to\infty} f(1/n) = f(0),$$

where the last equality follows from f being continuous. Define $\ell \in E'$ by $\ell(f) = f(0)$.

We just showed that $\ell_n \to \ell$ weakly. But we will now see that $\ell_n \nrightarrow \ell$ in the norm: pick $f \in E$ given by

$$f(x) = \begin{cases} 1, & x \in [-1, 0] \\ -nx + 1, & x \in (0, 1/n] \\ 0, & x \in (1/n, 1]. \end{cases}$$

Then $\|f\| = 1$ and by definition of the norm

$$\|\ell_n - \ell\| \geq |\ell_n(f) - \ell(f)| = |f(1/n) - f(0)| = 1 \quad \implies \quad \lim_{n\to\infty} \|\ell_n - \ell\| \neq 0.$$

Thus, we see that weak convergence does not imply convergence in the norm. □

Proposition 3.3.7

Let E be a normed space. E' is weakly complete, i.e., if $(\ell_n)_{n\in\mathbb{N}}$ is a sequence such that $(\ell_n(x))$ is Cauchy $\forall x \in E$, then $\exists \ell \in E'$ such that $\ell_n \to \ell$ weakly.

Proof. Let $\ell(x) = \lim_{n\to\infty} \ell_n(x)$ for all $x \in E$ (exists because $(\ell_n(x))$ is Cauchy and $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ is complete).

By Corollary, we know that $\ell \in E'$. And simply by definition, $\ell_n \to \ell$ weakly. □

Theorem 3.3.8

Let E be Banach and assume $M \subseteq E$ is a dense set. Then the sequence $(\ell_n)_{n\in\mathbb{N}} \subseteq E'$ converges weakly to $\ell \in E'$ if and only if the following hold:

1. $\ell_n(x) \to \ell(x)$ for all $x \in M$.
2. $\|\ell_n\| \leq c$ for some $c \in \mathbb{R}$ independent of $n \in \mathbb{N}$.

Proof. ($\Rightarrow$) (1) is clear and (2) is precisely the Banach–Steinhaus Theorem.

($\Leftarrow$) Let $y \in E$ and let us show that $\ell_n(y) \to \ell(y)$. $\forall x \in M$, we have

$$\begin{aligned} |\ell_n(y) - \ell(y)| &\leq |\ell_n(y) - \ell_n(x)| + |\ell_n(x) - \ell(x)| + |\ell(x) - \ell(y)| \\ &\leq \|\ell_n\| \cdot \|y - x\| + |\ell_n(x) - \ell(x)| + \|\ell\| \cdot \|x - y\| \\ &\leq (c + \|\ell\|) \cdot \|y - x\| + |\ell_n(x) - \ell(x)|, \end{aligned}$$

where the last inequality follows from (2).

Fix $\varepsilon > 0$. Since M is dense in E, there exists $x \in M$ such that

$$\|y - x\| \leq \frac{\varepsilon}{2(c + \|\ell\|)}.$$

Moreover, $\exists N \in \mathbb{N}$ such that

$$|\ell_n(x) - \ell(x)| < \varepsilon/2 \quad \text{for all} \quad n \geq N$$

(by (1)). Thus,

$$|\ell_n(y) - \ell(y)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \quad \forall n \geq N.$$

Therefore, $\ell_n(y) \to \ell(y)$ as $n \to \infty$ and $y \in E$ was arbitrary. □

Definition 3.3.9

Let E be a normed space. Weak convergence defines a topology (the *weak topology*) on E': a set $F \subseteq E'$ is closed if and only if $\forall (\ell_n)_{n\in\mathbb{N}} \subseteq F$ weakly convergent, its limit is in F.

A set $C \subseteq E'$ is weakly compact if it is compact under the weak topology.

Theorem 3.3.10: Banach–Alaoglu Theorem

If E is a separable Banach space, then the unit sphere $S_{E'} = \{\ell \in E' : \|\ell\|_{E'} = 1\}$ in E' is weakly compact.

Proof. Given a sequence $(\ell_n)_{n\in\mathbb{N}} \subseteq S_{E'} \subseteq E'$, we want to extract a convergent subsequence. Let $M = \{x_1, x_2, \ldots\}$ be a countable dense subset of E. Since $|\ell_n(x_1)| \leq \|\ell_n\| \cdot \|x_1\| \leq \|x_1\|$, there is a subsequence $(\ell_{k,1})_{k\in\mathbb{N}}$ of (ℓ_n) (by B-W) so that $(\ell_{k,1}(x_1))_{k\in\mathbb{N}}$ converges.

Applying the same reasoning to $(\ell_{k,1}(x_2))_{k\in\mathbb{N}}$, we extract a subsequence $(l_{k,2})_{k\in\mathbb{N}}$ of $(l_{k,1})_{k\in\mathbb{N}}$ such that $(\ell_{k,2}(x_2))_{k\in\mathbb{N}}$ is convergent. Thus, we have the following sequences:

$$\begin{aligned} &\ell_{1,1}(x_1), \ell_{2,1}(x_1), \ell_{3,1}(x_1), \ldots && \text{convergent} \\ &\ell_{1,2}(x_2), \ell_{2,2}(x_2), \ell_{3,2}(x_2), \ldots && \text{convergent} \\ &\ell_{1,3}(x_3), \ell_{2,2}(x_3), \ell_{3,3}(x_3), \ldots && \text{convergent.} \end{aligned}$$

Consider the subsequence $(\ell_{k,k})_{k\in\mathbb{N}}$ of $(\ell_n)_{n\in\mathbb{N}}$. By construction, $(\ell_{k,k}(x_i))_{k\in\mathbb{N}}$ converges $\forall i \in \mathbb{N}$. Indeed, for $k \geq i$, $(\ell_{k,k}(x_i))$ is a subsequence of $(\ell_{k,i}(x_i))_{k\geq i}$, which is convergent by construction.

Define

$$\ell(x_i) := \lim_{k\to\infty} \ell_{k,k}(x_i), \quad \forall i \in \mathbb{N},$$

and extend by continuity to all of E (recall that $\overline{M} = E$) to make the functionals defined on E.

Then by previous Theorem, $\ell \in E'$ and $\ell_{k,k} \to \ell$ weakly. □

Definition 3.3.11

A sequence $(x_n)_{n\in\mathbb{N}} \subseteq E$ **converges weakly** to $x \in E$ if $\ell(x_n) \to \ell(x)$ as $n \to \infty$ for all $\ell \in E'$.

Lemma 3.3.12

Recall that $\Phi : E \to E''$, $x \mapsto \phi_x$. Let $(x_n) \subseteq E$. Then $x_n \to x$ weakly if and only if $\Phi(x_n) \to \Phi(x)$ weakly.

Proof. ($\Rightarrow$) Suppose $(x_n) \subseteq E$ with $x_n \to x$ weakly. Then $\forall \ell \in E'$, $\ell(x_n) \to \ell(x)$. But this is the same as $\phi_{x_n}(\ell) \to \phi_x(\ell)$ for all $\ell \in E'$ (i.e., $\phi_{x_n} \to \phi_x$ weakly), by definition of ϕ_x, which, in turn, is the same as $\Phi(x_n) \to \Phi(x)$. Since this holds for all (x_n) such that $x_n \to x$ weakly, $\Phi(x_n) \to \Phi(x)$ weakly, as desired.

($\Leftarrow$) Suppose $\Phi(x_n) \to \Phi(x)$ weakly. Then $\forall \ell \in E'$, $\phi_{x_n}(\ell) \to \phi_x(\ell)$, which is precisely the definition of $x_n \to x$ weakly.

□

Notice that convergence in the norm (in E) implies weak convergence in E: if $\|x_n - x\| \to 0$, then given any $\ell \in E'$,

$$|\ell(x_n) - \ell(x)| \leq \|\ell\| \cdot \|x_n - x\| \to 0.$$

The converse does not hold

Proof. Take $E = \ell^2$ and $x_n = \ell_n = (0, \ldots, 0, 1, 0, \ldots)$, where the 1 is the n-th term in the sequence.

1. $x_n \to 0$ weakly. Indeed, any $f \in (\ell^2)' \simeq \ell^2$ can be obtained by

$$f((a_n)_{n\in\mathbb{N}}) = \sum_{n\geq 0} f_n a_n, \quad \forall (a_n) \in \ell^2$$

   for some sequence $(f_n)_{n\in\mathbb{N}} \in \ell^2$. Therefore, $f(\ell_n) = f_n \to 0$ as $n \to \infty$, because the series $\sum |f_n|^2$ converges.

2. $\|x_n - 0\|_2 \neq 0$, since $\|x_n\|_2 \equiv 1$ for all $n \in \mathbb{N}$.

□

Theorem 3.3.13

Let E be a separable, reflexive Banach space. Then $S_1^E(0) = \{x \in E : \|x\| = 1\} \subseteq E =$ is weakly compact.

Proof. Reflexivity means that $\Phi : E \to E'$ is an isometric linear isomorphism. Let $(x_n)_{n\in\mathbb{N}}$ be a sequence in $S_1^E(0)$. Then $(\Phi(x_n))_{n\in\mathbb{N}}$ is a sequence in $S_1^{E''}(0)$. By Banach-Alaoglu (applied to E'), $\mathcal{S}_1^{E''}(0)$ is weakly compact, so there exists a subsequence $(\Phi(x_{n_k}))_{k\in\mathbb{N}}$ converging weakly to $\ell = \Phi(x) \in E''$. By Lemma, $x_{n_k} \rightharpoonup x$ weakly □

3.4 The Riesz Representation Theorem

3.4.1 Reflexivity of Hilbert Spaces

Theorem 3.4.1

Let $\mathcal{H}$ be a Hilbert space and $\mathbb{F}$ a field. The inner product $\langle\cdot,\cdot\rangle : \mathcal{H} \times \mathcal{H} \to \mathbb{F}$ is a continuous function with respect to convergence in the norm.

The key here is Cauchy Schwarz.

Proof. Let $(x_n), (y_n) \subseteq \mathcal{H}$ with $x_n \to x$ and $y_n \to y$. Then (x_n) and (y_n) are both bounded, say by c, and

$$|\langle x_n, y_n\rangle - \langle x, y\rangle| \leq |\langle x_n, y\rangle - \langle x, y\rangle| + |\langle x_n, y_n\rangle - \langle x_n, y\rangle| \leq \|y\|\cdot\|x_n - x\| + \|x_n\|\cdot\|y_n - y\| \leq \|y\|\cdot\|x_n - x\| + c\cdot\|y_n - y\| \to 0,$$

so $\langle x_n, y_n\rangle \to \langle x, y\rangle$, as desired. □

Definition 3.4.2

Let $\mathcal{H}$ be a Hilbert space. Given a subspace $G \subseteq \mathcal{H}$, its **orthogonal complement** is the subspace

$$G^\perp := \{x \in \mathcal{H} : \langle x, y\rangle = 0 \ \forall y \in G\}$$

(sometimes also denoted by $\mathcal{H} \ominus G$). For $x \in \mathcal{H}$, we call $y \in G$ an **orthogonal projection** (of x onto G) if $x - y \in G^\perp$.

Theorem 3.4.3

Let $\mathcal{H}$ be a Hilbert space. Given a closed subspace $G \subseteq \mathcal{H}$ and any $x \in \mathcal{H}$, there exists a unique orthogonal projection $y = \mathrm{pr}_G x \in G$.

Proof. *Uniqueness.* If $y_1, y_2 \in G$ are such that $x - y_1, x - y_2 \in G^\perp$, then

$$\|y_1 - y_2\|^2 = \langle y_1 - y_2, y_1 - y_2\rangle = \langle (y_1 - x) + (x - y_2), y_1 - y_2\rangle = -\langle x - y_2, y_1 - y_2\rangle + \langle x - y_2, y_1 - y_2\rangle.$$

Since $x - y_1, x - y_2 \in G^\perp$ while $y_1 - y_2 \in G$, it follows that

$$\|y_1 - y_2\|^2 = 0 \iff \|y_1 - y_2\| = 0 \iff y_1 = y_2.$$

Existence. If $x \in G$, we are done ($y = x$). If not, then

$$d := \inf_{z\in G} \|x - z\| > 0$$

because G is closed. By completeness, the infimum is a minimum, realised by some $y \in G$. One then shows $y - x \in G^\perp$ as follows: y also minimises $\|x - z\|^2$, so $\forall v \in G$, $f(t) = \|x - y - tv\|^2$ attains a minimum at $t = 0$. Thus,

$$\|x - y - tv\|^2 = \|x - y\|^2 - 2t \cdot \langle x - y, v\rangle + t^2 \cdot \|v\|^2 \Longrightarrow 0 = f'(0) = -2\langle x - y, v\rangle.$$

□

Theorem 3.4.4: Riesz Representation Theorem

Let $\mathcal{H}$ be a Hilbert space and $\mathbb{F} \subseteq \mathbb{R}, \mathbb{C}$ a field. Given $y \in \mathcal{H}$, the mapping $\ell_y : \mathcal{H} \to \mathbb{F}$ defined by $\ell_y(x) := \langle x, y\rangle$ satisfies $\ell_y \in \mathcal{H}'$. Conversely, $\forall \ell \in \mathcal{H}'$, $\exists y \in \mathcal{H}$ such that $\ell = \ell_y$, and moreover $\|\ell_y\|_{\mathcal{H}'} = \|y\|_{\mathcal{H}}$.

Proof. It is clear that ℓ_y is a linear functional. To see that it is bounded, let $x \in \mathcal{H}$. Then

$$|\ell_y(x)| = |\langle x, y\rangle| \leq \|x\| \cdot \|y\| \quad \Longrightarrow \quad \|\ell_y\| \leq \|y\|.$$

(Cauchy-Schwarz)

By choosing $x = y$ we obtain equality, so $\|\ell_y\| = \|y\|$. Conversely, let $\ell \in \mathcal{H}'$. If $\ell = 0$, take $y = 0$, so assume that $\ell \neq 0$. Set $G := \ker \ell$, a closed subspace of $\mathcal{H}$. We have

$$\mathcal{H} = G \oplus G^\perp$$

since any $x \in \mathcal{H}$ can be written as $x = \mathrm{pr}_G x + (x - \mathrm{pr}_G x)$, where $x - \mathrm{pr}_G x \in G^\perp$ by definition of pr_G, and $\forall z_1 \in G$, $z_2 \in G^\perp$, $\langle z_1, z_2\rangle = 0$. We claim that $\dim G^\perp = 1$ (See A5). Let $e \in G^\perp$ with $\|e\| = 1$. Then $\forall x \in \mathcal{H}$, $x = g + \lambda e$, where $g \in \ker \ell$ and $\lambda \in \mathbb{F}$. Therefore, $\ell(x) = \ell(g) + \lambda\ell(e) = \lambda\ell(e)$. On the other hand,

$$\langle x, e\rangle = \langle g, e\rangle + \lambda \cdot \langle e, e\rangle = \lambda \quad \Longrightarrow \quad \ell(x) = \langle x, e\rangle \cdot \ell(e) = \left\langle x, \overline{\ell(e)}e\right\rangle.$$

Setting $y := \overline{\ell(e)}e$ gives $\ell(x) = \ell_y(x) = \langle x, y\rangle$ (remember that $\ell(e)$ is a number, so $\overline{\ell(e)}$ denotes the complex conjugate of $\ell(e)$). Uniqueness of such a y is clear. Indeed, suppose

$$\langle x, y_2\rangle = \langle x, y_1\rangle, \quad \forall x \in \mathcal{H} \quad \Longrightarrow \quad \langle x, y_1 - y_2\rangle = 0, \quad \forall x \in \mathcal{H}.$$

In particular, choosing $x = y_1 - y_2$ yields $y_1 = y_2$. □

Corollary 3.4.5

Hilbert spaces are reflexive.

Proof.

The map $\mathcal{H} \to \mathcal{H}'$, $y \mapsto \ell_y$ is an isometric linear isomorphism $\Longrightarrow \mathcal{H} \simeq \mathcal{H}'$ (isomorphic as Hilbert spaces).

Applying this to $\mathcal{H}'$ yields $\mathcal{H}' \simeq \mathcal{H}'' \Longrightarrow \mathcal{H} \simeq \mathcal{H}''$.

□

3.4.2 Orthonormal Bases

Definition 3.4.6

A basis of a vector space is a linearly independent system of vectors such that every other vector is a linear combination of theirs.

Definition 3.4.7

A collection of vectors $\{e_n\}_{n\in\mathbb{N}}$ in a Hilbert space $\mathcal{H}$ is said to be **orthonormal** if

$$\langle e_i, e_j\rangle_{\mathcal{H}} = \delta_{ij} := \begin{cases} 1, & i = j \\ 0, & i \neq j, \end{cases} \quad \forall i, j \in \mathbb{N}.$$

Example.

> In ℓ^2, we can take $e_i = (0, \dots, 0, 1, 0, \dots)$, with the 1 being the ith term in the sequence, $\forall i \in \mathbb{N}$, to form the orthonormal collection $\{e_i\}_{i\in\mathbb{N}}$.

Lemma 3.4.8

Let $\mathcal{H}$ be a Hilbert space and $\{e_k\}_{k\in\mathbb{N}}$ an orthonormal system in $\mathcal{H}$. The series $\sum_{k\geq 0} c_k e_k$ ($c_k \in \mathbb{K}$) then converges in $\mathcal{H}$ if and only if the number series $\sum_{k\geq 0} |c_k|^2$ is convergent.

Proof. Since $\mathcal{H}$ is complete, the series $\sum_{k\geq 0} c_k e_k$ converges if and only if the sequence of its partial sums is Cauchy. Let $n > m$. Then

$$\left\| \sum_{k=0}^{n} c_k e_k - \sum_{k=0}^{m} c_k e_k \right\|^2 = \left\| \sum_{k=m+1}^{n} c_k e_k \right\|^2 = \sum_{j,k=m+1}^{n} c_j \overline{c_k} \langle e_j, e_k\rangle = \sum_{k=m+1}^{n} |c_k|^2.$$

By the Cauchy criterion for the convergence of a series (MATH2400), the result follows. □

Theorem 3.4.9

Let $\mathcal{H}$ be a Hilbert space. If $x \in \mathcal{H}$ and $\{e_i\}_{i\in\mathbb{N}}$ is an orthonormal system, then the series $\sum_{i\geq 0}\langle x, e_i\rangle e_i$ converges in $\mathcal{H}$, and

$$\sum_{i\geq 0}\langle x, e_i\rangle e_i = \mathrm{pr}_G x, \quad G := \text{c.l.s.}(\{e_i : i \in \mathbb{N}\}).$$

Proof. $\forall N \in \mathbb{N}$,

$$0 \leq \left\| x - \sum_{i=0}^{N}\langle x, e_i\rangle e_i \right\|^2 = \left\langle x - \sum_{i=0}^{N}\langle x, e_i\rangle e_i,\, x - \sum_{j=0}^{N}\langle x, e_j\rangle e_j \right\rangle$$

$$= \|x\|^2 - \sum_{j=0}^{N}\langle x, e_j\rangle\langle x, e_j\rangle - \sum_{i=0}^{N}\langle x, e_i\rangle \cdot \overline{\langle e_i, x\rangle} + \sum_{i,j=0}^{N}\langle x, e_i\rangle \cdot \overline{\langle x, e_j\rangle} \cdot \langle e_i, e_j\rangle$$

$$= \|x\|^2 - \sum_{i=0}^{N}|\langle x, e_i\rangle|^2 \Rightarrow \sum_{i=0}^{N}|\langle x, e_i\rangle|^2 \leq \|x\|^2, \quad \forall N \in \mathbb{N}.$$

Therefore, $\sum_{i\geq 0}|\langle x, e_i\rangle|^2$ converges, meaning $\sum_{i\geq 0}\langle x, e_i\rangle e_i$ converges. For the last claim, we must show that

$$(y = x - \sum_{i\geq 0}\langle x, e_i\rangle e_i) \perp G.$$

Indeed, $\forall j \in \mathbb{N}$,

$$\left\langle x - \sum_{i\geq 0}\langle x, e_i\rangle e_i,\, e_j \right\rangle = \langle x, e_j\rangle - \sum_{i\geq 0}\langle x, e_i\rangle\delta_{ij} = \langle x, e_j\rangle - \langle x, e_j\rangle = 0.$$

Therefore, $y \perp e_j$ for all $j \in \mathbb{N}$, meaning $y \perp \operatorname{span}(\{e_j : j \in \mathbb{N}\})$, so by continuity, $y \perp G$.

□

Corollary 3.4.10

If $(e_j)_{j\in\mathbb{N}}$ is orthonormal, then $e_j \to 0$ weakly.

Proof. Every $\ell \in \mathcal{H}$ is of the form $\ell(\cdot) = \langle \cdot, x\rangle$, with $x \in \mathcal{H}$. Therefore, $\ell(e_j) = \langle e_j, x\rangle \to 0$ as $j \to \infty$, since $\sum_{j\geq 0} |\langle x, e_j\rangle|^2$ converges.

□

An orthonormal system is linearly independent.

Definition 3.4.11

A countable linearly independent set $(e_j)_{j\in\mathbb{N}}$ in a Banach space E is called a **Schauder basis** if $\forall x \in E$, $\exists (a_j) \subseteq \mathbb{K}$ such that

$$x = \sum_{j\geq 0} a_j e_j.$$

This is not a basis in the usual sense. Having a Schauder basis implies E is separable (but the converse does not hold).

Corollary 3.4.12

If an orthonormal system $(e_j)_{j\in\mathbb{N}}$ in a Hilbert space is total, i.e., c.l.s.$(\{e_j : j \in \mathbb{N}\}) = \mathcal{H}$, then it is a Schauder basis.

Proof. When $G = \mathcal{H}$, by previous theorem, $x = \mathrm{pr}_G x = \sum_{j\geq 0}\langle x, e_j\rangle e_j$. □

*From here on, if an orthonormal system (e_j) is a basis, we will abuse notation and simply call (e_j) an **orthonormal basis**.*

Example.

> (Fourier Series). Let $\mathcal{H} = L^2([0, 2\pi])$, with
>
> $$\langle f, g\rangle := \int_0^{2\pi} f(x) \cdot \overline{g(x)}\, dx, \quad e_n := \frac{1}{\sqrt{2\pi}} e^{inx}, \quad n \in \mathbb{Z}.$$

One sees that $(e_n)_{n\in\mathbb{Z}}$ is an orthonormal basis. Let

$$G := \{f \in \mathcal{C}([0, 2\pi]) : f(0) = f(2\pi)\} \subseteq \mathcal{H}.$$

Then Fourier's theorem says that every $f \in G$ can be written as

$$f(x) = \sum_{n\in\mathbb{Z}} f_n e^{inx}, \quad f_n = \frac{1}{\sqrt{2\pi}}\langle f, e_n\rangle = \frac{1}{2\pi}\int_0^{2\pi} f(x)e^{-inx}\, dx.$$

Example.

> (Orthogonal polynomials). Let $\mathcal{H} = L^2([-1, 1])$. The set $\{1, x, x^2, \ldots\}$ is total (i.e., its c.l.s. is $\mathcal{H}$).

By applying the Gram-Schmidt process (MATH2001), we obtain an orthonormal basis for $\mathcal{H}$ consisting of polynomials: these are the so-called **Legendre polynomials**:

$$L_n = c_n \cdot \frac{d^n}{dx^n}\left(x^2 - 1\right)^n.$$

Theorem 3.4.13

Every separable Hilbert space $\mathcal{H}$ is isometrically isomorphic to ℓ^2.

Proof. Let $D = (u_n)_{n\in\mathbb{N}}$ be dense in $\mathcal{H}$. Applying the Gram-Schmidt process, we obtain an orthonormal basis $(e_n)_{n\in\mathbb{N}}$. Define now $L : \mathcal{H} \to \ell^2$ by setting

$$L(e_n) := (0, \ldots, 0, 1, 0, \ldots),$$

where the 1 is the nth term in the sequence. Since we know that every $x \in \mathcal{H}$ can be written as

$$x = \sum_{n\geq 0} \langle x, e_n\rangle e_n$$

we may set

$$L(x) := \sum_{n\geq 0} \langle x, e_n\rangle L(e_n).$$

This clearly defines a linear map. It remains to be shown that L is

1. Bounded and isometric. We have

$$\|L(x)\|_{\ell^2}^2 = \lim_{N\to\infty} \left\| \sum_{n=0}^{N} \langle x, e_n\rangle L(e_n) \right\|^2 = \lim_{N\to\infty} \sum_{n=0}^{N} |\langle x, e_n\rangle|^2 = \|x\|^2,$$

   where we could change the limit because the $L(e_n)$ are orthonormal, while the final equality follows.

2. Surjective. $L(\mathcal{H})$ contains the "canonical basis" of ℓ^2, so it contains the closed linear span (by continuity), which is ℓ^2.

3. Injective. $L(x) = 0 \Rightarrow \langle x, e_n\rangle = 0$ for all $n \in \mathbb{N}$. This implies

$$x \in (\text{span}\{e_n : n \in \mathbb{N}\})^\perp \Rightarrow x \in (\text{c.l.s.}\{e_n : n \in \mathbb{N}\})^\perp = \mathcal{H}^\perp = \{0\}.$$

□

Recall.

$$\ell^p = \left\{ x = (x_k)_{k\geq 1} \subset \mathbb{K} \,\middle|\, \sum_{k=1}^{\infty} |x_k|^p < \infty \right\}, \quad \|x\|_p = \left(\sum_{k=1}^{\infty} |x_k|^p \right)^{1/p}, \quad 1 \leq p \leq \infty.$$

When $p = \infty$, define $\|x\|_\infty = \sup_k |x_k|$.

p	Space ℓ^p	Has a natural inner product?	Is Hilbert space?
$p = 2$	ℓ^2	Yes: $\langle x, y\rangle = \sum_k x_k \overline{y_k}$	Yes
$1 \leq p < \infty$, $p \neq 2$	ℓ^p	No inner product compatible with $\|\cdot\|_p$	No
$p = \infty$	ℓ^∞	No such inner product	No

Chapter 4

Bounded Linear Operators

4.1 Operator Norms and Convergence

Definition 4.1.1

Let E_1 and E_2 be normed spaces (over a field $\mathbb{F}$). A map $A : E_1 \to E_2$ is a **continuous linear operator** if it is continuous and

$$A(\alpha x + \beta y) = \alpha A(x) + \beta A(y), \quad \forall x, y \in E_1, \forall \alpha, \beta \in \mathbb{F}.$$

When $E_2 = \mathbb{F}$, A is just a linear functional.

Theorem 4.1.2

*Let E_1 and E_2 be normed spaces. A linear operator $A : E_1 \to E_2$ is continuous if and only if it is **bounded**, i.e. $\exists c > 0$ such that $\|Ax\|_{E_2} \leq c\|x\|_{E_1}$ for all $x \in E_1$.*

Proof. ($\Rightarrow$). Suppose A is continuous. Then by continuity at 0, $\exists \delta > 0$ such that if $\|x\| < \delta$, then

$$\|A(x) - A(0)\| = \|A(x)\| < 1.$$

Now $\forall y \in E \setminus \{0\}$, let $x := \frac{\delta y}{2\|y\|}$. Then $\|x\| = \frac{\delta}{2} < \delta$, so

$$\|Ax\| = \left\| A\left(\frac{\delta y}{2\|y\|}\right)\right\| = \frac{\delta}{2\|y\|}\|Ay\| < 1 \quad \Rightarrow \quad \|Ay\| < \frac{2}{\delta}\|y\|.$$

Taking $c = \frac{2}{\delta}$, we have the definition of boundedness.

($\Leftarrow$). Let $c \in \mathbb{R}$ be such that it bounds A, and $\varepsilon > 0$. Then $\forall x, y \in E_1$, we have

$$\|A(x) - A(y)\| = \|A(x - y)\| \leq c \cdot \|x - y\|.$$

Pick $\delta = \varepsilon/c$. Then if $\|x - y\| < \delta$, we have

$$\|A(x) - A(y)\| \leq c \cdot \|x - y\| < c \cdot \frac{\varepsilon}{c} = \varepsilon,$$

which is the definition of continuity. $\square$

Definition 4.1.3

The **(operator) norm** of an operator $A : E_1 \to E_2$ is defined to be

$$\|A\| := \sup_{x \neq 0} \frac{\|Ax\|_{E_2}}{\|x\|_{E_1}}.$$

Example.

Let $E_1 = E_2 = \mathcal{C}([a,b])$ and $h \in \mathcal{C}([a,b])$. We set $A_h : E_1 \to E_2$, defined by

$$(A_h f)(x) = h(x) \cdot f(x), \quad \forall x \in [a,b],\ f \in E_1.$$

We claim that $\|A_h\| = \|h\|_\infty$.

Proof. Indeed,

$$\|A_h f\|_\infty = \|h \cdot f\|_\infty \leq \|h\|_\infty \|f\|_\infty \quad \Rightarrow \quad \|A_h\| \leq \|h\|_\infty.$$

Taking $f \equiv 1$, we obtain equality. □

Example.

Let $A : C^1([a,b]) \to \mathcal{C}([a,b])$ be defined by $Af := f'$. If we take on C^1 the sup norm

$$\|f\|_\infty = \sup_{x \in [a,b]} |f(x)|$$

(which is not the most natural norm; that one is $\|f\|_{C^1} = \|f\|_\infty + \|f'\|_\infty$), then A is unbounded.

Proof. Indeed, take $f_n = \sin(n\pi x)$ for all $n \in \mathbb{N}$. Then $\|f_n\|_\infty = 1$, but

$$\|Af\| = \pi n \cdot \|\cos(n\pi x)\| = \pi n \to \infty.$$

□

Example.

Let $E_1 = E_2 = \mathcal{C}([a,b])$, $K \in \mathcal{C}([a,b] \times [a,b])$ and define

$$(Af)(x) := \int_a^b K(x,y) f(y)\, dy.$$

Definition 4.1.4

Let E_1 and E_2 be normed spaces. The **space of bounded operators** from E_1 to E_2 is denoted by $\mathcal{L}(E_1, E_2)$. If $E_1 = E_2 = E$, we simply write $\mathcal{L}(E)$.

Notice that if $\mathbb{F}$ is a field, $\mathcal{L}(E_1, \mathbb{F}) = E_1'$.

Theorem 4.1.5

With respect to the operator norm, the space $\mathcal{L}(E_1, E_2)$ is a Banach space, for any normed space E_1 and any Banach space E_2.

Definition 4.1.6

We say that a sequence of operators $(A_n)_{n \in \mathbb{N}} \subseteq \mathcal{L}(E_1, E_2)$ **converges** to $A \in \mathcal{L}(E_1, E_2)$:

1. **in the norm** (or *uniformly*), if $\|A_n - A\| \to 0$ as $n \to \infty$.
2. **strongly**, if $\|A_n x - Ax\|_{E_2} \to 0$ as $n \to \infty$, for each $x \in E_1$.
3. **weakly**, if $A_n x \to Ax$ as $n \to \infty$ weakly in E_2 (i.e., $\ell(A_n x) \to \ell(Ax)$ for all $\ell \in E_2'$, $\forall x \in E_1$).

Theorem 4.1.7

Let E_1 and E_2 be normed spaces. Then uniform convergence in $\mathcal{L}(E_1, E_2)$ implies strong convergence, and strong convergence implies weak convergence.

Not the other way around! example and exam.

Proof. Let $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{L}(E_1, E_2)$ be uniformly convergent to A. Then $\|A_n - A\| \to 0$ as $n \to \infty$. But

$$\|A_n - A\| = \sup_{x\neq 0} \frac{\|A_n x - Ax\|_{E_2}}{\|x\|_{E_1}} \quad \Rightarrow \quad \|A_n x - Ax\|_{E_2} \leq \|A_n - A\| \cdot \|x\|_{E_1} \to 0,$$

so $A_n \to A$ strongly.

Now let $(A_n) \subseteq \mathcal{L}(E_1, E_2)$ be strongly convergent to A. Then $\|A_n x - Ax\|_{E_2} \to 0$ as $n \to \infty$ for all $x \in E_1$. But

$$|\ell(A_n x) - \ell(Ax)| = |\ell(A_n x - Ax)| \leq \|\ell\| \cdot \|A_n x - Ax\|_{E_2} \to 0, \quad \forall \ell \in E_2',$$

since $\ell \in E_2'$ is bounded, so $A_n x \to Ax$ weakly. □

Example.

Let $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{L}(\ell^2(\mathbb{R}))$, with A_n given by

$$A_n x = (x_0, x_1, \dots, x_n, 0, 0, \dots), \quad \forall x = (x_n)_{n\in\mathbb{N}} \in \ell^2(\mathbb{R}).$$

Then $A_n \to \mathrm{Id}_{\ell^2(\mathbb{R})}$ strongly (Id is the "identity", i.e. the map $x \mapsto x$).

Proof. Indeed, observe that

$$\|A_n(x) - \mathrm{Id}(x)\|_2 = \|(x_0, \dots, x_n, 0, \dots) - (x_n)_{n\in\mathbb{N}}\|_2 = \|(0, \dots, 0, -x_{n+1}, -x_{n+2}, \dots)\|_2$$
$$= \left(\sum_{i=n+1}^{\infty} |x_i|^2\right)^{1/2}, \quad \forall x \in \ell^2(\mathbb{R}).$$

The full series in the final line is convergent by definition of $x \in \ell^2(\mathbb{R})$, so it follows that

$$\|A_n(x) - \mathrm{Id}(x)\|_2 = \left(\sum_{i=n+1}^{\infty} |x_i|^2\right)^{1/2} \to 0, \quad \forall x \in \ell^2(\mathbb{R}),$$

meaning $A_n \to \mathrm{Id}_{\ell^2(\mathbb{R})}$ strongly. □

Example.

However, $\|A_n - A_m\| = 1$ for all $n \neq m$, so $(A_n)_{n\in\mathbb{N}}$ does *not* converge uniformly.

The following theorems also hold for operators; their proofs are analogous to the case of functionals.

Theorem 4.1.8: Banach–Steinhaus Theorem

Let E_1 and E_2 be Banach spaces and $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{L}(E_1, E_2)$. Suppose that $\forall x \in E_1$, $\exists c_x > 0$ such that $\|A_n x\|_{E_2} \leq c_x$ for all $n \in \mathbb{N}$. Then $\exists c > 0$ such that $\|A_n\| \leq c$ for all $n \in \mathbb{N}$.

Theorem 4.1.9

Let E_1 and E_2 be Banach spaces. Then $\mathcal{L}(E_1, E_2)$ is complete with respect to strong convergence.

4.2 Inverse Operators

Definition 4.2.1

Let $A \in \mathcal{L}(E_1, E_2)$. From linear algebra, we know that A is injective if and only if $\ker A = 0$. In this case, we call the map $A^{-1} : \mathrm{im}(A) \to E_1$ the **algebraic inverse** of A.

$\mathrm{im}(A)$ denotes the image of A.

Definition 4.2.2

If $A \in \mathcal{L}(E_1, E_2)$ is bijective and $A^{-1} \in \mathcal{L}(E_2, E_1)$ (i.e., if it is bounded), we say that A is **invertible** and call A^{-1} the **inverse** of A.

Example.

Let $A \in \mathcal{L}(\mathcal{C}([a,b]))$ be defined by $Af = hf$ for some $h \in \mathcal{C}([a,b])$, $h \geq 0$. If $h \neq 0$ and $A^{-1}f = \frac{1}{h} \cdot f$, then A is invertible.

Example.

Let $A \in \mathcal{L}(\mathcal{C}([a,b]))$ be given by

$$(Af)(x) := \int_a^x f(t)\, dt.$$

Then $\mathrm{im}(A) = \{g \in C^1([a,b]) : g(a) = 0\}$. The algebraic inverse is $A^{-1} : \mathrm{im}(A) \to \mathcal{C}([a,b])$, defined by $A^{-1}f = f'$. We saw that this is unbounded. Thus, A is *not* invertible.

Theorem 4.2.3

Let $A \in \mathcal{L}(E_1, E_2)$ be surjective (i.e., $\mathrm{im}(A) = E_2$). Then A is invertible if and only if there exists $c > 0$ such that

$$\|Ax\|_{E_2} \geq c \cdot \|x\|_{E_1}, \quad \forall x \in E_1.$$

Proof. ($\Rightarrow$). Observe that

$$\|x\|_{E_1} = \|A^{-1}Ax\|_{E_1} \leq \|A^{-1}\| \cdot \|Ax\|_{E_2}, \quad \forall x \in E_1,$$

where the inequality holds because $A^{-1} \in \mathcal{L}(E_2, E_1)$, i.e., bounded. Thus, take $c = \|A^{-1}\|^{-1} > 0$; then the above implies that

$$\|Ax\|_{E_2} \geq \|A^{-1}\|^{-1}\|x\|_{E_1} = c \cdot \|x\|_{E_1}, \quad \forall x \in E_1.$$

($\Leftarrow$). We can see easily that $\ker A = 0$: $0 = \|Ax\| \geq c \cdot \|x\| \Rightarrow x = 0$. Since A is also surjective, there exists an algebraic inverse $A^{-1} : E_2 \to E_1$. To see boundedness, let $y \in E_2$. Then (first equality is by surjectivity of A)

$$\|A^{-1}y\|_{E_1} = \|A^{-1}Ax\|_{E_1} = \|x\|_{E_1} \leq c^{-1}\|Ax\| = c^{-1}\|y\| \Rightarrow A^{-1} \in \mathcal{L}(E_2, E_1).$$

□

Definition 4.2.4

Let E_1 and E_2 be normed spaces. We say $A : E_1 \to E_2$ is **open** if $\forall U \subseteq E_1$ open, $A(U)$ is open in E_2.

Theorem 4.2.5: Open Mapping Theorem

Let E_1 and E_2 be Banach spaces. Then $A \in \mathcal{L}(E_1, E_2)$ is open if and only if it is surjective.

Proof. We only prove the easier forwards ($\Rightarrow$) direction here. Let $U = B_1^{E_1}(0) \subseteq E_1$ be the unit open ball. By assumption, $A(U)$ is open, and $0 = A(0) \in A(U)$, so $\exists \varepsilon > 0$ such that $B_\varepsilon^{E_2}(0) \subseteq A(U)$. Now given $y \in E_2 \setminus \{0\}$, set $\bar{y} := \varepsilon y/(2\|y\|) \in B_\varepsilon^{E_2}(0)$. Then since $B_\varepsilon^{E_2}(0) \subseteq A(U)$, $\exists \bar{x} \in U$ with $A(\bar{x}) = \bar{y}$. If we define $x := 2\|y\|\bar{x}/\varepsilon$, then

$$A(x) = \frac{2\|y\|}{\varepsilon} \cdot A(\bar{x}) = \frac{2\|y\|}{\varepsilon} \cdot \bar{y} = y,$$

as desired. □

Theorem 4.2.6: Bounded Inverse Theorem

Let E_1 and E_2 be Banach spaces and $A \in \mathcal{L}(E_1, E_2)$ bijective. Then A is invertible, i.e., $A^{-1} : E_2 \to E_1$ is bounded.

Proof. By the Open Mapping Theorem, A is an open map. Equivalently, A^{-1} is continuous (i.e., bounded). □

Definition 4.2.7

Let E_1 and E_2 be normed spaces. The **graph** of a map $A : E_1 \to E_2$ is the set

$$\Gamma(A) := \{(x, Ax) : x \in E_1\} \subseteq E_1 \times E_2.$$

If $A \in \mathcal{L}(E_1, E_2)$, then $\Gamma(A)$ is closed in $E_1 \times E_2$. Indeed, if $((x_n, Ax_n))_{n\in\mathbb{N}} \subseteq \Gamma(A)$ is a sequence converging to (x, y), then $x_n \to x$, which by continuity of A implies that $Ax_n \to Ax$. Therefore, $y = Ax$ by uniqueness of the limit, meaning $(x, y) = (x, Ax) \in \Gamma(A)$.

Theorem 4.2.8: Closed Graph Theorem

If E_1 and E_2 are Banach and $A : E_1 \to E_2$ is a linear operator with $\Gamma(A)$ closed, then A is bounded.

Proof. $\Gamma(A)$ is a Banach space (since it is closed) with respect to the norm

$$\|(x, Ax)\|_{\Gamma(A)} := \|x\|_{E_1} + \|Ax\|_{E_2}, \quad \forall x \in E_1.$$

Let $\pi_1 : \Gamma(A) \to E_1$ and $\pi_2 : \Gamma(A) \to E_2$ denote the corresponding projections. Clearly, they are bounded, and π_1 is bijective, so by the bounded inverse theorem, π_1 is invertible. Hence,

$$A = \pi_2 \circ \pi_1^{-1} : E_1 \to E_2 \text{ is bounded.}$$

□

4.3 Dual Operators

Definition 4.3.1

Let E_1 and E_2 be normed spaces. Given $A \in \mathcal{L}(E_1, E_2)$, the operator $A^\star : E_2' \to E_1'$ defined by

$$(A^\star \ell)(x) = \ell(Ax), \quad \forall \ell \in E_2', x \in E_1,$$

is called the *dual* of A.

The resulting map $A^\star \ell : E_1 \to \mathbb{F}$ is still a linear functional, and hence belongs to E_1'. $A^\star$ maps E_2' to E_1'.

Theorem 4.3.2

$A^\star \in \mathcal{L}(E_2', E_1')$ and $\|A^\star\| = \|A\|$.

Proof. It is clear that $A^\star$ is linear ($(A^*(\alpha\ell_1 + \beta\ell_2))(x) = \alpha\ell_1(Ax) + \beta\ell_2(Ax)$). Now, $\forall \ell \in E_2'$, $A^\star\ell$ is bounded: indeed,

$$|(A^\star\ell)(x)| = |\ell(Ax)| \leq \|\ell\| \cdot \|Ax\| \leq \underbrace{\|\ell\| \cdot \|A\|}_{c} \cdot \|x\| \quad \Longrightarrow \quad A^\star\ell \in E_1'.$$

The above computation also shows that $\|A^\star\ell\| \leq \|A\| \cdot \|\ell\|$, meaning $A^\star$ is bounded and $\|A^\star\| \leq \|A\|$. To see equality, let $x \in E_2 \setminus \{0\}$. By Hahn-Banach , $\exists \ell \in E_2'$ such that $\|\ell\| = 1$ and $\ell(Ax) = \|Ax\|$. Thus,

$$\|Ax\| = \ell(Ax) = |(A^\star\ell)(x)| \leq \|A^\star\ell\| \cdot \|x\| \leq \|A^\star\| \cdot \underbrace{\|\ell\|}_{=1} \cdot \|x\| = \|A^\star\| \cdot \|x\| \quad \Longrightarrow \quad \|A\| \leq \|A^\star\|.$$

Therefore, $\|A^\star\| = \|A\|$. □

Example.

> Let $A \in \mathcal{L}(\ell^1)$ be given by
>
> $$A(x_0, x_1, \ldots) = (\alpha_0 x_0, \alpha_1 x_1, \ldots), \quad \forall x = (x_n)_{n\in\mathbb{N}} \in \ell^1$$
>
> and some fixed $\alpha = (\alpha_n)_{n\in\mathbb{N}} \in \ell^\infty$.

Take $f \in (\ell^1)' \simeq \ell^\infty$, i.e. $f = (f_n)_{n\in\mathbb{N}} \in \ell^\infty$. Then

$$(A^\star f)(x) = f(Ax) = \sum_{i\geq 0} f_i \alpha_i x_i = g(x), \quad g = (g_n)_{n\in\mathbb{N}} \in \ell^\infty \simeq (\ell^1)'.$$

Then

$$A^\star f = (f_0\alpha_0, f_1\alpha_1, \ldots)$$

(in a sense, the same as A, only that it acts on ℓ^∞).

Theorem 4.3.3

If $A \in \mathcal{L}(E_1, E_2)$ and $B \in \mathcal{L}(E_2, E_3)$, then $BA \in \mathcal{L}(E_1, E_3)$ and $(BA)^\star = A^\star B^\star$.

Proof. By definition of dual operator, $\forall \ell \in E_3'$ and $x \in E_1$, we have

$$((BA)^\star\ell)(x) = \ell(BAx) = \ell(B(Ax)) = (B^\star\ell)(Ax).$$

Denote the functional $B^\star\ell \in E_2'$ by m. Then, by definition of dual operator, we conclude that $m(Ax) = (A^\star m)(x)$ for all $x \in E_1$, whence

$$((BA)^\star\ell)(x) = (A^\star B^\star\ell)(x), \quad \forall \ell \in E_3', \forall x \in E_1.$$

□

Theorem 4.3.4

Let E_1 and E_2 be reflexive Banach spaces. Then $\forall A \in \mathcal{L}(E_1, E_2)$, $(A^\star)^\star = A$.

Proof. Since $A : E_1 \to E_2$, $A^\star : E_2' \to E_1'$ and $(A^\star)^\star : E_1'' \to E_2''$. Recall that the isomorphism between E_i and E_i'' is given by

$$\Phi^i : E_i \to E_i'', \quad x \mapsto \phi_x^i \in E_i'', \quad \phi_x^i(\ell) = \ell(x), \quad \forall \ell \in E_i'.$$

Now, by definition of $A^\star$,

$$((A^\star)^\star \phi_x^1)(\ell) = \phi_x^1(A^\star \ell) = (A^\star \ell)(x) = \ell(Ax) = \phi_{Ax}^2(\ell).$$

Thus, $(A^\star)^\star = A$ up to composition with Φ^i. □

4.4 Operators in Hilbert Spaces

4.4.1 The Adjoint Operator

Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be Hilbert spaces over $\mathbb{C}$. Consider $A \in \mathcal{L}(\mathcal{H}_1, \mathcal{H}_2)$. By Riesz, we may identify $\mathcal{H}_i \simeq \mathcal{H}_i'$ via $x \in \mathcal{H}_i \mapsto \ell_x \in \mathcal{H}_i'$ given by $\ell_x(y) = \langle x, y \rangle$ (recall that this identification is *not* $\mathbb{C}$-linear). Then $A^\star$ becomes a map $A^\star : \mathcal{H}_2 \to \mathcal{H}_1$ defined by

$$(A^\star x)(y) = \ell_x(Ay) = \langle Ay, x \rangle_{\mathcal{H}_2} \quad x \in \mathcal{H}_2, y \in \mathcal{H}_1$$

Also, if $A^\star x = \ell_z \in \mathcal{H}_1'$, then

$$(A^\star x)(y) = \ell_z(y) = \langle y, z \rangle = \langle y, A^\star x \rangle_{\mathcal{H}_1} \quad y, z \in \mathcal{H}_1$$

Definition 4.4.1

Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be Hilbert spaces. Given $A \in \mathcal{L}(\mathcal{H}_1, \mathcal{H}_2)$, the **adjoint operator** $A^* \in \mathcal{L}(\mathcal{H}_2, \mathcal{H}_1)$ is defined implicitly by

$$\langle Ax, y \rangle_{\mathcal{H}_2} = \langle x, A^* y \rangle_{\mathcal{H}_1}, \quad \forall x \in \mathcal{H}_1, y \in \mathcal{H}_2.$$

For the dual operator we have $(\lambda A)^\star = \lambda A^\star$ for all $\lambda \in \mathbb{C}$, while for the adjoint we have $(\lambda A)^* = \overline{\lambda} A^*$. Indeed, for the dual operator,

$$((\lambda A)^\star \ell)(x) = \ell(\lambda A x) = \lambda \ell(Ax) = \lambda (A^\star \ell)(x) = ((\lambda A^\star)\ell)(x), \quad \forall \ell \in E_2', x \in E_1,$$

meaning $(\lambda A)^\star = \lambda A^\star$. Meanwhile, for the adjoint operator,

$$\langle \lambda A x, y \rangle = \lambda \langle Ax, y \rangle = \lambda \langle x, A^* y \rangle = \langle x, \overline{\lambda} A^* y \rangle, \quad \forall x \in \mathcal{H}_1, y \in \mathcal{H}_2,$$

so $(\lambda A)^* = \overline{\lambda} A^*$. From now on we work with adjoint operators.

Definition 4.4.2

We say $A \in \mathcal{L}(\mathcal{H})$ is **self-adjoint** if $A = A^*$.

Example.

> An operator on $\mathbb{C}^n$ is self-adjoint if and only if its corresponding matrix M is **Hermitian**, i.e., $M = \overline{M}^T$.

Theorem 4.4.3

Any $A \in \mathcal{L}(\mathcal{H})$ may be written as $A = \mathrm{Re}(A) + i \cdot \mathrm{Im}(A)$, where $\mathrm{Re}(A), \mathrm{Im}(A) \in \mathcal{L}(\mathcal{H})$ are self-adjoint.

Proof. Set

$$\mathrm{Re}(A) := \frac{A + A^*}{2}, \quad \mathrm{Im}(A) := \frac{A - A^*}{2i}.$$

□

Definition 4.4.4

We say $A \in \mathcal{L}(\mathcal{H})$ is **normal** if and only if $AA^* = A^*A$.

Theorem 4.4.5

$A \in \mathcal{L}(\mathcal{H})$ is *normal if and only if*

$$(\mathrm{Re}(A))(\mathrm{Im}(A)) = (\mathrm{Im}(A))(\mathrm{Re}(A)).$$

Proof. ($\Rightarrow$). Suppose A is normal. Then

$$\begin{aligned}
(\mathrm{Re}(A))(\mathrm{Im}(A)) &= \frac{(A + A^*)(A - A^*)}{4i} \\
&= \frac{A^2 - AA^* + A^*A - (A^*)^2}{4i} \\
&= \frac{A^2 - A^*A + AA^* - (A^*)^2}{4i} \\
&= \frac{(A - A^*)A + (A - A^*)A^*}{4i} \\
&= \frac{(A - A^*)(A + A^*)}{4i} \\
&= (\mathrm{Im}(A))(\mathrm{Re}(A)).
\end{aligned}$$

($\Leftarrow$). We have

$$(\mathrm{Re}(A))(\mathrm{Im}(A)) = (\mathrm{Im}(A))(\mathrm{Re}(A)) \Longrightarrow \frac{A^2 - AA^* + A^*A - (A^*)^2}{4i} = \frac{A^2 + AA^* - A^*A - (A^*)^2}{4i}.$$

Cancelling the common terms, we're left with

$$-AA^* + A^*A = AA^* - A^*A \Longrightarrow 2A^*A = 2AA^* \Longrightarrow A^*A = AA^*,$$

meaning A is normal. □

Example.

> Let $A \in \mathcal{L}(\ell^2(\mathbb{C}))$ be defined by
>
> $$A((x_n)_{n\in\mathbb{N}}) = (\alpha_n x_n)_{n\in\mathbb{N}}, \quad \alpha \in \ell^\infty.$$
>
> We wish to find A^*.

Let $x, y \in \ell^2(\mathbb{C})$. Then

$$\langle x, A^*y\rangle = \langle Ax, y\rangle = \sum_{n\geq 0} \alpha_n x_n \overline{y_n} = \sum_{n\geq 0} x_n \overline{(\alpha_n y_n)} = \langle (x_n), (\overline{\alpha_n} y_n)\rangle \Longrightarrow A^*y = (\overline{\alpha_n} y_n)_{n\in\mathbb{N}}.$$

A is normal, since $\alpha_n \overline{\alpha_n} = \overline{\alpha_n}\alpha_n$ for all $n \in \mathbb{N}$. It is self-adjoint if and only if $\alpha_n \in \mathbb{R}$ for all $n \in \mathbb{N}$.

Theorem 4.4.6: Hellinger-Toeplitz Theorem

Let $\mathcal{H}$ be a Hilbert space and $A : \mathcal{H} \to \mathcal{H}$ a linear operator such that $\langle Ax, y\rangle = \langle x, Ay\rangle$ for all $x, y \in \mathcal{H}$. Then A is bounded.

Proof. By the closed graph theorem, it suffices to show that $\Gamma(A)$ is closed. Assume that

$$(x_n, Ax_n)_{n\in\mathbb{N}} \to (x, y) \in \mathcal{H} \times \mathcal{H}. \quad \forall z \in \mathcal{H}, \text{ we have}$$

$$\langle z, y\rangle = \lim_{n\to\infty} \langle z, Ax_n\rangle = \lim_{n\to\infty} \langle Az, x_n\rangle = \langle Az, x\rangle = \langle z, Ax\rangle,$$

which implies $y = Ax$ by the non-degeneracy of $\langle\cdot,\cdot\rangle$ (map from $\mathcal{H}$ to $\mathcal{H}'$ given by $y \mapsto (x \mapsto \langle x, y\rangle)$). Therefore, A is bounded. □

4.4.2 Types of Operators on Hilbert Spaces

Theorem 4.4.7

The operator $\mathcal{L}(H)$ satisfies

$$A = A^* \iff \langle Ax, x\rangle \in \mathbb{R}.$$

Proof. ($\Rightarrow$) $\langle Ax, x\rangle = \langle x, A^*x\rangle = \langle x, Ax\rangle$. Also, $\overline{\langle x, Ax\rangle} = \langle Ax, x\rangle$.

($\Leftarrow$) We have $A = \operatorname{re} A + i \operatorname{im} A$, and

$$\langle Ax, x\rangle = \langle(\operatorname{re} A)x, x\rangle + i\langle(\operatorname{im} A)x, x\rangle \in \mathbb{R}.$$

Thus $\langle(\operatorname{im} A)x, x\rangle = 0$.

Now, for $x, y \in H$,

$$\begin{aligned} 0 &= \langle(\operatorname{im} A)(x + y),\, x + y\rangle \\ &= \langle(\operatorname{im} A)x, x\rangle + \langle(\operatorname{im} A)y, y\rangle + \langle(\operatorname{im} A)x, y\rangle + \langle y, (\operatorname{im} A)x\rangle \\ &= 2\ \operatorname{Re}\langle(\operatorname{im} A)x, y\rangle. \end{aligned}$$

Setting $y = (\operatorname{im} A)x$ we get

$$2\ \operatorname{Re}\|(\operatorname{im} A)x\|^2 = 0.$$

Thus $\operatorname{Im} A = 0$, and $A = \operatorname{re} A = A^*$. □

Definition 4.4.8

An operator $A \in \mathcal{L}(\mathcal{H})$ is called **positive (non-negative)**, denoted by $A > 0$ ($A \geq 0$), if $\langle Ax, x\rangle > 0$ (≥ 0) $\forall x \in \mathcal{H} \setminus \{0\}$. We also write $A > B$ if $A - B > 0$.

Lemma 4.4.9

Let $B \in \mathcal{L}(\mathcal{H})$ with $B = B^*$. Then $\langle Bx, x\rangle = 0$ for all $x \in \mathcal{H}$ implies $B = 0$.

Proof. Let $x, y \in \mathcal{H}$. Then

$$\begin{aligned} 0 &= \langle B(x + y), x + y\rangle \\ &= \underbrace{\langle Bx, x\rangle}_{=0} + \underbrace{\langle By, y\rangle}_{=0} + \langle Bx, y\rangle + \langle By, x\rangle \\ &= \langle Bx, y\rangle + \langle y, Bx\rangle = 2\operatorname{Re}(\langle Bx, y\rangle) \end{aligned}$$

implies that $y = Bx$, in which case the above says that

$$0 = 2\operatorname{Re}(\|Bx\|^2) = \|Bx\|^2, \quad \forall x \in \mathcal{H} \implies Bx = 0, \quad \forall x \in \mathcal{H} \implies B = 0.$$

□

Theorem 4.4.10

If $A \in \mathcal{L}(\mathcal{H})$ is non-negative, then A is self-adjoint.

Proof. From Theorem, we know that $A = \mathrm{Re}(A) + i \cdot \mathrm{Im}(A)$, and

$$0 \leq \langle Ax, x\rangle = \underbrace{\langle \mathrm{Re}(A)x, x\rangle}_{\in\mathbb{R}} + i \cdot \underbrace{\langle \mathrm{Im}(A)x, x\rangle}_{\in\mathbb{R}} \implies \langle \mathrm{Im}(A)x, x\rangle = 0, \quad \forall x \in \mathcal{H}.$$

Thus, $\mathrm{Im}(A) = 0$ by Lemma, so $A = \mathrm{Re}(A)$, which is self-adjoint.

□

Definition 4.4.11

Let $\mathcal{H}$ be a Hilbert space and $G \subseteq \mathcal{H}$. Then $\mathrm{pr}_G : \mathcal{H} \to G$, $x \mapsto \mathrm{pr}_G(x)$, is called a projection operator.

Some remarks:

1. pr_G *is linear.*
2. $\mathrm{pr}_G \in \mathcal{L}(\mathcal{H})$ *and* $\|\mathrm{pr}_G\| = 1$ *if* $G \neq 0$. *Indeed,*

$$x = \mathrm{pr}_G x + x - \mathrm{pr}_G x \quad \Rightarrow \quad \|x\|^2 = \|\mathrm{pr}_G x\|^2 + \|x - \mathrm{pr}_G x\|^2 \geq \|\mathrm{pr}_G x\|^2,$$

where we have equality if and only if $x \in G$. *Therefore,*

$$\|\mathrm{pr}_G\| = \sup_{x \neq 0} \frac{\|\mathrm{pr}_G x\|}{\|x\|} = 1.$$

3. $\mathrm{pr}_G^2 = \mathrm{pr}_G$. *If* $x \in \mathcal{H}$, *then* $\mathrm{pr}_G x \in G$, *so*

$$\mathrm{pr}_G(\underbrace{\mathrm{pr}_G x}_{\in G}) = \mathrm{pr}_G x.$$

4. $\mathrm{pr}_G \geq 0$. *Indeed,*

$$\langle \mathrm{pr}_G x, x\rangle = \langle \mathrm{pr}_G x, \mathrm{pr}_G x + x - \mathrm{pr}_G x\rangle = \|\mathrm{pr}_G x\|^2 + \underbrace{\langle \mathrm{pr}_G x, x - \mathrm{pr}_G x\rangle}_{=0} = \|\mathrm{pr}_G x\|^2 \geq 0.$$

Theorem 4.4.12

Let $A \in \mathcal{L}(\mathcal{H})$ be self-adjoint, with $A^2 = A$. Then $A = \mathrm{pr}_G$ for some subspace $G \subseteq \mathcal{H}$.

Proof. Define $G := \ker(A - \mathrm{Id}) = \{x \in \mathcal{H} : Ax = x\}$, a closed subspace. Since $A^2 = A$, we have $(A - \mathrm{Id})(Ax) = 0$ for all $x \in \mathcal{H}$, so $Ax \in G$ for all $x \in \mathcal{H}$. By definition, we also have $Ay = y$ for all $y \in G$. Now, given $y \in G$,

$$\langle Ax, y\rangle = \langle x, Ay\rangle = \langle x, y\rangle = \langle x, \mathrm{pr}_G y\rangle = \langle \mathrm{pr}_G x, y\rangle \quad \Rightarrow \quad \langle Ax - \mathrm{pr}_G x, y\rangle = 0, \quad \forall y \in G.$$

Taking $y = Ax - \mathrm{pr}_G x$ yields $Ax = \mathrm{pr}_G x$ for all $x \in \mathcal{H}$.

□

Definition 4.4.13

An operator $A \in \mathcal{L}(\mathcal{H})$ is **unitary** if it is surjective and an isometry, i.e.

$$\langle Ax, Ay\rangle = \langle x, y\rangle, \quad \forall x, y \in \mathcal{H} \quad \Longleftrightarrow \quad \|Ax\| = \|x\|, \quad \forall x \in \mathcal{H}.$$

If A is an isometry, then $\|A\| = 1$.

Example.

Some examples. Let $\mathcal{H} = \ell^2$.

1. An operator A such that

$$Ax = A((x_n)_{n\in\mathbb{N}}) := (0, x_0, x_1, \dots), \quad \forall x \in \ell^2$$

is an isometry but *not* surjective and so not unitary.

2. An operator A defined by

$$A((x_n)) := (\xi_0 x_0, \xi_1 x_1, \dots),$$

where $\xi_n \in \mathbb{C}$ such that $|\xi_n| = 1$ for all $n \in \mathbb{N}$, is a unitary operator.

Theorem 4.4.14

An operator $A \in \mathcal{L}(\mathcal{H})$ is unitary if and only if it is invertible and $A^{-1} = A^*$.

Proof. Recall that a surjective operator A is invertible if and only if $\exists c > 0$ such that $\|Ax\| \geq c\|x\|$ for all $x \in \mathcal{H}$

($\Rightarrow$). If A is unitary, then $\|Ax\| = \|x\|$, so take $c = 1$ and A is invertible. Let $y = Az \in \mathcal{H}$. Then $z = A^{-1}y$ and

$$\langle Ax, y\rangle = \langle Ax, Az\rangle = \langle x, z\rangle = \langle x, A^{-1}y\rangle \quad \Rightarrow \quad A^* = A^{-1}.$$

($\Leftarrow$). Invertibility implies surjectivity. A is isometric since

$$\langle Ax, Ay\rangle = \langle x, A^*Ay\rangle = \langle x, A^{-1}Ay\rangle = \langle x, y\rangle, \quad \forall x, y \in \mathcal{H}.$$

□

4.4.3 Sesquilinear Forms

Definition 4.4.15

he map $b : H \times H \to \mathbb{C}$ is a sesquilinear form if

1. $b(x + y, z) = b(x, z) + b(y, z)$,
2. $b(x, y + z) = b(x, y) + b(x, z)$,
3. $b(\lambda x, y) = \lambda\, b(x, y)$,
4. $b(x, \lambda y) = \overline{\lambda}\, b(x, y), \quad \lambda \in \mathbb{C}$.

The form b is Hermitian if $b(x, y) = \overline{b(y, x)}$.

Definition 4.4.16

The map $B(x) = b(x, x)$ is the quadratic form associated with b.

Take $A \in \mathcal{L}(H)$. Then $b_A(x,y) = (Ax, y)$ is a sesquilinear form, and

$$|b_A(x,y)| \leq \|A\| \, \|x\| \, \|y\|.$$

Theorem 4.4.17

Assume b is a sesquilinear form on H such that $|b(x,y)| \leq C \, \|x\| \, \|y\|$ for some $C > 0$. Then there exists $A \in L(H)$ such that $b(x,y) = (Ax, y)$.

Proof. Fix $x \in H$. Then $\ell(y) = \overline{b(x,y)}$ is a bounded linear functional on H. By Riesz, there exists $a_x \in H$ such that

$$\overline{b(x,y)} = \ell(y) = (y, a_x), \qquad \text{i.e.} \qquad b(x,y) = (a_x, y).$$

Now set $Ax = a_x$.

Why is A linear? For every y,

$$(A(\alpha x_1 + \beta x_2), y) = b(\alpha x_1 + \beta x_2, y) = \alpha \, b(x_1, y) + \beta \, b(x_2, y) = (\alpha A x_1 + \beta A x_2, y).$$

Thus

$$\alpha A x_1 + \beta A x_2 = A(\alpha x_1 + \beta x_2).$$

Why is A bounded? For $x \in H$,

$$\|Ax\|^2 = b(x, Ax) \leq C\|x\| \, \|Ax\|, \qquad \text{so} \qquad \|Ax\| \leq C\|x\|.$$

□

Lemma 4.4.18: Polarisation Identity

For all $x, y \in H$,

$$4 \, b_A(x,y) = b_A(x+y, x+y) - b_A(x-y, x-y) + i\big(b_A(x+iy, x+iy) - b_A(x-iy, x-iy)\big).$$

Proof. Direct computation. □

Theorem 4.4.19

Given $A \in \mathcal{L}(H)$, the following statements are equivalent:

1. A is self-adjoint;
2. b_A is Hermitian;
3. $B_A(x) = b_A(x,x)$ maps H to $\mathbb{R}$.

Proof. (1) $\Leftrightarrow$ (2): Because

$$b_A(x,y) = (Ax, y) \quad \text{and} \quad \overline{b_A(y,x)} = \overline{(Ay, x)} = (x, Ay).$$

(2) $\Rightarrow$ (3): Since $b_A(x,x) = \overline{b_A(x,x)}$.
(3) $\Rightarrow$ (2): As $b_A(z,z) \in \mathbb{R}$ for every $z \in H$, the polarisation identity gives

$$4 \, b_A(x,y) = b_A(y+x, y+x) - b_A(y-x, y-x) + i\big(b_A(ix-y, ix-y) - b_A(ix+y, ix+y)\big) = 4\,\overline{b_A(y,x)}.$$

□

4.5 Spectral Theory

Let E be a normed space. Consider $A \in \mathcal{L}(E)$.

Definition 4.5.1

A number $\lambda \in \mathbb{C}$ is an eigenvalue of A if there exists $v \neq 0$ such that $Av = \lambda v$. Such a v is an eigenvector of A.

Observe that $v \in \ker(A - \lambda I)$, so $\ker(A - \lambda I) \neq \{0\}$, which means $A - \lambda I$ is not invertible since not injective.

Definition 4.5.2

The *spectrum* of A, denoted $\operatorname{spec}(A)$ or $\sigma(A)$, is the set of all $z \in \mathbb{C}$ for which $A - zI$ is not invertible.

If λ is an eigenvalue, then $\lambda \in \sigma(A)$. If $\dim E < \infty$, then $\sigma(A)$ is the set of eigenvalues; but this need not hold when $\dim E = \infty$.

Example.

> Let $E = C([0,1])$ and define $(Af)(x) = x\,f(x)$. Assume that $Af = \lambda f$. Then
>
> $$x\,f(x) = \lambda f(x) \quad \Longrightarrow \quad (x - \lambda) f(x) = 0,$$
>
> hence $f(x) = 0$ except at $x = \lambda$. By continuity, f is identically 0. Thus A has *no eigenvalues*. What is $(A - \lambda I)^{-1}$?

$$(A - \lambda I)^{-1}(A - \lambda I)f = (A - \lambda I)^{-1}(x - \lambda)f = f,$$

so

$$(A - \lambda I)^{-1} f = \frac{1}{x - \lambda} f.$$

This only makes sense if $\lambda \notin [0,1]$. Thus $\sigma(A) = [0,1]$.

Definition 4.5.3

A point $z \in \mathbb{C}$ is a *regular point* of A if $z \notin \sigma(A)$.

Lemma 4.5.4

If $\|A\| < 1$ then $I - A$ is invertible and

$$(I - A)^{-1} = I + A + A^2 + \cdots .$$

Also, if $\|B\| \leq \|A^{-1}\|^{-1}$ then $A + B$ is invertible.

Theorem 4.5.5

$\sigma(A) \subset \overline{B_{\|A\|}(0)} = \{\, z \in \mathbb{C} : |z| \leq \|A\| \}$. (the ball in $\mathbb{C}$).

Proof. Assume $|z| > \|A\|$. Then

$$A - zI = -z\Big(I - \frac{A}{z}\Big) \quad \text{and} \quad \Big\|\frac{A}{z}\Big\| < 1,$$

so $A - zI$ is invertible by the lemma. □

Theorem 4.5.6

The spectrum $\sigma(A)$ is closed.

Proof. Let z_0 be a regular point. For $z \in \mathbb{C}$,

$$A - zI = A + z_0 I - z_0 I - zI = A - z_0 I + (z_0 - z)I.$$

Choose z so close to z_0 that

$$\|(z_0 - z)I\| < \|(A - z_0 I)^{-1}\|^{-1}.$$

Then the sum is invertible by the lemma. Thus, the compliment set $\sigma(A)$ is closed. □

Definition 4.5.7

If $z \in \mathbb{C}$ is a regular point of A, then $(A - zI)^{-1}$ is the *resolvent* of A, denoted $R_z(A)$ or $R_z \in \mathcal{L}(E)$.

Theorem 4.5.8: Hilbert Identity

If z_1 and z_2 are regular, then

$$R_{z_1}(A) - R_{z_2}(A) = (z_1 - z_2)\, R_{z_1}(A) R_{z_2}(A).$$

Proof. Observe that

$$(A - z_2 I) - (A - z_1 I) = (z_1 - z_2)I.$$

Multiply this by $R_{z_1}(A)$ on the left and by $R_{z_2}(A)$ on the right:

$$R_{z_1}(A)(A - z_2 I)R_{z_2}(A) - R_{z_1}(A)(A - z_1 I)R_{z_2}(A) = (z_1 - z_2)R_{z_1}(A)R_{z_2}(A).$$

Since $R_{z_1}(A)(A - z_1 I) = I = (A - z_2 I)R_{z_2}(A)$, we get

$$R_{z_1}(A) - R_{z_2}(A) = (z_1 - z_2)R_{z_1}(A)R_{z_2}(A).$$

□

Lemma 4.5.9

If $z_0 \in \mathbb{C} \setminus \sigma(A)$ and $z \in \overline{B_r}(z_0)$ with $r < \|R_{z_0}(A)\|^{-1}$, then $z \in \mathbb{C} \setminus \sigma(A)$ and

$$\|R_z(A)\| \leq \frac{\|R_{z_0}(A)\|}{1 - r\|R_{z_0}(A)\|}.$$

Proof. Write

$$A - zI = A + z_0 I - z_0 I - zI = (A - z_0 I)\big(I + (z_0 - z)R_{z_0}(A)\big).$$

Set $B := (z_0 - z)R_{z_0}(A)$; then $\|B\| < r\|R_{z_0}(A)\| < 1$, so $I + B$ is invertible and hence $A - zI$ is invertible. Moreover,

$$\begin{aligned}
\|R_z(A)\| &= \|(A - zI)^{-1}\| = \big\|(A - z_0 I)^{-1}\big(I - (z - z_0)R_{z_0}(A)\big)^{-1}\big\| \\
&= \|R_{z_0}(A)\| \, \big\|\big(I + B\big)^{-1}\big\| \leq \|R_{z_0}(A)\| \, \big\|I + B + B^2 + \cdots\big\| \\
&\leq \|R_{z_0}(A)\| \, \frac{1}{1 - \|B\|} \leq \frac{\|R_{z_0}(A)\|}{1 - r\|R_{z_0}(A)\|}.
\end{aligned}$$

□

Theorem 4.5.10

The function

$$\mathbb{C} \setminus \sigma(A) \longrightarrow \mathcal{L}(E), \qquad z \longmapsto R_z(A),$$

is continuous.

Proof. If $z \to z_0$ in $\mathbb{C}$, then

$$\|R_z - R_{z_0}\| = |z - z_0|\,\|R_z\|\,\|R_{z_0}\|.$$

By the lemma, the right-hand side is bounded by some $C|z - z_0|$ for $C > 0$, and therefore goes to 0. □

Theorem 4.5.11

The function

$$\mathbb{C} \setminus \sigma(A) \longrightarrow \mathcal{L}(E), \qquad z \longmapsto R_z(A),$$

is differentiable with derivative $(R_z(A))^2$; that is,

$$\lim_{h\to 0} \frac{1}{h}\big(R_{z+h}(A) - R_z(A)\big) = (R_z(A))^2.$$

If $\dim E = 1$ this is obvious: $z \mapsto (a - z)^{-1}$, so the derivative is

$$\frac{1}{(a-z)^2} = \big((a-z)^{-1}\big)^2.$$

Proof.

$$\lim_{h\to 0} \frac{1}{h}\big(R_{z+h}(A) - R_z(A)\big) = \lim_{h\to 0} \frac{z+h-z}{h}\, R_{z+h}(A)\, R_z(A) = \lim_{h\to 0} R_{z+h}(A)\, R_z(A) = R_z(A)^2.$$

□

Lemma 4.5.12

The function $z \mapsto \|R_z\|$ is bounded on $\mathbb{C} \setminus \overline{B_{2\|A\|}(0)}$. Moreover,

$$\lim_{|z|\to\infty} \|R_z\| = 0.$$

Proof. For $|z| > 2\|A\|$,

$$\|R_z\| = \Big\|(-z)^{-1}\big(I - A/z\big)^{-1}\Big\| = |z|^{-1}\Big\|\big(I - A/z\big)^{-1}\Big\| \le \frac{1}{|z|}\,\frac{1}{1 - \|A\|/|z|} \le \frac{1}{|z|}\,\frac{1}{1 - \|A\|/(2\|A\|)} = \frac{2}{|z|} \le \frac{1}{\|A\|}.$$

□

Lemma 4.5.13

Fix $x \in E$ and $\ell \in E'$. Define $f_{x,\ell} : \mathbb{C} \setminus \sigma(A) \to \mathbb{C}$ by

$$f_{x,\ell}(z) = \ell\big(R_z x\big).$$

Then $f_{x,\ell}$ is analytic.

Proof. For $z \in \mathbb{C} \setminus \sigma(A)$,

$$\lim_{h\to 0} \frac{f_{x,\ell}(z+h) - f_{x,\ell}(z)}{h} = \lim_{h\to 0} \frac{1}{h}\,\ell\big((R_{z+h} - R_z)x\big) = \ell\left(\lim_{h\to 0} \frac{R_{z+h} - R_z}{h}\, x\right) = \ell\big(R_z^2 x\big).$$

□

Lemma 4.5.14: Liouville's Theorem

A complex-differentiable (analytic) function $f : \mathbb{C} \to \mathbb{C}$ which is bounded must be constant.

Theorem 4.5.15

$\sigma(A) \neq \varnothing$.

The map $z \mapsto R_z(A)$ is differentiable, hence analytic, and it is bounded. If $\sigma(A) = \varnothing$, the map is defined on all of $\mathbb{C}$ and must be constant (by Liouville), which is impossible. We assume $A \neq 0$; otherwise the theorem is obvious.

Proof. If $A = 0$, the result is obvious. Assume $\sigma(A) = \varnothing$. The map $f_{x,\ell}$ is analytic on $\mathbb{C}$. Moreover, it is bounded. Indeed, by Lemma 1, $\|R_z\|$ is bounded on $\mathbb{C} \setminus B_{2\|A\|}(0)$ and on $B_{2\|A\|}(0)$ by continuity. Thus $\sup_{z\in\mathbb{C}} \|R_z\| = d < \infty$. Therefore

$$|f_{x,\ell}(z)| = \big|\ell(R_z x)\big| \leq \|\ell\|\,\|x\|\,d.$$

By Liouville's theorem, $f_{x,\ell}$ is constant. Since $\lim_{|z|\to\infty} \|R_z\| = 0$, it follows that $f_{x,\ell} \equiv 0$.

Consequently $\ell(R_z x) = 0$ for all ℓ, so $R_z x = 0$ for every x, hence $R_z = 0$. This is impossible because R_z is invertible. □

Let H be a Hilbert space and $A \in \mathcal{L}(H)$ be self-adjoint.

Lemma 4.5.16

Given $B \in \mathcal{L}(H)$, the closure of the image satisfies

$$\overline{B(H)} = (\ker B^*)^\perp.$$

Proof. If $x \in B(H)$ and $v \in \ker B^*$, then $x = By$ for some $y \in H$ and

$$(x, v) = (By, v) = (y, B^* v) = 0,$$

so $x \in (\ker B^*)^\perp$. Hence $B(H) \subset (\ker B^*)^\perp$, and, since $(\ker B^*)^\perp$ is closed,

$$\overline{B(H)} \subset (\ker B^*)^\perp.$$

For the reverse inclusion, take $h \in \big(\overline{B(H)}\big)^\perp$. Then $h \perp BB^*h$, and

$$0 = (h, BB^*h) = \|B^*h\|^2,$$

so $B^*h = 0$; thus $h \in \ker B^*$. Hence $\big(\overline{B(H)}\big)^\perp \subset \ker B^*$, and taking orthogonal complements gives $\overline{B(H)} \supset (\ker B^*)^\perp$. □

Lemma 4.5.17

For every $z \in \mathbb{C}$ and $x \in H$,

$$\|(A - zI)x\| \geq |\operatorname{Im} z|\,\|x\|.$$

Proof. Observe that

$$((A - zI)x, x) = (Ax, x) - z\|x\|^2$$

and

$$(Ax, x) = (x, Ax) = \overline{(Ax, x)} \in \mathbb{R},$$

Consequently,

$$\operatorname{Im}((A - zI)x, x) = -\operatorname{Im} z\,\|x\|^2, \qquad |\operatorname{Im} z|\,\|x\|^2 \leq |((A - zI)x, x)| \leq \|(A - zI)x\|\,\|x\|.$$

Thus,

$$\|(A - zI)x\| \geq |\operatorname{Im} z|\,\|x\|.$$

□

Theorem 4.5.18

Let H be a Hilbert space. If $A \in \mathcal{L}(H)$ is self-adjoint, then

$$\sigma(A) \subset \mathbb{R}.$$

Proof. Assume $z \in \mathbb{C} \setminus \mathbb{R}$. We need to show that $A - zI$ is invertible. By Lemma 2 it suffices to prove that $(A - zI)(H) = H$.

First, we claim that this image is closed. Indeed, if

$$\|(A - zI)x_n - (A - zI)x_m\| \to 0,$$

then

$$\|x_n - x_m\| \leq \frac{1}{|\operatorname{Im} z|} \|(A - zI)x_n - (A - zI)x_m\| \to 0,$$

so $(x_n)_{n=1}^{\infty}$ converges to some $x \in H$, and $(A - zI)x_n \to (A - zI)x$; hence $(A - zI)(H)$ is closed.

By Lemma 1,

$$\overline{(A - zI)(H)} = (\ker(A^* - \bar{z}I))^{\perp} = (\ker(A - \bar{z}I))^{\perp}.$$

Since $\operatorname{Im} z \neq 0$, the last kernel is $\{0\}$ by Lemma 2. Thus

$$(A - zI)(H) = \overline{(A - zI)(H)} = \{0\}^{\perp} = H.$$

□

Chapter 5

Unbounded Operators

Example.

> $H = L^2([0,1])$. For $f \in C^1([0,1])$, define $Af = \frac{d}{dx} f$. A is unbounded.

Take $f_n = e^{inx}$. Then

$$\|Af_n\| = \|nf_n\| = n\|f_n\|.$$

Of course, A is not defined on all of H. Indeed,

$$C^1(0,1) \subset L^2([0,1]) = H,$$

but $\dfrac{d}{dx}$ does not make sense for certain functions in general.

Definition 5.0.1

Let H be a Hilbert space and $D(A)$ a (linear) subset of H and $A\colon D(A) \to H$ a (linear) map. This A is called an operator with domain $D(A)$.

Definition 5.0.2

$A = B$ if $D(A) = D(B)$ and $Ah = Bh$ for all $h \in D(A) = D(B)$. Also, $A \subset B$ if $D(A) \subset D(B)$ and $Ah = Bh$ for $h \in D(A)$. Then B is an extension of A.

We can add and compose operators, but we must keep track of the domains:

$$\mathcal{D}(A+B) = \mathcal{D}(A) \cap \mathcal{D}(B) \quad \textit{and} \quad \mathcal{D}(AB) = \{\, h \in \mathcal{D}(B) \mid Bh \in \mathcal{D}(A) \,\}.$$

Definition 5.0.3

The graph of A is

$$\Gamma_A = \{\, (x, Ax) \mid x \in \mathcal{D}(A) \,\}.$$

Definition 5.0.4

We say A is *closed* if Γ_A is closed in $H \times H$.

Closed graph theorem: If Γ_A is closed and $\mathcal{D}(A) = H$, then A is bounded.

Theorem 5.0.5

A is closed if and only if $\mathcal{D}(A)$ is complete with respect to the graph scalar product

$$(f,g)_{\Gamma_A} = (f,g)_H + (Af, Ag)_H.$$

What if A is not closed? Can we "close" it?
If $h_n \to f$ and $Ah_n \to g$, try defining $Af = g$. Thus we try to take the closure of the graph.
Problem. What if there exist (h_n) and (h'_n) with $h_n \to f$ and $h'_n \to f$ such that

$$\lim_{n\to\infty} Ah_n \neq \lim_{n\to\infty} Ah'_n \ ?$$

Definition 5.0.6

A is *closable* if $\overline{\Gamma_A}$ is the graph of some operator $\bar{A}$, called the *closure* of A.

Theorem 5.0.7

The following statements are equivalent:

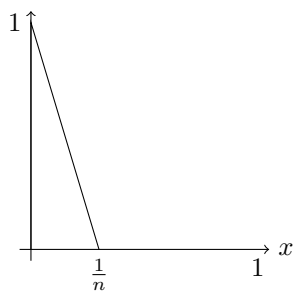
1. A is closable.
2. If $(h_n) \subset \mathcal{D}(A)$ and $\lim_{n\to\infty} h_n = 0$ and $\lim_{n\to\infty} Ah_n = g$, then $g = 0$.

1. *$\mathcal{D}(\bar{A})$ is the completion of $\mathcal{D}(A)$ in the graph norm $\|\cdot\|_{\Gamma_A}$.*
2. *If $A \subset B$ and B is closable, then so is A. Indeed, $\Gamma_A \subset \Gamma_B$, so $\overline{\Gamma}_A \subset \overline{\Gamma}_B = \Gamma_{\bar{B}}$, which shows $\overline{\Gamma}_A$ is a graph.*

Example.

> $H = L^2([0,1])$, $\mathcal{D}(A) = C([0,1])$. Define Af to be the constant function
>
> $$(Af)(x) = f(0).$$

Take f_n like this:



Then

$$\|f_n\|_{L^2} \longrightarrow 0 \quad (n \to \infty), \qquad \|Af_n\|_{L^2} = 1 \neq 0.$$

By the theorem this operator is not closable.

5.1 Adjoint Operators

If A is bounded, then for every $g \in H$ there exists $g^* \in H$ such that

$$(Af, g) = (f, g^*)$$

by the Riesz representation theorem. Then $A^*g = g^*$.

Now, if A is *not* bounded, define

$$\mathcal{D}(A^*) = \big\{\, g \in H \mid \exists\, g^* \in H \text{ s.t. } (Af, g) = (f, g^*) \text{ for all } f \in \mathcal{D}(A) \big\}.$$

It is easy to see that $\mathcal{D}(A^*)$ is a linear set. We define A^* by $A^* g = g^*$. Clearly A^* is linear.

Note. A^* is well-defined if and only if $\mathcal{D}(A)$ is dense in H. Indeed, this is the only case where g^* can be determined uniquely.

Fact 5.1.1

1. A^* is closed;
2. $(\bar{A})^* = A^*$;
3. $(A + B)^* \supset A^* + B^*$;
4. $(AB)^* \supset B^* A^*$;
   one has equalities in (3) and (4) if A is bounded;
5. $(A^*)^* = \bar{A}$. Not every operator has $(A^*)^*$.

Definition 5.1.2

A is *self-adjoint* if $A = A^*$.

5.2 Differential Operators

Definition 5.2.1

The expression

$$\mathcal{L} = \sum_{i=0}^{n} a_i(x) \frac{\partial^i}{\partial x^i}$$

is a *differential expression.*

We want to turn it into an operator. Take $u : [a, b] \to \mathbb{C}$. To apply $\mathcal{L}$ we need $u \in C_0^n([a, b])$, i.e. compactly supported, n-times differentiable functions. Also, we want the "adjoint" of $\mathcal{L}$.

Given $u, v \in C_0^n([a, b])$, we have

$$\int_a^b (\mathcal{L}u)(x)\, \overline{v(x)}\, dx = \sum_i \int_a^b a_i(x) \frac{\partial^i}{\partial x^i} u(x)\, \overline{v(x)}\, dx = \sum_i (-1)^i \int_a^b u(x)\, \overline{\frac{\partial^i}{\partial x^i} \big(\overline{a_i(x)}\, v(x)\big)}\, dx = \int_a^b u(x)\, \overline{(\mathcal{L}^+ v)(x)}\, dx,$$

where

$$(\mathcal{L}^+ v)(x) = \sum_i (-1)^i \frac{\partial^i}{\partial x^i} \big(\overline{a_i(x)}\, v(x)\big).$$

This is a differential expression.

Definition 5.2.2

$\mathcal{L}^+$ is the *formal adjoint* of $\mathcal{L}$. $\mathcal{L}$ is *formally self-adjoint* if $\mathcal{L} = \mathcal{L}^+$.

Example.

1. Schrödinger "operator"
$$(\mathcal{L}u)(x) = -\frac{\partial^2}{\partial x^2}u(x) + q(x)\,u(x).$$
2. Momentum operator
$$(\mathcal{L}u)(x) = -i\frac{\partial}{\partial x}u(x).$$

Theorem 5.2.3

Define $H = L^2([a,b])$, $\mathcal{D}(A) = C_0^\infty([a,b])$ and set $Au = \mathcal{L}u$. Then A is closable. The closure of A is the *minimal operator* generated by $\mathcal{L}$.

What is the domain of $\overline{A}$? It is the completion of $C_0^\infty([a,b])$ with respect to the norm

$$\|x\|_{\Gamma_A} = \|x\|_H + \|Ax\|_H.$$

When $A = \dfrac{d}{dx}$, this closure is trace 0 Sobolev space.